

Topics on Electrical Engineering

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The document is designed with no intention of publication and has only been designed for education purposes.

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Part I

Magnetic Circuits

Is it a fact - or have I dreamt it - that, by means of electricity, the world of matter has become a great nerve, vibrating thousands of miles in a breathless point of time?

(Nathaniel Hawthorne)

Chapter 1

Magnetic Circuits and Materials

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1.1 Introduction

The primary objective of this lecture is to study devices used in the conversion between **electric** and **mechanical** energy. Of course, as this lecture is about electric drives,¹ emphasis is placed on electromagnetic rotating machinery, by means of which the majority of this energy conversion takes place.

The techniques developed here are generally applicable to a wide range of additional devices including linear machines, actuators, and sensors.

Although **NOT** an electromechanical-energy-conversion device by strictest definitions, the **transformer** is an important component of the overall energy-conversion process and is discussed later in the course. The techniques developed for transformer analysis will form the basis for the ensuing discussion of electric machinery.

Practically, all transformers and electric machinery use **ferromagnetic** material² for shaping and directing the magnetic fields which act as the medium for transferring and converting energy, with Permanent Magnet (PM) materials also being widely used. Without these materials, practical implementations of most familiar electro-mechanical energy-conversion devices would **NOT** be possible.

¹On paper.

²a property of certain materials (such as iron) which results in a significant, observable magnetic permeability, and in many cases, a significant magnetic coercivity, allowing the material to become a permanent magnet. Ferromagnetic materials are noticeably attracted to a magnet, which is a consequence of their substantial magnetic permeability [1].

The ability to analyse and describe systems containing these materials is paramount for designing and understanding these devices.

In this chapter we will develop the necessary tools for analysing of magnetic field systems and will provide a brief introduction to the properties of practical magnetic materials.

1.1.1 Maxwell's Equations

The complete, detailed solution for magnetic fields in most situations of practical engineering interest involves the solution of Maxwell's equations along with various constitutive relationships which describe material properties. While, in practice, **exact** solutions are often **unattainable**, various simplifying assumptions allow us to have approximate solutions.³

We start our journey with the assumption that, for the systems treated in this lecture, the frequencies and sizes involved are such that the displacement-current (**D**) term in Maxwell's equations can be neglected. This term accounts for magnetic fields being produced in space by time-varying electric fields and is associated with electromagnetic radiation. Neglecting this term results in the magneto-quasistatic form of the relevant Maxwell's equations which relate magnetic fields to the currents which produce them [2]:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a} \quad (1.1)$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0 \quad (1.2)$$

Let's have a look at these two (2) equations and try to understand what they mean;

1. Eq. (1.1) states that the line integral of the tangential component of the magnetic field intensity (**H**) around a closed contour (**C**) is equal to the total current passing through any surface (**S**) linking that contour. From Eq. (1.1) we see that the source of **H** is the current density **J**.
2. Eq. (1.2) states that the magnetic flux density (**B**) is conserved.⁴

From these equations we see that the magnetic field quantities can be determined solely from the instantaneous values of the source currents and time variations of the magnetic fields follow directly from time variations of the sources.

A second simplifying assumption involves the concept of the **magnetic circuit**. The general solution for the magnetic field intensity **H** and the magnetic flux density **B** in a structure of complex geometry is extremely difficult. However, a three-dimensional field problem can often be reduced to what is essentially a one dimensional circuit equivalent, giving solutions of **within acceptable engineering accuracy** [3].

³ Although exact analytical solutions cannot be obtained, numerical solutions are quite common and have become indispensable tools for analysis and design. However, their use contributes little to a fundamental understanding of the principles and basic performance of electric machines and therefore should be used after learning the fundamentals.

⁴ i.e., that no net flux enters or leaves a closed surface.

A magnetic circuit consists of a structure composed for the most part of **high-permeability** ($\mu \gg 0$) magnetic material. The presence of high-permeability material tends to cause magnetic flux to be **confined** to the paths defined by the structure, similar to currents being confined to the conductors of an electric circuit.

A simple example of a magnetic circuit is shown in **Fig. 1.2**. The core is assumed to be composed of magnetic material whose permeability is much greater than that of the surrounding air ($\mu \gg \mu_0$). The core is of **uniform** cross section and is excited by a winding of N turns carrying a current of i amperes. This winding produces a magnetic field in the core. Due to the high permeability of the magnetic core, an exact solution would show the magnetic flux is confined **almost entirely to the core**, the field lines follow the path defined by the core, and the flux density is **essentially uniform** over a cross section due to the cross-sectional area being uniform.

The magnetic field can be visualised in terms of flux lines which form closed loops interlinked with the winding.

As applied to the magnetic circuit of **Fig. 1.2**, the source of the magnetic field in the core is the ampere-turn product of Ni . In magnetic circuit terminology Ni is defined as the Magneto-motive Force (MMF) ($\mathcal{F}$) acting on the magnetic circuit.

Although **Fig. 1.2** shows only a single coil, transformers and most rotating machines have at least two (2) windings, and Ni therefore needs to be replaced by the algebraic sum of the ampere-turns of all the windings.

The magnetic flux (ϕ) crossing a surface (S) is the surface integral of the normal component of (B). Therefore

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a} \quad (1.3)$$

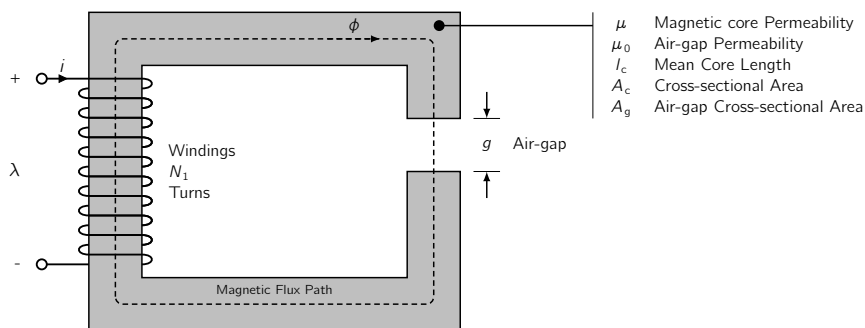


Figure 1.2: Simple magnetic circuit. λ is the winding flux linkage.

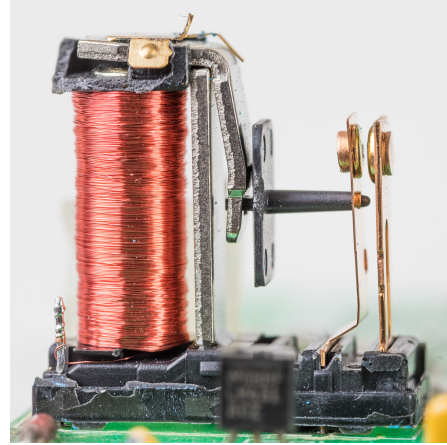


Figure 1.1: An example of a magnetic circuit in action. Relays are used in places where a signal needs to be cut-off at will. These electromechanical relays use magnetism to control the flow of electricity [4].

In SI units⁵, the unit of ϕ is the weber (Wb).

As we can see, Eq. (1.2) states the net magnetic flux entering or leaving a closed surface, equal to the surface integral of $\mathbf{B}$ over that closed surface, is zero.

This is equivalent to saying all flux entering the surface enclosing a volume must leave that volume over some other portion of that surface as magnetic flux lines form closed loops.

These facts can be used to justify the assumption that the magnetic flux density is uniform across the cross section of a magnetic circuit such as the core of **Fig. 1.2**. In this case Eq. (1.3) reduces to the simple scalar equation:

$$\phi_c = B_c A_c, \quad (1.4)$$

where, ϕ_c is the core flux (Wb), B_c is the core flux density ($\text{Wb} \cdot \text{m}^{-2}$), and A_c is the cross-sectional area (m^2). Using Eq. (1.1), the relationship between the MMF acting on a magnetic circuit and the magnetic field intensity in that circuit is:

$$\mathcal{F} = Ni = \oint \mathbf{H} d\mathbf{l}. \quad (1.5)$$

The core dimensions are such that the path length of any flux line is close to the mean core length l_c . As a result, the line integral of Eq. (1.5) becomes simply the scalar product $H_c l_c$ of the magnitude of $\mathbf{H}$ and the mean flux path length l_c . Therefore, the relationship between the MMF and the magnetic field intensity can be written in magnetic circuit terminology as:

$$\mathcal{F} = Ni = H_c l_c \quad (1.6)$$

where H_c is **average magnitude** of $\mathbf{H}$ in the core.

The direction of H_c in the core can be found from the right-hand rule.⁶

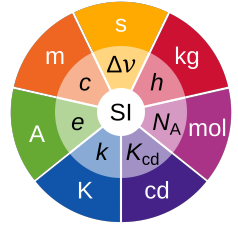
This can be stated in two (2) equivalent but slightly different ways:

- Imagine a current-carrying conductor held in the right hand with the thumb pointing in the direction of current flow; the fingers then point in the direction of the magnetic field created by that current.
- Equivalently, if the coil in **Fig. 1.2** is grasped in the right hand with the fingers pointing in the direction of the current, the thumb will point in the direction of the magnetic fields.

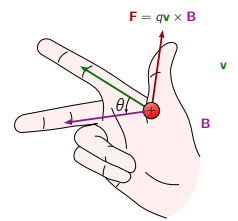
The relationship between the magnetic field intensity $\mathbf{H}$ and the magnetic flux density $\mathbf{B}$ is a property of the material in which the field exists. It is common to assume a **linear relationship**, which means:

$$\mathbf{B} = \mu \mathbf{H} \quad (1.7)$$

where μ is known as the **magnetic permeability**.



⁵ The International System of Units, internationally known by the abbreviation SI (from French *Système international d'unités*), is the modern form of the metric system and the world's most widely used system of measurement. It is the only system of measurement with official status in nearly every country in the world, employed in science, technology, industry, and everyday commerce.

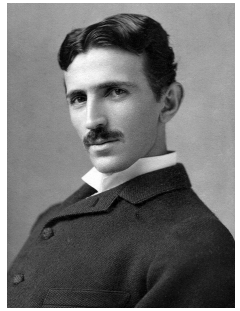


⁶ A mnemonic, used to define the orientation of axes in three-dimensional space and to determine the direction of the cross product of two vectors, as well as to establish the direction of the force on a current-carrying conductor in a magnetic field.

In SI units, $\mathbf{H}$ is measured in units of $\text{A} \cdot \text{m}^{-1}$, $\mathbf{B}$ is in $\text{Wb} \cdot \text{m}^{-2}$, also known as teslas⁷ (T), and μ is in $\text{Wb} \cdot \text{A}^{-1} \cdot \text{Turn}^{-1} \cdot \text{m}^{-1}$, or equivalently ($\text{H} \cdot \text{m}^{-1}$). In SI units, the permeability of free space is $\mu_0 = 4 \times 10^{-7} \text{ H} \cdot \text{m}^{-1}$ and is a **physical constant**.

The permeability of linear magnetic material can be expressed in terms of its **relative permeability**, its value relative to that of free space, or $\mu = \mu_r \mu_0$. Typical values of μ_r range from $2,000 \times 10^3$ to 80,000 for materials used in transformers and rotating machines.

For now we will assume μ_r is a **known** constant, although it actually varies appreciably with the magnitude of the magnetic flux density.



⁷Nikola Tesla
(1856 - 1943)

A Serbian-American engineer, futurist, and inventor. He is known for his contributions to the design of the modern alternating current (AC) electricity supply system. The SI unit is magnetic flux density is named after him.

Transformers are wound on closed cores similar to one we have seen in **Fig. 1.2**. However, energy conversion devices which incorporate a moving element must have air gaps in their magnetic circuits which is also shown in **Fig. 1.2**. When the air-gap length (g) is much smaller than the dimensions of the adjacent core faces, the magnetic flux will follow the path defined by the core and the air gap and the techniques of magnetic-circuit analysis can be used.

If the air-gap length were to become excessively large, the flux will be observed to **leak out** of the sides of the air gap and the techniques of magnetic-circuit analysis will no longer be strictly applicable. Therefore, provided the air-gap length g is sufficiently small, the configuration of **Fig. 1.2** can be analysed as a magnetic circuit with two (2) series components both carrying the same flux ϕ :

- a magnetic core of permeability μ , cross-sectional area A_c and mean length l_c , and
- an air gap of permeability μ_0 , cross-sectional area A_g and length g .

In the core, the magnetic field density is approximated to be:

$$B_c = \frac{\phi}{A_c} \quad (1.8)$$

and in the air gap, the value is:

$$B_g = \frac{\phi}{A_g} \quad (1.9)$$

If we apply Eq. (1.5) to aforementioned equations, this gives:

$$\mathcal{F} = H_c l_c + H_g g \quad (1.10)$$

and using the **linear B - H** relationship⁸ of Eq. (1.7) gives

$$\mathcal{F} = \frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} g \quad (1.11)$$

Here the $\mathcal{F} = Ni$ is the MMF applied to the magnetic circuit. From Eq. (1.10) we can see a portion of the MMF, namely $F_c = H_c l_c$, is required to produce magnetic field in the core whereas the remainder, $F_g = H_g g$ produces magnetic field in the air gap.

For practical applications, B_c and H_c are related **NOT** only to constant permeability μ .

⁸This is, again, just an assumption of simplification.

In fact, B_c is often a **non-linear**, multi-valued function of H_c . Therefore, although Eq. (1.10) continues to hold, it does not lead directly to a simple expression relating the MMF and the flux densities, such as that of Eq. (1.11).

Instead, the specifics of the nonlinear $B_c - H_c$ relation must be used, either graphically or analytically. However, in many cases, the concept of constant material permeability gives results of acceptable engineering accuracy and is frequently used.

Using both Eq. (1.8) and Eq. (1.9), Eq. (1.11) can be rewritten in terms of the flux ϕ_c as:

$$\mathcal{F} = \phi \left(\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g} \right) \quad (1.12)$$

The terms that multiply the flux in Eq. (1.12), are known as the reluctance⁹ ($\mathcal{R}$) of the core and air gap, respectively,

$$\mathcal{R}_c = \frac{l_c}{\mu A_c} \quad (1.13)$$

$$\mathcal{R}_g = \frac{g}{\mu_0 A_g} \quad (1.14)$$

and therefore we arrive at:

$$\mathcal{F} = \phi(\mathcal{R}_c + \mathcal{R}_g) \quad (1.15)$$

Finally, Eq. (1.15) can be inverted to solve for the flux

$$\phi = \frac{\mathcal{F}}{\mathcal{R}_c + \mathcal{R}_g} \quad (1.16)$$

$$\phi = \frac{\mathcal{F}}{\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g}} \quad (1.17)$$

In general, for any magnetic circuit of total reluctance ($\mathcal{R}_{\text{tot}}$), the flux can be found as

$$\phi = \frac{\mathcal{F}}{\mathcal{P}} \quad (1.18)$$

The term which multiplies the MMF is known as the permeance¹⁰ ($\mathcal{P}$) and is the inverse of the reluctance. With that in mind, the total permeance of a magnetic circuit is:

$$\mathcal{P}_{\text{tot}} = \frac{1}{\mathcal{R}_{\text{tot}}} \quad (1.19)$$

As can be seen, Eq. (1.15) and Eq. (1.16) are analogous to the relationships between the current and voltage in an electric circuit.

This analogy is illustrated in **Fig. 1.3**. In the figure (a) shows an electric circuit in which a voltage V drives a current I through resistors R_1 and R_2 . (b) shows the schematic equivalent representation of the magnetic circuit of **Fig. 1.2**. Here we see that the MMF $\mathcal{F}$, analogous to voltage in the electric circuit, drives a flux ϕ , analogous to the current in the electric circuit, through the combination of the reluctances of the core $\mathcal{R}_c$ and the air gap $\mathcal{R}_g$. This analogy between the solution of electric

⁹A concept used in the analysis of magnetic circuits and defined as the ratio of MMF to magnetic flux. It represents the opposition to magnetic flux, and depends on the geometry and composition of an object.

It is analogous to electrical resistance in an electrical circuit in that resistance is a measure of the opposition to the electric current. The definition of magnetic reluctance is analogous to Ohm's law in this respect. However, magnetic flux passing through a reluctance does **NOT** give rise to dissipation of heat as it does for current through a resistance.

¹⁰The inverse of reluctance and is a measure of the quantity of magnetic flux for a number of current-turns. A magnetic circuit almost acts as though the flux is conducted, therefore permeance is larger for large cross-sections of a material and smaller for smaller cross section lengths. This concept is analogous to electrical conductance in the electric circuit.

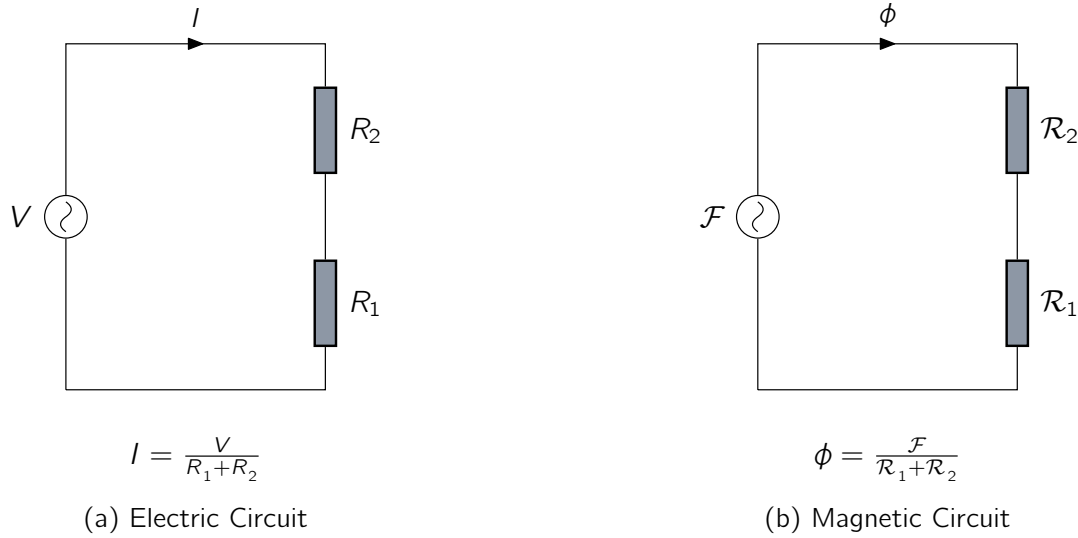


Figure 1.3: A visual comparison to show the conceptual similarities between an electric and magnetic circuit.

and magnetic circuits can often be exploited to produce simple solutions for the fluxes in magnetic circuits of considerable complexity.

The fraction of the MMF required to drive flux through each portion of the magnetic circuit, commonly referred to as the MMF **drop** across that portion of the magnetic circuit, varies in proportion to its reluctance.¹¹

Consider the magnetic circuit of **Fig. 1.2**. From Eq. (1.13) we see that high material permeability can result in low core reluctance, which can often be made much smaller than that of the air gap. In this case, the reluctance of the core can be neglected and the flux can be found from Eq. (1.16) in terms of $\mathcal{F}$ and the air-gap properties alone:

$$\phi \approx \frac{\mathcal{F}}{\mathcal{R}_s} = \frac{\mathcal{F}\mu_0 A_g}{g} = Ni \frac{\mu_0 A_g}{g} \quad (1.20)$$

While we treat permeability as constant here, it is worth remembering that practical magnetic materials have permeabilities which are not constant but vary with the flux level. However, from Eq. (1.13) to Eq. (1.16) we see that as long as this permeability remains sufficiently large, its variation will not significantly affect the performance of a magnetic circuit in which the dominant reluctance is that of an air gap.

In practical systems, the magnetic field lines “fringe” outward somewhat as they cross the air gap. Provided this fringing effect is **NOT** excessive, the magnetic-circuit concept remains applicable. The effect of these fringing fields is to increase the effective cross-sectional area A_g of the air gap. Various empirical methods have been developed to account for this effect.

A correction for fringing fields in short air gaps can be made by adding the gap length to each of the two dimensions making up its cross-sectional area. For simplicity, we shall ignore fringing effects.

¹¹This is directly analogous to the voltage drop across a resistive element in an electric circuit.

1.2 Flux Linkage, Inductance, and Energy

When a magnetic field varies with time, an electric field is produced in space as per Faraday's¹² law:

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a} \quad (1.21)$$

Eq. (1.22) states:

The line integral of the electric field intensity $\mathbf{E}$ around a closed contour $\mathcal{C}$ is equal to the time rate of change of the magnetic flux linking, passing through, that contour.

In magnetic structures with windings of high electrical conductivity, such as in **Fig. 1.2**, it can be shown that the $\mathbf{E}$ field in the wire is extremely small and can be neglected, so that the left-hand side of Eq. (1.22) reduces to the negative of the induced voltage e at the winding terminals. In addition, the flux on the right-hand side of Eq. (1.22) is dominated by the core flux ϕ_c . Since the winding, and hence the contour $\mathcal{C}$, links the core flux N times, Eq. (1.22) reduces to:

$$e = N \frac{d\phi}{dt} = \frac{d\lambda}{dt} \quad (1.22)$$

where λ is the flux linkage of the winding and is defined as:

$$\lambda = N\phi \quad (1.23)$$

Flux linkage is measured in units of Wb (or equivalently Wb · Turn). The symbol ϕ is used to indicate the **instantaneous value** of a time-varying flux. In general the flux linkage of a coil is equal to the surface integral of the normal component of the magnetic flux density integrated over any surface spanned by that coil.

Direction of the induced voltage (e) is defined by Eq. (1.22) so if the winding terminals were short-circuited, a current would flow in such a direction as to oppose the change of flux linkage.

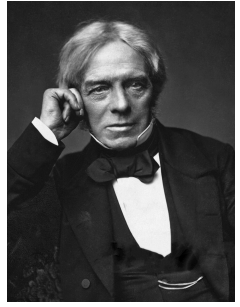
For a magnetic circuit composed of magnetic material of constant magnetic permeability or which includes a dominating air gap, the relationship between λ and i will be linear and we can define the inductance L as:

$$L = \frac{\lambda}{i} \quad (1.24)$$

Substitution of Eq. (1.5), Eq. (1.18) and Eq. (1.23) into Eq. (1.24) gives:

$$L = \frac{N^2}{\mathcal{R}_{\text{total}}}$$

From which we see that the inductance of a winding in a magnetic circuit is proportional to the square of the turns and inversely proportional to the reluctance of the magnetic circuit associated with that winding.



¹²Michael Faraday
(1791 - 1867)

An English chemist and physicist who contributed to the study of electro-chemistry and electromagnetism. His main discoveries include the principles underlying electromagnetic induction, diamagnetism, and electrolysis. Although Faraday received little formal education, as a self-made man, he was one of the most influential scientists in history [5]. It was by his research on the magnetic field around a conductor carrying a direct current Faraday established the concept of the electromagnetic field in physics. Faraday also established magnetism could affect rays of light and that there was an underlying relationship between the two phenomena. He similarly discovered the principles of electromagnetic induction, diamagnetism, and the laws of electrolysis. His inventions of electromagnetic rotary devices formed the foundation of electric motor technology, and it was largely due to his efforts that electricity became practical for use in technology. The SI unit of capacitance, the farad, is named after him.

As an example, from Eq. (1.20), under the assumption that the reluctance of the core is negligible as compared to that of the air gap, the inductance of the winding in **Fig. 1.2** is equal to

$$L = \frac{N^2}{(g/\mu_0 A_g)} = \frac{N^2 \mu_0 A_g}{g} \quad (1.25)$$

Inductance is measured in henrys¹³ (H) or weber-turns per ampere.

Eq. (1.25) shows the dimensional form of expressions for inductance;

Inductance is proportional to the square of the number of turns, to a magnetic permeability, and to a cross-sectional area and is inversely proportional to a length.

It must be emphasised that, strictly speaking, the concept of inductance requires a linear relationship between flux and MMF. Therefore, it cannot be rigorously applied in situations where the nonlinear characteristics of magnetic materials, dominate the magnetic system.

However, for many situations of practical interest, the reluctance of the system is dominated by that of an air gap, which is of course linear, and the non-linear effects of the magnetic material can be ignored. In other cases it may be perfectly acceptable to assume an average value of magnetic permeability for the core material and to calculate a corresponding average inductance which can be used for calculations of reasonable engineering accuracy.



¹³ Joseph Henry
(1797 - 1878)

An American physicist and inventor who served as the first secretary of the Smithsonian Institution. While building electromagnets, discovered self-inductance. He also discovered mutual inductance independently of Michael Faraday, though Faraday was the first to make the discovery and publish his results and work on the electromagnetic relay was the basis of the practical electrical telegraph. In his honour, the SI unit of inductance is named the henry.

Exercise 1.1 Plotting Permeability

Using Python, plot the inductance of the magnetic circuit given in (**TutBook**) **Ex. 1** as a function of core permeability over the range $100 \leq \mu_r \leq 1,000.00 \times 10^5$.

SOLUTION The script for the plot is as follows:

```
import numpy as np
import matplotlib.pyplot as plt

mu0 = 4 * np.pi * 1e-7          # (H/m) permeability of free space
lc = 0.3                        # (m) length
Ac = 9e-4                       # (m2) cross-sectional area
Ag = 9e-4                       # (m2) cross-sectional area
g = 5e-4                        # (m) air gap length
N = 500                         # (-) number of turns
Bc = 1                          # (T) Core magnetic field density

Rel_g = (g) / (mu0 * Ag)        # (Ampere-Turns/Weber) air-gap reluctance

mur = np.zeros(1000)           # Relative Permeability empty array for looping

for i in range(len(mur)):
    mur[i] = 100 + (100000 - 100) * (i - 1)/1000

Rel_c = (lc) / (mur * mu0 * Ac) # (Ampere-Turns/Weber) core reluctance
R_tot = Rel_c + Rel_g          # (Ampere-Turns/Weber) total reluctance

L = N**2 / R_tot               # (H) Inductance

# -- Plotting
```

```
plt.xlabel('Core Relative Permeability ($\mu_r$) [H/m]')
plt.ylabel('Inductance [H]')
plt.xlim(0, 0.8e5)
plt.ylim(0, 1)
plt.title("Core Relative Permeability vs. Inductance Graph", color='#c6d0f5')
plt.plot(mur, L, '#ef9f76', linewidth=5)
fig.tight_layout()
plt.savefig('images/python-matplot-fig.svg')
```

Please observe the resultant plot. The figure clearly confirms that, for the magnetic circuit of this example, the inductance is quite **insensitive** to relative permeability until the relative permeability drops to on the order of 1000. Therefore, as long as the effective relative permeability of the core is "large", which in this case greater than 1000, any non-linearities in the properties of the core material will have little effect on the terminal properties of the inductor.

Fig. 1.4 shows a magnetic circuit with an air gap and two (2) windings. In this case please observe the MMF acting on the magnetic circuit is given by the total ampere-turns acting on the magnetic circuit¹⁴ and that the reference directions for the currents have been chosen to produce flux in the same direction.

¹⁴i.e., the net ampere turns of both windings

The total MMF is therefore calculated to be:

$$\mathcal{F} = N_1 i_1 + N_2 i_2 \quad (1.26)$$

and from Eq. (1.20), with the reluctance of the core neglected and assuming $A_c = A_g$, the core flux ϕ is:

$$\phi = (N_1 i_1 + N_2 i_2) \frac{\mu_0 A_c}{g} \quad (1.27)$$

In Eq. (1.27), ϕ is the resultant core flux produced by the total MMF of the two windings.

It is this resultant ϕ which determines the operating point of the core material.

If Eq. (1.27) is broken up into individual currents, the resultant flux linkages of coil 1 can be expressed

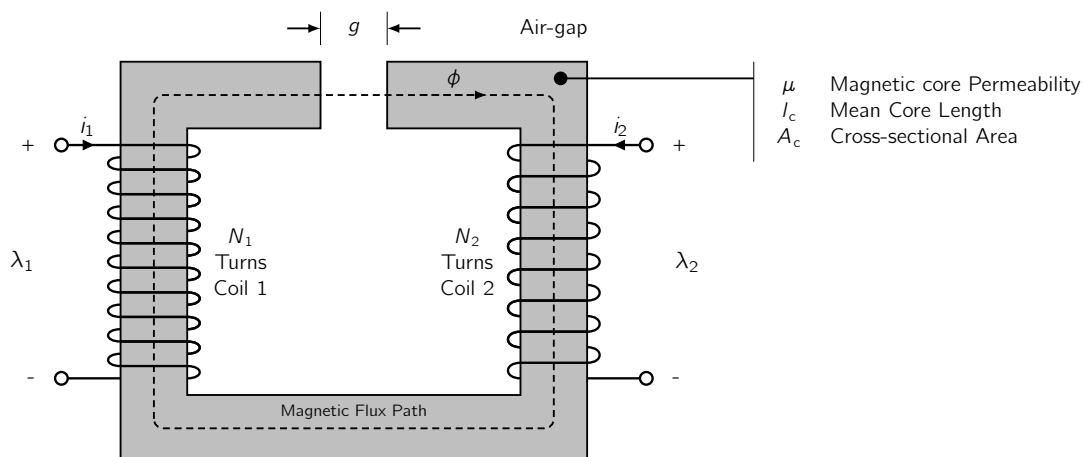


Figure 1.4: An example of a magnetic circuit, excited with two windings.

as

$$\lambda_1 = N_1 \phi = N_1^2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_2 \quad (1.28)$$

which can be written in a simpler form as:

$$\lambda_1 = L_{11} i_1 + L_{12} i_2 \quad (1.29)$$

where:

$$L_{11} = N_1^2 \frac{\mu_0 A_c}{g} \quad (1.30)$$

is the **self-inductance** of Coil 1, and $L_{11} i_1$ is the flux linkage of Coil 1 due to its own current i_1 . The **mutual inductance** between coils 1 and 2 is:

$$L_{12} = N_1 N_2 \frac{\mu_0 A_c}{g} \quad (1.31)$$

and $L_{12} i_2$ is the flux linkage of coil 1 due to current i_2 in the other coil. Similarly, the flux linkage of coil 2 is

$$\lambda_2 = N_2 \phi = N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_2^2 \left(\frac{\mu_0 A_c}{g} \right) i_2 \quad (1.32)$$

or

$$\lambda_2 = L_{21} i_1 + L_{22} i_2 \quad (1.33)$$

where $L_{21} = L_{12}$ is the mutual inductance and

$$L_{22} = N_2^2 \frac{\mu_0 A_c}{g} \quad (1.34)$$

is the self-inductance of coil 2.

It is important to note that the resolution of the resultant flux linkages into the components produced by i_1 and i_2 is based on superposition of the individual effects and therefore implies a linear flux-MMF relationship.¹⁵

¹⁵characteristic of materials of constant permeability.

Substitution of Eq. (1.24) in Eq. (1.22) gives:

$$e = \frac{d(Li)}{dt} \quad (1.35)$$

for a magnetic circuit with a single winding. For a static magnetic circuit, the inductance is fixed,¹⁶ and this equation reduces to the familiar circuit-theory form:

¹⁶assuming that material non-linearities do not cause the inductance to vary

$$e = L \frac{di}{dt} \quad (1.36)$$

However, in electromechanical energy conversion devices, inductances are often time-varying, and Eq. (1.35) must be written as

$$e = L \frac{di}{dt} + i \frac{dL}{dt} \quad (1.37)$$

Note that in situations with multiple windings, the total flux linkage of each winding must be used in Eq. (1.22) to find the winding-terminal voltage. The power at the terminals of a winding on

a magnetic circuit is a measure of the rate of energy flow into the circuit through that particular winding. The **power**, p , is determined from the product of the voltage and the current

$$p = ie = i \frac{d\lambda}{dt} : \quad (1.38)$$

and its unit is watts (W), or ($\text{J} \cdot \text{s}^{-1}$). Therefore, the change in magnetic stored **energy** ΔW in the magnetic circuit in the time interval t_1 to t_2 is

$$\Delta W = \int_0^{t_2} p dt = \int_0^{\lambda_2} i d\lambda \quad (1.39)$$

In SI units, the magnetic stored energy W is measured in joules (J). For a single-winding system of constant inductance, the change in magnetic stored energy as the flux level is changed from λ_1 to λ_2 can be written as

$$\Delta W = \int_{\lambda_1}^{\lambda_2} i d\lambda = \int_{\lambda_1}^{\lambda_2} \frac{\lambda}{L} d\lambda = \frac{1}{2L} (\lambda_2^2 - \lambda_1^2) \quad (1.40)$$

The total magnetic stored energy at any given value of λ can be found from setting λ_1 equal to zero:

$$W = \frac{1}{2L} \lambda^2 = \frac{L}{2} i^2 \quad (1.41)$$

Information: The First Electrical Motor

Before their modern versions, motors using electrostatic forces were investigated. The first electric motors were simple electrostatic devices described in experiments by **Andrew Gordon** and **Benjamin Franklin** in the 1740s [7]. However, due to the difficulty of generating the high voltages they required, electrostatic motors were never used for practical purposes.

The invention of the electro-chemical battery by **Alessandro Volta** in 1799 [8] made persistent electric currents possible. **Hans Christian Ørsted** discovered in 1820 that an electric current creates a magnetic field, which can exert a force on a magnet. It only took a few weeks for **André-Marie Ampère** to develop the first formulation of the electromagnetic interaction and present the Ampère's force law, that described the production of mechanical force by the interaction of an electric current and a magnetic field [9].

Michael Faraday gave the first demonstration of the effect with a rotary motion in 1821 [10]. A free-hanging wire was dipped into a pool of mercury, on which a PM was placed. When a current was passed through the wire, the wire rotated around the magnet, showing that the current gave rise to a close circular magnetic field around the wire [11].

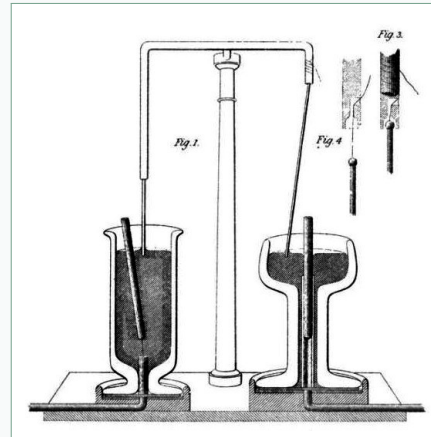


Figure 1.5: Faraday's electromagnetic experiment, 1821, the first demonstration of the conversion of electrical energy into motion [6].

1.3 Magnetic Material Properties

In the context of electromechanical energy conversion devices, the importance of magnetic materials is twofold. Through their use it is possible to obtain **large magnetic flux densities** with relatively **low levels of magnetising force**.

As magnetic forces and energy density increase with increasing flux density, this effect plays a large role in the performance of energy-conversion devices.

In addition, magnetic materials can be used to **constrain and direct** magnetic fields in well-defined paths.

For example, in a transformer, they are used to maximise the coupling between the windings as well as to lower the excitation current¹⁷ required for transformer operation. In electric machinery, magnetic materials are used to shape the fields to obtain desired torque-production and electrical terminal characteristics.

Therefore a trained engineer can use magnetic materials to achieve specific desirable device characteristics.

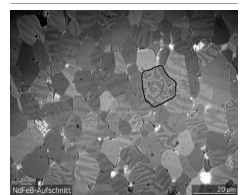
¹⁷ The current or amperes required to energise the core. Even with zero load, a transformer will draw a small amount of current due to internal loss. The excitation current is made up of two components. The real component in the form of losses that are commonly referred to as no-load losses. The second form is reactive power measured in KVAR.

1.3.1 Ferromagnetic Materials

These materials are typically composed of iron and alloys of iron with cobalt, tungsten, nickel, aluminium, and other metals, are by far the most common magnetic materials. Although these materials are characterised by a wide range of properties, the basic phenomena responsible for their properties are common to them all.

Ferromagnetic materials are found to be composed of a large number of domains.¹⁸ In a non-magnetised sample of material, the domain magnetic moments are randomly oriented, and the net resulting magnetic flux in the material is zero. When an external magnetising force is applied to this material, the domain magnetic moments tend to **align with the applied magnetic field**. As a result, the domain magnetic moments add to the applied field, producing a much larger value of flux density than would exist due to the magnetising force alone. Therefore, the effective permeability (μ), equal to the ratio of the total magnetic flux density to the applied magnetic-field intensity, is large compared with the permeability of free space (μ_0). As the magnetising force is increased, this behaviour continues until all the magnetic moments are aligned with the applied field at which point they can no longer contribute to increasing the magnetic flux density, and the material is said to be **fully saturated**.

In the absence of an externally applied magnetising force, the domain magnetic moments naturally align along certain directions associated with the **crystal structure of the domain**, known as **axes of easy magnetisation**. Therefore if the applied magnetising force is reduced, the domain magnetic



¹⁸ i.e., regions in which the magnetic moments of all the atoms are parallel, giving rise to a net magnetic moment for that domain [12].



Figure 1.6: The periodic table of elements. Of all the elements that exist only a few of them have the ability to generate a magnetic field without any external need. The elements which exhibit ferromagnetism has been highlighted.

moments relax to the direction of easy magnetism nearest to that of the applied field. As a result, when the applied field is reduced to zero, although they will tend to relax towards their initial orientation, the magnetic dipole moments will no longer be totally random in their orientation; they will retain a net magnetisation component along the applied field direction. It is this effect which is responsible for the phenomenon known as magnetic hysteresis.

Due to this hysteresis effect, the relationship between B and H for a ferromagnetic material is **both non-linear and multi-valued**.

In general, material properties cannot be described analytically. They are commonly presented in graphical form as a set of empirically determined curves based on test samples.

The most common curve used to describe a magnetic material is the B-H curve or hysteresis loop which an example is shown in **Fig. 1.7**. Here, only the first quadrant is shown as it is the most important part when designing a machine.

In practice, each curve is obtained while cyclically varying the applied magnetising force between equal positive and negative values of fixed magnitude. Hysteresis causes these curves to be multi-valued. After several cycles the B-H curves form closed loops, which is shown in **Fig. 1.9**.

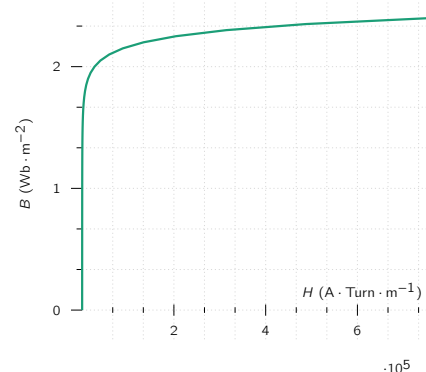


Figure 1.7: An example of a B-H curve showing only the first quadrant.

1.4 AC Excitation

In Alternating Current (AC) power systems, the wave-forms of voltage and flux **closely approximate** sinusoidal functions of time.¹⁹ We will now look at the excitation characteristics and losses associated with steady-state AC operation of magnetic materials under such operating conditions. We use as our model a closed-core magnetic circuit,²⁰ such as that shown in **Fig. 1.2**. The magnetic path length is l_c , and the cross-sectional area is A_c throughout the length of the core.

¹⁹We say here closely as we must take into account environmental factors, such as noise and interference, which distorts the waveform.

²⁰This means our circuit has no air gap.

We further assume a sinusoidal variation of the core flux $\phi(t)$, which gives:

$$\phi(t) = \phi_{\max} \sin \omega t = A_c B_{\max} \sin \omega t \quad (1.42)$$

where,

- $\phi_{\max}$ is the amplitude of core flux ϕ (Wb),
- $B_{\max}$ is the amplitude of flux density B_c (T),
- ω is the angular frequency defined as $\omega = 2\pi f$, and
- f is the frequency (Hz).

From Eq. (1.22), the voltage induced in the N -turn winding is:

$$e(t) = \omega N \phi_{\max} \cos \omega t = E_{\max} \cos \omega t \quad (1.43)$$

where

$$E_{\max} = \omega N \phi_{\max} = E_{\max} \cos \omega t \quad (1.44)$$

In steady-state AC operation, we usually are more interested in the Root-Mean Square (RMS) values of voltages and currents than their instantaneous or maximum values. In general, the RMS value of a periodic function of time, say $f(t)$, of period T is defined as:

$$F_{\text{rms}} = \sqrt{\left(\frac{1}{T} \int_0^T f^2(t) dt \right)} \quad (1.45)$$

From Eq. (1.45), the RMS value of a sine wave can be shown to be $1/\sqrt{2}$ times its peak value. Therefore, the RMS value of the induced voltage is:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N A_c B_{\max} = \sqrt{2} \pi f N A_c B_{\max} \quad (1.46)$$

An excitation current i_ϕ , corresponding to an excitation MMF $Ni_\phi(t)$, is **required** to produce the flux $i_\phi(t)$ in the core.

Because of the non-linear magnetic properties of the core, the excitation current corresponding to a sinusoidal core flux will be **non-sinusoidal**.

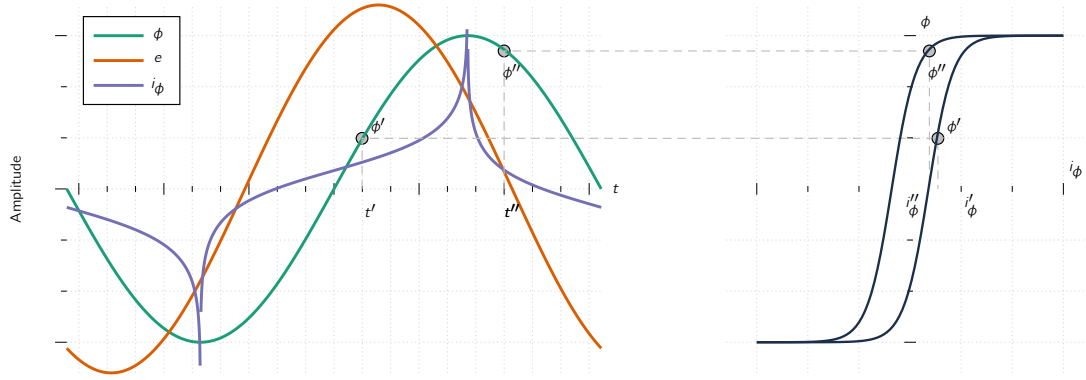


Figure 1.8: Excitation phenomena where left side shows the Voltage, flux, and exciting current and right shows the corresponding hysteresis loop. The current is not to scale.

As B_c and H_c are related to ϕ and i_ϕ by known **geometric constants**, the AC hysteresis loop of **Fig. 1.8** has been drawn in terms of $\phi = B_c A_c$ and is $i_\phi = H_c l_c / N$. The sine waves of induced voltage (e), and flux (ϕ), are shown in the left side of **Fig. 1.8**.

At any given time, the value of i_ϕ corresponding to the given value of flux can be found directly from the hysteresis loop.

As an example, at time t' the flux is ϕ' and the current is i' , and at time t'' the corresponding values are ϕ'' and i'' . Please observe that since the hysteresis loop is multi-valued, it is necessary to be careful to pick the rising-flux values from the **rising-flux portion** of the hysteresis loop. In the same context, the falling-flux portion of the hysteresis loop must be selected for the falling-flux values.

A second point worth mentioning is, due to the hysteresis loop “flattening out” from the **saturation effects**, the waveform of the exciting current is sharply peaked. Its RMS value $I_{\phi, \text{rms}}$ defined by Eq. (1.45), where T is the period of a cycle. It is related to the corresponding RMS value $H_{c, \text{rms}}$ of H_c by the relationship:

$$I_{\phi, \text{rms}} = \frac{l_c H_{c, \text{rms}}}{N} \quad (1.47)$$

The AC excitation characteristics of core materials are often described in terms of RMS volt-amperes (V·A) rather than a magnetisation curve relating $\mathbf{B}$ and $\mathbf{H}$. The theory behind this representation can be explained by combining Eq. (1.47) and Eq. (1.48). Therefore, from Eq. (1.47), and Eq. (1.48) the RMS volt-amperes required to excite the core of **Fig. 1.2** to a specified flux density is equal to

$$\begin{aligned} E_{\text{rms}} I_{\phi, \text{rms}} &= \sqrt{2} \pi f N A_c B_{\text{max}} \frac{l_c H_{c, \text{rms}}}{N} \\ &= \sqrt{2} f B_{\text{max}} H_{\text{rms}} (A_c l_c) \end{aligned} \quad (1.48)$$

In Eq. (1.48), the product $A_c l_c$ can be seen to be equal to the volume of the core and hence the RMS exciting volt-amperes required to excite the core with sinusoidal can be seen to be proportional to the frequency of excitation, the core volume and the product of the peak flux density and the RMS magnetic field intensity.

For a magnetic material of mass density ρ_c , the mass of the core is $A_c l_c \rho_c$ and the exciting RMS voltamperes per unit mass, S_a , can be expressed as

$$S_a = \frac{E_{rms} l_{\phi, rms}}{\text{mass}} = \sqrt{2} \pi f \left(\frac{B_{max} H_{rms}}{\rho_c} \right) \quad (1.49)$$

Observe that excitation of the material really depends on frequency and B_{max} . As a result, the ac excitation requirements for a magnetic material are often supplied by manufacturers in terms of RMS voltamperes per unit weight as determined by laboratory tests on closed-core samples of the material.

The exciting current supplies the MMF required to produce the core flux and the power input associated with the energy in the magnetic field in the core. Part of this energy is dissipated as losses and results in heating of the core. The rest appears as reactive power associated with energy storage in the magnetic field. This reactive power is not dissipated in the core; it is cyclically supplied and absorbed by the excitation source.

There are two (2) loss mechanisms are associated with time-varying fluxes in magnetic materials.

Eddy Current Losses The first is ohmic ($I^2 R$) heating, associated with induced currents in the core material. From Faraday's law (Eq. (1.22)) we see that time-varying magnetic fields give rise to electric fields. In magnetic materials, these electric fields result in induced currents, commonly referred to as eddy currents, which circulate in the core material and **oppose** changes in flux density in the material. To counteract the corresponding demagnetising effect, the current in the exciting winding must increase.

This causes the resultant "dynamic" B-H loop under ac operation to become somewhat "fatter" than the hysteresis loop for slowly varying conditions, and this effect increases as the excitation frequency is increased.

It is for this reason that the characteristics of electrical steels vary with frequency and therefore manufacturers typically supply characteristics over the expected operating frequency range of a particular electrical steel.

To reduce the effects of eddy currents, magnetic structures are usually built of **thin sheets of laminations** of the magnetic material.²¹ These laminations, which are **aligned in the direction of the field lines**, are insulated from each other by an oxide layer on their surfaces or by a thin coat of insulating enamel or varnish. This greatly reduces the magnitude of the eddy currents since the layers of insulation interrupt the current paths.

The thinner the laminations, the lower the losses.

In general, eddy-current loss tends to increase as the **square of the excitation frequency** and also as the square of the peak flux density.



²¹The primary reason is to reduce the volume in which the current flows. Smaller the volume, smaller the losses [13].

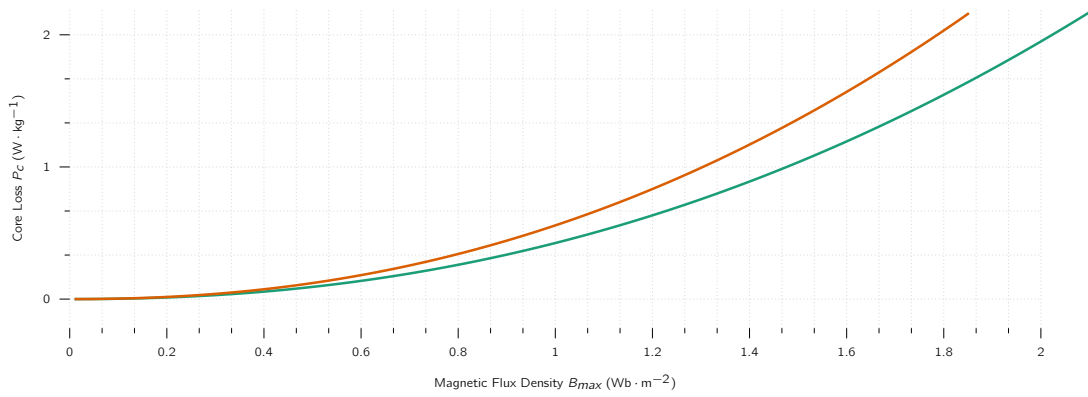


Figure 1.10: Core-loss density of M130-27S, Electric Grain steel under 50 Hz and 60 Hz. The following plot has been generated using the Steinmetz equation.

Hysteresis Losses The second loss mechanism is due to the **hysteretic nature of magnetic material**. In a magnetic circuit similar to **Fig. 1.2**, a time-varying excitation will cause the magnetic material to undergo a cyclic variation described by a hysteresis loop such as that shown in **Fig. 1.9**.

We can use Eq. (1.40) to calculate the energy input W to the magnetic core of **Fig. 1.2** as the material undergoes a single cycle:

$$W = \oint i_{\phi} d\lambda = \oint \left(\frac{H_c l_c}{N} \right) (A_c N dB_c) = A_c l_c \oint H_c dB_c \quad (1.50)$$

Looking at Eq. (1.50), we see that $A_c l_c$ is the volume of the core and the integral is the area of the AC hysteresis loop, and we see each time the magnetic material undergoes a cycle, there is a net energy input into the material. This energy is required to move around the magnetic dipoles in the material and is dissipated as heat in the material.

Therefore for a given flux level, the corresponding hysteresis losses are proportional to the area of the hysteresis loop and to the total volume of material.

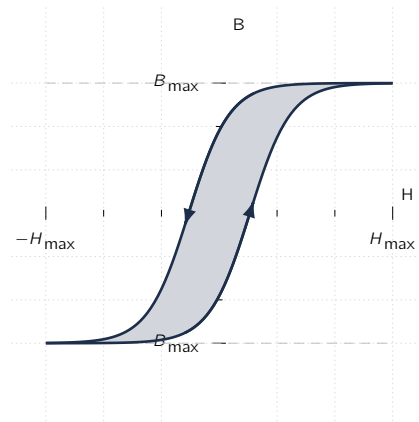


Figure 1.9: Hysteresis loop; hysteresis loss is proportional to the loop area (shaded).

Since there is an energy loss per cycle, hysteresis power loss is proportional to the frequency of the applied excitation.

In general, these losses depend on the metallurgy of the material as well as the flux density and frequency. Information on core loss is typically presented in graphical form. It is plotted in terms of watts per unit weight as a function of flux density; often a family of curves for different frequencies are given. **Fig. 1.10** shows the core loss (P_c) for M130-27S grain-oriented electrical steel at 60 Hz and 50 Hz.

Nearly all transformers and certain sections of electric machines use sheet-steel material that has **highly favourable directions of magnetisation** along which the core loss is low and the permeability is high. This material is termed grain-oriented steel.

Exercise 1.2 Time Varying Voltage in Magnetic Circuit

For the magnetic circuit of (TutBook) Ex. 1 with the following diagram:

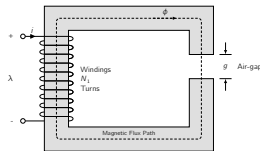


Figure 1.11: The circuit in question.

Find:

1. the inductance (L),
2. the magnetic stored energy W for $B_c = 1 \text{ T}$.
3. the induced voltage e for a 60 Hz time-varying core flux of the form $B_c = 1 \sin \omega t$.

SOLUTION

- a. From the flux linkage relationship, we know that:

$$L = \frac{\lambda}{i} = \frac{N\phi}{i} = \frac{N^2}{\mathcal{R}_\alpha + \mathcal{R}_\alpha}$$

$$= \frac{500^2}{4.46 \times 10^5} = 0.56 \text{ H}$$

Please observe the core reluctance is much smaller than that of the gap ($\mathcal{R}_c \ll \mathcal{R}_g$). Therefore, to a good approximation the inductance is dominated by the gap reluctance, i.e.,

$$L \approx \frac{N^2}{R} = 0.57 \text{ H}$$

- b. In (TutBook) Ex. 1 we found that when $B_c = 1 \text{ T}$, $i = 0.80 \text{ A}$. Thus from Eq. 1.46,

$$W = \frac{1}{2} Li^2 = \frac{1}{2} (0.56) (0.80)^2 = 0.18 \text{ J}$$

- c. The induced voltage is calculated as follows:

$$\begin{aligned} e &= \frac{d\lambda}{dt} = N \frac{d\phi}{dt} = NA_c \frac{dB_c}{dt} \\ &= 500 \times (0.0009) \\ &\times (377 \times 1.0 \cos(377t)) \\ &= 170 \cos(377t) \quad \blacksquare \end{aligned}$$

Information: Orientation of Atoms

The reason for this property lies in the atomic structure of a crystal of the silicon-iron alloy, which is a body-centred cube. Each cube has an atom at each corner as well as one in the centre of the cube. In the cube, the easiest axis of magnetisation is the cube edge; the diagonal across the cube face is more difficult, and the diagonal through the cube is the most difficult. By suitable manufacturing techniques most of the crystalline cube edges are aligned in the rolling direction to make it the favourable direction of magnetisation. The behaviour in this direction is superior in core loss and permeability to non-oriented steels in which the crystals are randomly oriented to produce a material with characteristics which are uniform in all directions. As a result, oriented steels can be operated at higher flux densities than the non-oriented grades.

Non-oriented electrical steels are used in applications where the flux does not follow a path which can be oriented with the rolling direction or where low cost is of importance. In these steels the losses are somewhat higher and the permeability is very much lower than in grain-oriented steels.

Chapter 2

Transformers

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2.1 The Stationary Electric Machine

Before we continue on with electric drives, it is worthwhile to discuss particular aspects of the theory of magnetically-coupled circuits, with a special emphasis on the **transformer**. Although the transformer is **NOT** an energy conversion device,¹ by the strictest of the definition, it is nevertheless an indispensable component in many energy conversion systems and used heavily by the power generation, distribution, and transmission companies as it allows the:

- electric generation at the most economical generator voltage,
- power transfer at the most economical transmission voltage, and
- power utilization at the most suitable voltage,

for the particular utilisation device. The transformer is also widely used in low-power, low-current electronic and control circuits for performing such functions as:

¹In the end all it does is to convert one level of AC voltage to another level of AC voltage. It does **NOT** convert energy.

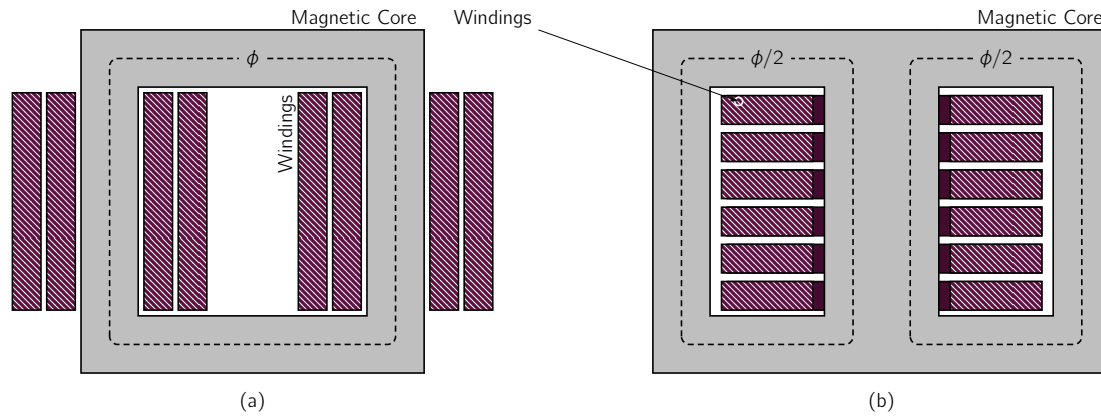


Figure 2.1: Schematic views of (a) core-type and (b) shell-type transformers.

- matching the impedances of a source and its load for maximum power transfer,
- isolating one circuit from another, or
- isolating direct current while maintaining ac continuity between two circuits.

The transformer is one of the simpler devices made from two (2) or more electric circuits coupled by a **common magnetic circuit**. Its analysis involves many of the principles essential to the study of electric drives.

Therefore, our study of the transformer will serve as a bridge between the introduction to magnetic-circuit analysis we conducted previously and the more detailed study of electric drives to follow.²



²In 1884, Zipernowsky, Bláthy and Déri (ZBD), associated with the Ganz Works, determined open-core devices were impracticable, as they were incapable of reliably regulating voltage. In their joint 1885 patent applications for novel transformers (later called ZBD transformers), they described two designs with closed magnetic circuits where copper windings were either wound around an iron wire ring core or surrounded by an iron wire core. The two designs were the first application of the two basic transformer constructions in common use to this day, termed “core form” or “shell form”.

2.1.1 An Overview of Transformers

To give a brief definition, a transformer consists of two (2) or more windings coupled by **mutual magnetic flux**. If one of these windings, **the primary**, is connected to an AC source, alternating flux will be produced whose amplitude will depend on the primary voltage, the frequency of the applied voltage, and finally, the number of turns. The mutual flux will link the other winding, **the secondary**, and will induce a voltage in it whose value will depend on the number of secondary turns as well as the magnitude of the mutual flux and the frequency. By properly proportioning the number of primary and secondary turns, almost any desired voltage ratio, or ratio of transformation, can be obtained.

The essence of transformer action requires only the existence of a **time-varying mutual flux** linking the two (2) windings. Such action can occur for two (2) windings coupled through air, but coupling between the windings can be made much more effectively using a core of iron or other ferromagnetic material, as most of the flux is then confined to a definite, high-permeability path, linking both windings.

Such a transformer is commonly called an iron-core transformer which makes majority of the market.³

The chapter is concerned primarily with iron-core transformers. As we looked at the topic previously, to reduce the losses caused by eddy currents in the core, the magnetic circuit usually consists of a **stack of thin laminations**. Two common construction types are shown in **Fig. 2.1**:

Core Shown in **Fig. 2.1a**, the windings are **wound around two legs** of a rectangular magnetic core.

Shell Shown in **Fig. 2.1b** the windings are wound **around the centre leg** of a three-legged core.

Silicon-steel laminations (0,55 mm) thick are generally used for transformers operating at frequencies below a few hundred Hz. Silicon steel has the desirable properties of low cost, low core loss, and high permeability at high flux densities.⁴

The cores of small transformers used in communication circuits at high frequencies and low energy levels are sometimes made of compressed powdered ferromagnetic alloys known as ferrites.⁵

In each of these configurations, most of the flux is confined to the core and therefore links both windings. The windings also produce additional flux, known as **leakage flux**, which links one winding without linking the other.

Although leakage flux is a small fraction of the total flux, it plays an important role in determining the overall behaviour of the transformer which will be looked later.

In practice, leakage is reduced by subdividing the windings into sections placed as close together as possible. In the core-type construction, each winding consists of two sections, one section on each of the two legs of the core, the primary and secondary windings being concentric coils. In the shell-type construction, variations of the **concentric-winding** arrangement may be used, or the windings may consist of a number of thin “pancake” coils assembled in a stack with primary and secondary coils interleaved.

Information: Distribution Transformers

Provides a final voltage reduction in the electric power distribution system, stepping down the voltage used in the distribution lines to the level used by the customer.⁶ If mounted on a utility pole, they are called pole-mount transformers. When placed either at ground level or underground, distribution transformers are mounted on concrete pads and locked in steel cases, thus known as distribution tap pad-mounted transformers.

Distribution transformers typically have ratings less than 200 kVA, although some national standards allow units up to 5000 kVA to be described as distribution transformers. Since distribution transformers are energized 24 hours a day (even when they don't carry any load), reducing iron losses is vital in their design. They usually don't operate at full load, so they are designed to have maximum efficiency at lower loads. To have better efficiency, voltage regulation in these transformers is kept to a minimum. Therefore, they are designed to have small leakage reactance.



³ An example of an iron-core transformer.

⁴ This value ranges between 1 T to 1,5 T.



⁵ A family of iron oxide-containing magnetic ceramic materials. They are ferrimagnetic, meaning they are attracted by magnetic fields and can be magnetized to become permanent magnets. Unlike many ferromagnetic materials, most ferrites are not electrically conductive, making them useful in applications like magnetic cores for transformers to suppress eddy currents.



⁶ A single-phase distribution transformer.

2.2 Transformers under No-Load Conditions

Fig. 2.2 shows in schematic form a transformer with its secondary circuit open and an alternating voltage v_1 applied to its primary terminals.

To simplify the drawings, it is common on schematic diagrams of transformers to show the primary and secondary windings as if they were on **separate legs** of the core, as in **Fig. 2.2**, even though the windings are actually interleaved in practice.

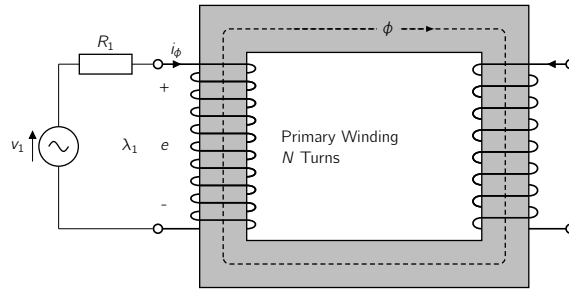


Figure 2.2: Transformer with open secondary

As discussed previously, a small steady-state current (i_ϕ), called the **exciting current**, flows in the primary and establishes an alternating flux in the magnetic circuit.⁷ This flux induces an Electro-motive Force (EMF) in the primary equal to:

$$e_1 = \frac{d\lambda_1}{dt} = N_1 \frac{d\phi}{dt} \quad (2.1)$$

where λ_1 is flux linkage of the primary winding (Wb),⁸ ϕ is the flux in the core linking both windings (Wb), N_1 is the number of turns in the primary winding, the voltage e_1 is in volts when ϕ is in webers. This EMF, together with the voltage drop in the primary resistance R_1 , must balance the applied voltage v_1 , therefore we can write the relation:

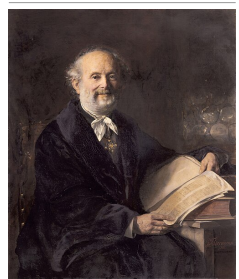
$$v_1 = R_1 i_\phi + e_1 \quad (2.2)$$

At the moment we are neglecting the effects of primary leakage flux as it would create unnecessary complications. For all practical purposes, the flux contributes a small value to the core-flux, therefore we can neglect it. It does however play an important role in the behavior of transformers in real-life applications and taken into consideration.

If the transformer is sufficiently large, the effect of the no-load resistance can be neglected which would make the induced EMF e_1 very nearly equals the applied voltage v_1 . Furthermore, the waveforms of voltage and flux are very **nearly** sinusoidal. Therefore, if the instantaneous flux is:

$$\phi = \phi_{\max} \sin \omega t \quad (2.3)$$

⁷In general, the exciting current corresponds to the net ampere-turns (MMF) acting on the magnetic circuit, and it is not possible to distinguish whether it flows in the primary or secondary winding or partially in each winding.



⁸**Wilhelm Eduard Weber**
(1804 - 1891)

a German physicist and, together with Carl Friedrich Gauss, inventor of the first electromagnetic telegraph. The unit of magnetic flux is named after him.

the induced voltage (e_1) is therefore:

$$e_1 = N_1 \frac{d\phi}{dt} = N_1 \frac{d\phi_{\max} \sin \omega t}{dt} = \omega N_1 \phi_{\max} \cos \omega t \quad (2.4)$$

where $\phi_{\max}$ is the maximum value of the flux (Wb) and $\omega = 2\pi f$, with frequency being f (Hz). For the current and voltage reference directions shown in **Fig. 2.2**, the induced EMF leads the flux by 90° . The RMS value of the induced EMF (e_1) is:

$$E_1 = \frac{2\pi}{\sqrt{2}} f N_1 \phi_{\max} = \sqrt{2} \pi f N_1 \phi_{\max} \quad (2.5)$$

If the resistive voltage drop is assumed negligible, the counter EMF equals the applied voltage. Under these conditions, if a sinusoidal voltage is applied to a winding, a sinusoidally varying core flux must be established whose maximum value $\phi_{\max}$ satisfies the requirement that E_1 in Eq. (2.5) equal the rms value V_1 of the applied voltage. Therefore we can write:

$$\phi_{\max} = \frac{V_1}{\sqrt{2} \pi f N_1} \quad (2.6)$$

Under these conditions, the core flux is determined solely by the applied voltage (V_1), its frequency (f), and the number of turns (N) in the winding.

This relation applies not only to transformers but to any device with a sinusoidally-alternating voltage, as long as the resistance and leakage-inductance voltage drops are negligible.

The core flux is fixed by the applied voltage, and the required exciting current is determined by the magnetic properties of the core. The exciting current must adjust itself so as to produce the MMF required to create the flux demanded by Eq. (2.6).

Due to the non-linear magnetic properties of iron, the waveform of the exciting current differs from the waveform of the flux, which we have looked at **Fig. 1.8** in Chapter 1.

If the exciting current is analysed using Fourier-series, it is found to consist of a fundamental component and a series of **odd harmonics**⁹. The fundamental component can, in turn, be resolved into two (2) components:

1. one in phase with the counter EMF, and
2. the other **lagging** the counter EMF by 90° .

The in-phase component supplies the power absorbed by hysteresis and eddy-current losses in the core. It is referred to as **core-loss component of the exciting current**. When the core-loss component is subtracted from the total exciting current, the remainder is called the **magnetizing current**. It comprises a fundamental component lagging the counter EMF by 90° , together with all the harmonics. The principal harmonic is the third.¹⁰

⁹This is caused by the function being a sine function which classify it as a Fourier odd-series.

¹⁰For typical power transformers, the third harmonic usually is about 40 percent of the exciting current.

Information: Effect of Harmonics

Except in problems concerned directly with the effects of harmonic currents, the exciting-current waveform distortions can be assumed negligible, as the exciting current itself is small, especially in large transformers.

For example, the exciting current of a typical power transformer is about 1 % to 2 % of full-load current. In the end, the effects of the harmonics usually are dominated out by the sinusoidal-currents supplied to other linear elements in the circuit.

The exciting current can then be represented by an equivalent sinusoidal current which has the same RMS value and frequency and produces the same average power as the actual exciting current. Such representation is essential to the construction of a **phasor diagram**, which represents the phase relationship between the various voltages and currents in a system in vector form. Each signal is represented by a phasor whose length is proportional to the amplitude of the signal and whose angle is equal to the phase angle of that signal as measured with respect to a chosen reference signal.

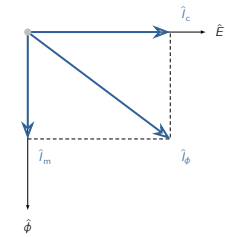


Figure 2.3: No-load phasor diagram

In **Fig. 2.3**, the phasors $\hat{E}_1$ and $\hat{\Phi}$ respectively, represent the RMS values of the induced EMF and the flux. The phasor $\hat{I}_\phi$ represents the RMS value of the equivalent sinusoidal exciting current. It lags the induced emf $\hat{E}_1$ by a phase angle θ_c . The core loss P_c , equal to the product of the in-phase components of the $\hat{E}_1$ and $\hat{I}_\phi$, is given by:

$$P_c = E_1 I_\phi \cos \theta_c \quad (2.7)$$

Information: Electric Arc Furnace

An electric arc furnace (EAF) is a furnace that heats material by means of an electric arc.

An Industrial EAF range in size from small units of approximately one-tonne capacity, used in foundries for producing cast iron products, up to about 400-tonne units used for secondary steelmaking. Industrial EAF temperatures can reach 1,800 °C, while laboratory units can exceed 3,000 °C.

As an example, a mid-sized modern steelmaking furnace would have a transformer rated about 60,000,000 volt-amperes (60 MVA), with a secondary voltage between 400 and 900 volts and a secondary current in excess of 44,000 amperes.

In electric arc furnaces, the material inside the furnace (referred to as a charge) is directly exposed to an electric arc, and the current from the electrode terminals passes through the charge material. Arc furnaces differ from induction furnaces, which use eddy currents to heat the charge.



Figure 2.4: An electric arc furnace during operation.

2.3 The Effect of Secondary Current on an Ideal Transformer

As a first approximation, consider a transformer with a primary winding of N_1 turns and a secondary winding of N_2 turns, as shown schematically in **Fig. 2.5**. Please observe the secondary current is defined as positive out of the winding, therefore positive secondary current produces an MMF in the **opposite** direction from that created by positive primary current.

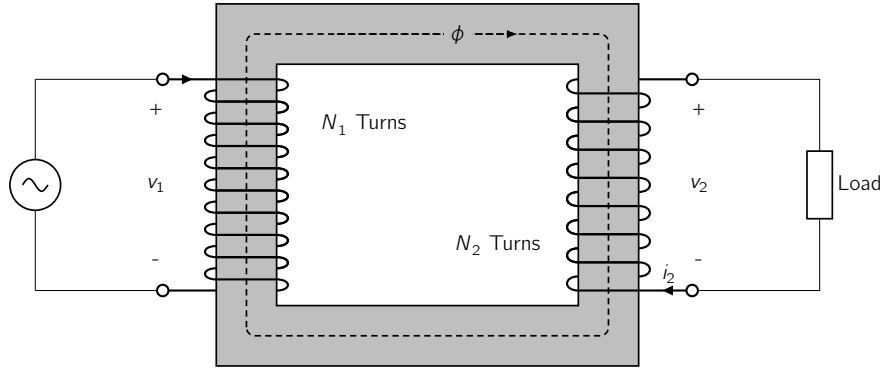


Figure 2.5: Ideal transformer and load.

Let us assume the properties of this transformer are idealised under the assumption that winding resistances are negligible, and all the flux is confined to the core and links both windings,¹¹ that there are no losses in the core, and that the permeability of the core is so high that only a negligible exciting MMF is required to establish the flux.

¹¹This means the leakage flux is assumed negligible.

These properties are closely approached but never actually attained in practical transformers.

A hypothetical transformer having these properties is often called an **ideal transformer**. Under the above assumptions, when a time-varying voltage v_1 is given on the primary terminals, a core flux ϕ must be established such that the counter emf e_1 equals the impressed voltage. Therefore:

$$v_1 = e_1 = N_1 \frac{d\phi}{dt} \quad (2.8)$$

The core flux also links the secondary and produces an induced emf e_2 , and an equal secondary terminal voltage v_2 , given by

$$v_2 = e_2 = N_2 \frac{d\phi}{dt} \quad (2.9)$$

From the ratio of Eq. (2.8) and Eq. (2.9):

$$\frac{v_1}{v_2} = \frac{N_1}{N_2} \quad (2.10)$$

An ideal transformer transforms voltages in the direct ratio of the turns in its windings.

Now let's have a load and connect it to the secondary. A current i_2 and an MMF $N_2 i_2$ are then present in the secondary. Given that the core permeability is assumed very large and since the

impressed primary voltage sets the core flux as specified by Eq. (2.8), the core flux is unchanged by the presence of a load on the secondary, and hence the net exciting mmf acting on the core¹² will **NOT** change and hence will remain negligible. Therefore

¹²which is equal to $N_1 i_1 - N_2 i_2$.

$$N_1 i_1 - N_2 i_2 = 0 \quad (2.11)$$

From Eq. (2.11) we see that a compensating primary MMF must result to cancel that of the secondary. We can write it as:

$$N_1 i_1 = N_2 i_2 \quad (2.12)$$

Therefore we see that the requirement that the net MMF **remain unchanged** is the means by which the primary “knows” of the presence of load current in the secondary; any change in mmf flowing in the secondary as the result of a load must be accompanied by a corresponding change in the primary mmf.

For the reference directions shown in **Fig. 2.5** the mmf’s of i_1 and i_2 are in opposite directions and therefore compensate.

The net mmf acting on the core therefore is zero, in accordance with the assumption that the exciting current of an ideal transformer is zero. From Eq. (2.12):

$$\frac{i_1}{i_2} = \frac{N_2}{N_1} \quad (2.13)$$

Therefore an ideal transformer transforms currents in the inverse ratio of the turns in its windings. Also notice from Eq. (2.10) and Eq. (2.13) that:

$$v_1 i_1 = v_2 i_2 \quad (2.14)$$

This means, the instantaneous power input to the primary equals the instantaneous power output from the secondary, a necessary condition because all dissipative and energy storage mechanisms in the transformer have been neglected.

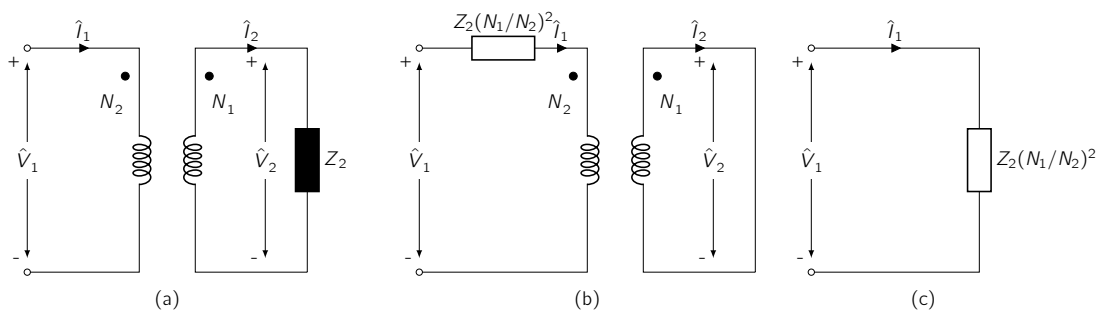


Figure 2.6: Three circuits which are identical at the primary terminals when the transformer is ideal.

An additional property of the ideal transformer can be seen by considering the case of a sinusoidal applied voltage and an impedance load. Phasor symbolism can be used. The circuit is shown in simplified form in **Fig. 2.6a**, in which the dotmarked terminals of the transformer correspond to the similarly marked terminals in **Fig. 2.5**.

The dot markings indicate terminals of corresponding polarity; i.e., if one follows through the primary and secondary windings of **Fig. 2.5**, beginning at their dot-marked terminals, one will find that both windings encircle the core in the same direction with respect to the flux. Therefore, if one compares the voltages of the two windings, the voltages from a dot-marked to an unmarked terminal will be of the same instantaneous polarity for primary and secondary.

In other words, the voltages $\hat{V}_1$ and $\hat{V}_2$ in **Fig. 2.6a** are in phase. Also currents $\hat{I}_1$ and $\hat{I}_2$ are in phase as seen from Eq. (2.12).

The polarity of $\hat{I}_1$ is defined as into the dotted terminal and the polarity of $\hat{I}_2$ is defined as out of the dotted terminal.

We next investigate the impedance transformation properties of the ideal transformer. In phasor form, Eq. (2.10) and Eq. (2.13) can be expressed as

$$\hat{V}_1 = \frac{N_1}{N_2} \hat{V}_2 \quad \text{and} \quad \hat{V}_2 = \frac{N_2}{N_1} \hat{V}_1 \quad (2.15)$$

$$\hat{I}_1 = \frac{N_2}{N_1} \hat{I}_2 \quad \text{and} \quad \hat{I}_2 = \frac{N_1}{N_2} \hat{I}_1 \quad (2.16)$$

From these equations we can derive the following relation:

$$\frac{\hat{V}_1}{\hat{I}_1} = \left(\frac{N_1}{N_2} \right)^2 \frac{\hat{V}_2}{\hat{I}_2} \quad (2.17)$$

Noting that the load impedance Z_2 is related to the secondary voltages and currents

$$Z_2 = \frac{\hat{V}_2}{\hat{I}_2} \quad (2.18)$$

where Z_2 is the complex impedance of the load. Consequently, as far as its effect is concerned, an impedance Z_2 in the secondary circuit can be replaced by an equivalent impedance Z_1 in the primary circuit, provided the following condition is assumed true:

$$Z_1 = \left(\frac{N_1}{N_2} \right)^2 Z_2 \quad (2.19)$$

Therefore, the three circuits of **Fig. 2.6** are indistinguishable as far as their performance viewed from terminals ab is concerned. Transferring an impedance from one side of a transformer to the other in this fashion is called **referring the impedance** to the other side; impedances transform as the square of the turns ratio.

In a similar manner, voltages and currents can be **referred** to one side or the other by using Eq. (2.15) to evaluate the equivalent voltage and current on that side.

2.4 Reactances and Types of Equivalent Circuits

The departures of an actual transformer from those of an ideal transformer must be included to a greater or lesser degree in most analyses of transformer performance. A more complete model must take into account the effects of winding resistances, leakage fluxes, and finite exciting current due to the finite¹³ permeability of the core. In some cases, the capacitances of the windings also have important effects, notably in problems involving transformer behavior at frequencies above the audio range or during rapidly changing transient conditions such as those encountered in power system transformers as a result of voltage surges caused by lightning or switching transients.

¹³and indeed non-linear

The analysis of these high-frequency problems is beyond the scope of the lecture. In addition, to keep thing simple, the capacitances of the windings will be neglected.

Two methods of analysis by which departures from the ideal can be taken into account:

1. An equivalent-circuit technique based on physical reasoning,
2. A mathematical approach based on the classical theory of magnetically coupled circuits.

Both methods are in everyday use, and both have very close parallels in the theories of rotating machines.

As it offers an excellent example of the thought process involved in translating physical concepts to a quantitative theory, the equivalent circuit technique is presented here.

To begin the development of a transformer equivalent circuit, we first consider the primary winding. The total flux linking the primary winding can be divided into two (2) components:

1. The resultant mutual flux, confined essentially to the iron core and produced by the combined effect of the primary and secondary currents, and
2. The primary leakage flux, which links only the primary.

These components are identified in the schematic transformer shown in Fig. 2.9, where for simplicity the primary and secondary windings are shown on opposite legs of the core.¹⁴

¹⁴In an actual transformer with interleaved windings, the details of the flux distribution are more complicated, but the essential features remain the same.

The leakage flux induces voltage in the primary winding which adds to the mutual flux. Given the leakage path is largely in air, this flux and the voltage induced by it vary linearly with primary current i_1 . It can therefore be represented by a **primary leakage inductance** L_1 .¹⁵ The corresponding **primary leakage reactance** X_{l_1} is found as

$$X_{l_1} = 2\pi f L_1 \quad (2.20)$$

¹⁵equal to the leakage-flux linkages with the primary per unit of primary current.

Of course, it is also worth mentioning, there will be a voltage drop in the primary resistance R_1 .

We now see the primary terminal voltage $\hat{V}_1$ consists of three components: the $\hat{i}_1 R_1$ drop in the primary resistance, the $\hat{i}_1 X_1$ drop arising from primary leakage flux, and the emf $\hat{E}_1$ induced in the

primary by the resultant mutual flux. Fig. 2.10a shows an equivalent circuit for the primary winding which includes each of these voltages.

The resultant mutual flux links both the primary and secondary windings and is created by their combined MMF's. It is convenient to treat these mmf's by considering the primary current must meet two requirements of the magnetic circuit:

1. It must not only produce the mmf required to produce the resultant mutual flux,
2. It must also counteract the effect of the secondary MMF which acts to demagnetize the core.

An alternative viewpoint is that the primary current must not only magnetize the core, it must also supply current to the load connected to the secondary.

According to this picture, it is convenient to resolve the primary current into two (2) components:

1. an exciting component, and
2. a load component.

The exciting component $\hat{I}_\phi$ is defined as the additional primary current required to produce the resultant mutual flux. It is a nonsinusoidal current of the nature described in Section 2.2.¹⁶ The load component $\hat{I}'_2$ is defined as the component current in the primary which would exactly counteract the mmf of secondary current $\hat{I}_2$. Given it is the exciting component which produces the core flux, the net mmf must equal $N_1 \hat{I}_\phi$ and thus we see that

$$N_1 \hat{I}_\phi = N_1 \hat{I}_1 - N_2 \hat{I}_2 = N_1 (\hat{I}_\phi + \hat{I}'_2) - N_2 \hat{I}_2 \quad (2.21)$$

and from Eq. (2.21) we see that

$$\hat{I}'_2 = \frac{N_2}{N_1} \hat{I}_2 \quad (2.22)$$

From Eq. (2.22), we see that the load component of the primary current equals the secondary

¹⁶In fact, the exciting current corresponds to the net mmf acting on the transformer core and cannot, in general, be considered to flow in the primary alone. However, for the purposes of this discussion, this distinction is not significant.

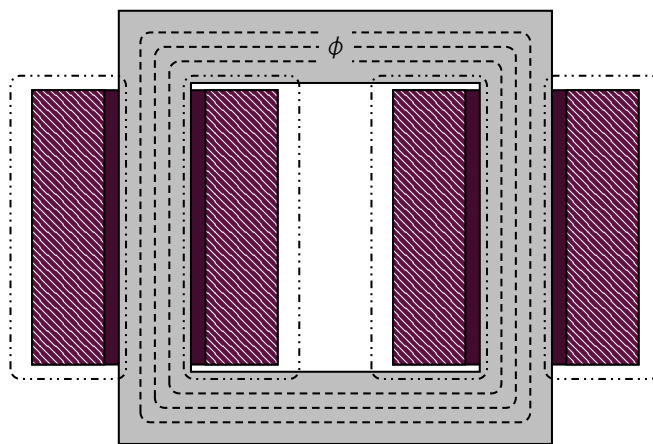


Figure 2.7: Schematic view of mutual and leakage fluxes in a transformer. The paths which are drawn densely dashed are the resultant magnetic flux paths, and paths drawn using dashed dot are leakage path on the primary (LHS) and secondary (RHS) respectively.

current referred to the primary exactly as in an ideal transformer.

The exciting current can be treated as an equivalent sinusoidal current $\hat{I}_\phi$, and can be resolved into a core-loss component $\hat{I}_c$ in phase with the emf $\hat{E}_1$ and a magnetizing component $\hat{I}_m$ lagging $\hat{E}_1$ by 90° . In the equivalent circuit (Fig. 2.10b) the equivalent sinusoidal exciting current is accounted for by means of a shunt branch connected across $\hat{E}_1$, comprising a coreloss resistance R_c in parallel with a magnetizing inductance L_m whose reactance, known as the magnetizing reactance, is given by

$$X_m = 2\pi f L_m \quad (2.23)$$

In the equivalent circuit of (Fig. 2.10b) the power E_1^2/R_c accounts for the core loss due to the resultant mutual flux. R_c is referred to as the magnetizing resistance or **core loss** resistance and together with X_m forms the excitation branch of the equivalent circuit, and we will refer to the parallel combination of R_c and X_m as the exciting impedance Z_ϕ .

When R_c is assumed constant, the core loss is thereby assumed to vary as E_1^2 or as $\phi_{\max}^2 f^2$, where $\phi_{\max}$ is the maximum value of the resultant mutual flux. Strictly speaking, the magnetizing reactance (X_m) varies with the saturation of the iron.

When X_m is assumed constant, the magnetizing current is thereby assumed to be independent of frequency and directly proportional to the resultant mutual flux. Both R_c and X_m are usually determined at rated voltage and frequency; they are then assumed to remain constant for the small departures from rated values associated with normal operation.

We will next add to our equivalent circuit a representation of the secondary winding. We begin by recognizing that the resultant mutual flux (Φ) induces an EMF E_2 in the secondary, and given this flux links both windings, the induced-emf ratio must equal the winding turns ratio, i.e.,

$$\frac{\hat{E}_1}{\hat{E}_2} = \frac{N_1}{N_2} \quad (2.24)$$

just as in an ideal transformer. This voltage transformation and the current transformation of Eq. (2.22) can be accounted for by introducing an ideal transformer in the equivalent circuit, as in Fig. 2.10c. Just as is the case for the primary winding, the EMF ($\hat{E}_2$) is not the secondary terminal voltage, however, because of the secondary resistance R_2 and because the secondary current ($\hat{I}_2$) creates secondary leakage flux (see Fig. 2.9). The secondary terminal voltage ($\hat{V}_2$) differs from the induced voltage $\hat{E}_2$ by the voltage drops due to secondary resistance (R_2) and secondary leakage reactance (X_{12}),¹⁷ as in the portion of the complete transformer equivalent circuit (Fig. 2.10c) to the right of E_2 .

¹⁷ This corresponds to the secondary leakage inductance L_{12} .

From the equivalent circuit of Fig. 2.10, the actual transformer therefore can be seen to be equivalent to an ideal transformer plus external impedances. By referring all quantities to the primary or secondary, the ideal transformer in Fig. 2.10c can be moved out to the right or left, respectively, of the equivalent circuit. This is almost invariably done, and the equivalent circuit is usually drawn as in Fig. 2.10d, with the ideal transformer not shown and all voltages, currents, and impedances referred to either the primary or secondary winding. Specifically, for Fig. 2.10d,

$$X'_{12} = \left(\frac{N_1}{N_2} \right)^2 X_{12} \quad (2.25)$$

$$R'_2 = \left(\frac{N_1}{N_2} \right)^2 R_2 \quad (2.26)$$

and

$$V'_2 = \frac{N_1}{N_2} V_2 \quad (2.27)$$

The circuit of Fig. 2.10d is called the equivalent-T circuit for a transformer. In Fig. 2.10d, in which the secondary quantities are referred to the primary, the referred secondary values are indicated with primes, for example, X'_1 , and R'_2 , to distinguish them from the actual values of Fig. 2.10c.

In the discussion that follows we almost always deal with referred values, and the primes will be omitted. One must simply keep in mind the side of the transformers to which all quantities have been referred.

2.5 Understanding Transformer Behaviour

In analysis involving the transformer as a circuit element, it is useful to adopt one of several approximate forms of the equivalent circuit¹⁸ of Fig. 2.10 rather than the full circuit. The approximations chosen in a particular case depend largely on physical reasoning based on orders of magnitude of the neglected quantities.

The approximate equivalent circuits commonly used for constant-frequency power transformer analyses are summarized for comparison in Fig. 2.12. All quantities in these circuits are referred to either the primary or the secondary, and the ideal transformer is **NOT** shown.

Calculations can often be greatly simplified by moving the shunt branch representing the exciting current out from the middle of the T circuit to either the primary or the secondary terminals, as in Fig. 2.12a and b.

These forms of the equivalent circuit are referred to as cantilever circuits.

The series branch is the combined resistance and leakage reactance of the primary and secondary, referred to the same side. This impedance is sometimes called the equivalent series impedance (Z_{eq}) and its components are:

- the equivalent series resistance (R_{eq}), and
- equivalent series reactance (X_{eq}),

are shown in Fig. 2.12a and b.

As compared to the equivalent-T circuit of Fig. 2.10d, the cantilever circuit introduces additional error in that it **neglects the voltage drop** in the primary or secondary leakage impedance caused by the exciting current. As the impedance of the exciting branch is typically quite large in large power transformers, the corresponding exciting current is quite small.

This error becomes insignificant in most situations involving large transformers.

Further analytical simplification can be achieved by neglecting the exciting current **entirely**, as in Fig. 2.12c, in which the transformer is represented as an equivalent series impedance. If the transformer is large,¹⁹ the equivalent resistance (R_{eq}) becomes measurably insignificant compared to the equivalent reactance (X_{eq}) and therefore can generally be neglected, giving the equivalent circuit of Fig. 2.12d.

For most power system problems and analysis the circuits shown in Fig. 2.12c and d are sufficiently accurate.

As a final note, in situations where the currents and voltages are determined **almost wholly by the circuits external to the transformer** or when a high degree of accuracy is **NOT** required, the entire transformer impedance can be neglected and the transformer considered to be ideal, as mentioned

¹⁸A theoretical circuit that retains all of the electrical characteristics of a given circuit. Often, an equivalent circuit is sought that simplifies calculation, and more broadly, that is a simplest form of a more complex circuit in order to aid analysis.

¹⁹In this context we would be talking about several hundred kilovoltamperes or more

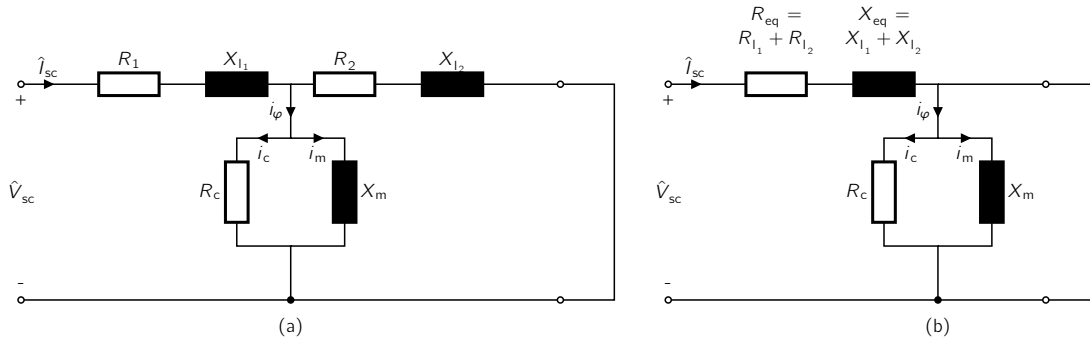


Figure 2.8: Equivalent circuit with short-circuited secondary. (a) Complete equivalent circuit. (b) Cantilever equivalent circuit with the exciting branch at the transformer secondary.

previously.

The circuits of Fig. 2.12 have the additional advantage being the total equivalent resistance (R_{eq}) and equivalent reactance (X_{eq}) can be found from a very simple test in which one terminal is short-circuited. Whereas, the process of determining the individual leakage reactances (X_{l1} and X_{l2}) and a complete set of parameters for the equivalent-T circuit of Fig. 2.10c is, as can be expected, more difficult.

Without some prior knowledge of the turns ratio,²⁰ it is not possible to make a set of measurements which uniquely determine the turns ratio, the magnetizing inductance, and the individual leakage impedances.

²⁰based for example upon knowledge of the internal construction of the transformer.

It can be shown that, simply from terminal measurements, neither the turns ratio, the magnetizing reactance, or the leakage reactances are unique characteristics of a transformer equivalent circuit. As an example, the turns ratio can be chosen arbitrarily and for each choice of turns ratio, there will be a corresponding set of values for the leakage and magnetizing reactances which matches the measured characteristic. Each of the resultant equivalent circuits will have the same electrical terminal characteristics, a fact which has the fortunate consequence that any self-consistent set of empirically determined parameters will adequately represent the transformer.

There are two (2) simple tests employed to determine the parameters of the equivalent circuits of Fig. 2.10 and 2.12. These consist of measuring the input voltage, current, and power to the primary:

- first with the secondary short-circuited, and
- then with the secondary open-circuited,

which are looked in detail below.

Short-Circuit Test The short-circuit test can be used to find the equivalent series impedance:

$$Z_{eq} = R_{eq} + jX_{eq}.$$

Although the choice of winding to short-circuit is arbitrary, for the sake of this discussion we shall consider the short circuit to be applied to the transformer secondary and voltage applied to primary.

For convenience, the high-voltage side is usually taken as the primary in this test due to the equivalent series impedance is relatively small, typically an applied primary voltage on the order of 10 to 15 percent or less of the rated value will result in rated current.

Figure 2.15a shows the equivalent circuit with transformer secondary impedance referred to the primary side and a short circuit applied to the secondary. The short-circuit impedance (Z_{sc}) looking into the primary under these conditions is:

$$Z_{sc} = R_1 + jX_{l1} + \frac{Z_\phi(R_2 + jX_{l2})}{Z_\phi + R_2 + jX_{l2}} \quad (2.28)$$

Because the impedance of the exciting branch (Z_ϕ) is much larger than that of the secondary leakage impedance,²¹ the short-circuit impedance can be approximated as

$$Z_{sc} \approx R_1 + jX_{l1} + R_2 + jX_{l2} = R_{eq} + jX_{eq} \quad (2.29)$$

²¹which will be true unless the core is heavily saturated by excessive voltage applied to the primary; certainly not the case here

The approximation made here is equivalent to the approximation made in reducing the equivalent-T circuit to the cantilever equivalent.

This can be seen from Fig. 2.15b; the impedance seen at the input of this equivalent circuit is clearly:

$$Z_{sc} = Z_{eq} = R_{eq} + jX_{eq}$$

given the exciting branch is directly shorted out by the short on the secondary.

Typically the instrumentation used for this test will measure the RMS magnitude of the applied voltage (V_{sc}), the short-circuit current (I_{sc}), and the power (P_{sc}). Based upon these three (3) measurements, the equivalent resistance and reactance²² can be found from:

$$|Z_{eq}| = |Z_{sc}| = \frac{V_{sc}}{I_{sc}} \quad (2.30)$$

$$R_{eq} = R_{sc} = \frac{P_{sc}}{I_{sc}^2} \quad (2.31)$$

$$X_{eq} = X_{sc} = \sqrt{|Z_{sc}|^2 - R_{sc}^2} \quad (2.32)$$

²²Both being referred to the primary.

The equivalent impedance can, of course, be referred from one side to the other in the usual manner.

On the rare occasions when the equivalent-T circuit in Fig. 2.10d must be resorted to, approximate values of the individual primary and secondary resistances and leakage reactances can be obtained by assuming:

$$R_1 = R_2 = 0.5R_{eq} \quad \text{and} \quad X_{l1} = X_{l2} = 0.5X_{eq}$$

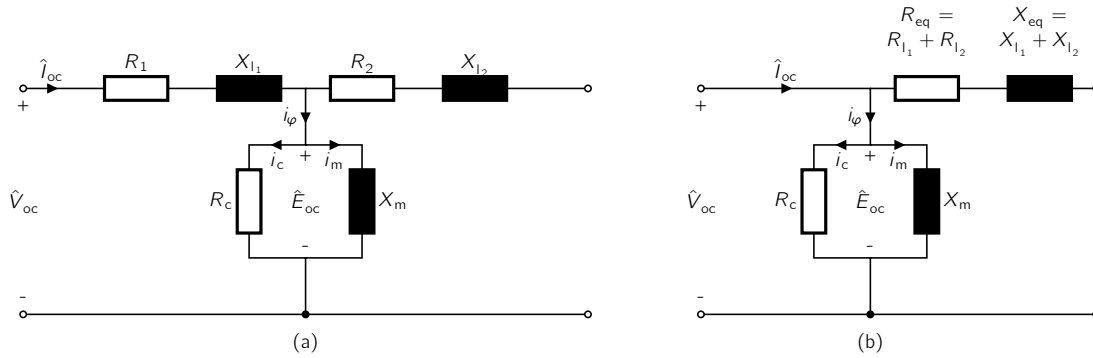


Figure 2.9: Equivalent circuit with open-circuited secondary. (a) Complete equivalent circuit. (b) Cantilever equivalent circuit with the exciting branch at the transformer primary.

when all impedances are referred to the same side.

Strictly speaking, of course, it is possible to measure R_1 and R_2 directly by a dc resistance measurement on each winding.²³

²³and then referring one or the other to the other side of the idea transformer

However, no such simple test exists for the leakage reactances X_{l1} and X_{l2} .

Open-Circuit Test The open-circuit test is performed with the **secondary open-circuited** and rated voltage impressed on the primary. Under this condition an exciting current of a few percent of full-load current²⁴ is obtained. Rated voltage is chosen to ensure that the magnetizing reactance will be operating at a flux level close to that which will exist under normal operating conditions. If the transformer is to be used at other than its rated voltage, the test should be done at that voltage.

²⁴less on large transformers and more on smaller ones.

For convenience, the low-voltage side is usually taken as the primary in this test.

If the primary in this test is chosen to be the opposite winding from that of the short-circuit test, one must of course be careful to refer the various measured impedances to the same side of the transformer in order to obtain a self-consistent set of parameter values. Figure 2.16a shows the equivalent circuit with the transformer secondary impedance referred to the primary side and the secondary open-circuited. The open-circuit impedance Z_{oc} looking into the primary under these conditions is

$$Z_{oc} = R_1 + jX_{l1} + Z_\phi = R_1 + jX_{l1} + \frac{R_c(jX_m)}{R_c + jX_m} \quad (2.33)$$

Because the impedance of the exciting branch is quite large, the voltage drop in the primary leakage impedance caused by the exciting current is typically negligible, and the primary impressed voltage $\hat{V}_{oc}$ very nearly equals the emf $\hat{E}_{oc}$ induced by the resultant core flux.

Similarly, the primary $I_{oc}^2 R_1$ loss caused by the exciting current is negligible, so the power input P_{oc} very nearly equals the core loss E_{oc}^2/R_c . As a result, it is common to ignore the primary leakage impedance and to approximate the open-circuit impedance as being equal to the magnetizing

impedance

$$Z_{oc} \approx Z_{\varphi} = \frac{R_c(jX_m)}{R_c + jX_m} \quad (2.34)$$

The approximation made here is equivalent to the approximation made in reducing the equivalent-T circuit to the cantilever equivalent circuit of Fig. 2.16b

The impedance seen at the input of this equivalent circuit is clearly Z_{φ} since no current will flow in the open-circuited secondary. As with the short-circuit test, typically the instrumentation used for this test will measure the rms magnitude of the applied voltage, V_{oc} , the open-circuit current I_{oc} and the power P_{oc} . Neglecting the primary leakage impedance and based upon these three measurements, the magnetizing resistance and reactance²⁵ can be found from

²⁵referred to the primary.

$$\begin{aligned} R_c &= \frac{V_{oc}^2}{P_{oc}} \\ |Z_{\varphi}| &= \frac{V_{oc}}{I_{oc}} \\ X_m &= \frac{1}{\sqrt{(1/|Z_{\varphi}|)^2 - (1/R_c)^2}} \end{aligned}$$

The values obtained are, of course, referred to the side used as the primary in this test. The open-circuit test can be used to obtain the core loss for efficiency computations and to check the magnitude of the exciting current. Sometimes the voltage at the terminals of the open-circuited secondary is measured as a check on the turns ratio.

Information: Measure from the Other Side

If desired, a slightly more accurate calculation of X_m and R_c by retaining the measurements of R_1 and X_1 obtained from the short-circuit test (referred to the proper side of the transformer) and basing the derivation on Eq. (2.33). However, such additional effort is rarely necessary for the purposes of engineering accuracy.

2.6 Auto and Multiwinding Transformers

The core concept we have developed were for two-winding transformers only. However, they are also applicable to transformers with other winding configurations such as auto-transformers and multi-winding transformers which will be the focus here.

2.6.1 AutoTransformers

In **Fig. 2.10a**, a two-winding transformer is shown with N_1 and N_2 turns on the primary and secondary windings respectively. The same transformation effect on voltages, currents, and impedances can also be obtained when these windings are connected as shown in **Fig. 2.10b**.

In **Fig. 2.10b**, winding bc is common to both the primary and secondary circuits.

This type of transformer is called an **autotransformer**. From construction point of view, it is little more than a normal transformer connected in a special way.

However, an important difference is the windings of the two-winding transformer are **electrically isolated** whereas those of the autotransformer are connected directly together. In addition, in the autotransformer connection, winding ab must be provided with **extra insulation** as it must be insulated against the full maximum voltage of the autotransformer.

They have lower leakage reactances, lower losses, and smaller exciting current and cost less than two-winding transformers when the voltage ratio does not differ too greatly from 1 : 1.

The following example illustrates the benefits of an autotransformer for those situations where electrical isolation between the primary and secondary windings is **NOT** an important consideration.

When a transformer is connected as an autotransformer as shown in **Fig. 2.10**, the rated voltages of the autotransformer can be expressed in terms of those of the two-winding transformer as:

$$\begin{aligned} \text{Low-Voltage} \quad V_{L_{\text{rated}}} &= V_{1_{\text{rated}}} \\ \text{High-Voltage} \quad V_{H_{\text{rated}}} &= V_{1_{\text{rated}}} + V_{2_{\text{rated}}} = \left(\frac{N_1 + N_2}{N_1} \right) V_{L_{\text{rated}}} \end{aligned}$$

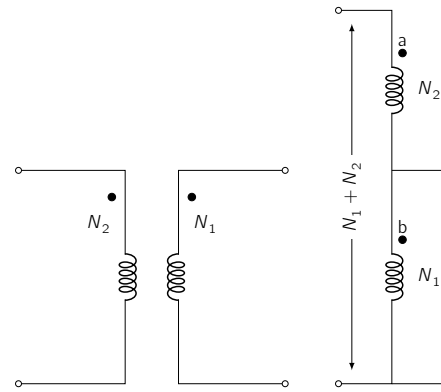


Figure 2.10: The schematics of an autotransformer

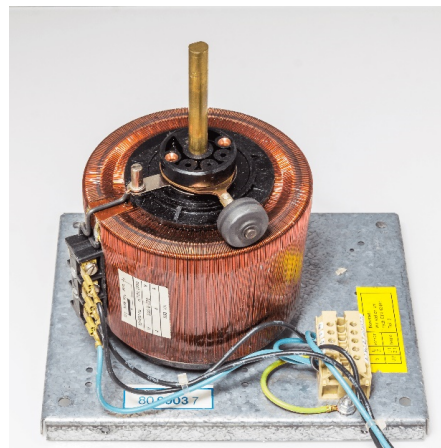


Figure 2.11: An example of an auto transformer.

The effective turns ratio of the autotransformer is $(N_1 + N_2)/N_1$ and the power rating of the autotransformer is equal to $(N_1 + N_2)/N_2$ times that of the two-winding transformer, although the actual power processed by the transformer will **NOT** increase over that of the standard two-winding connection.

2.6.2 Multiwinding Transformers

Transformers having three (3) or more windings, known as **multi-winding** or **multi-circuit** transformers, are often used to interconnect three or more circuits which may have different voltages.²⁶

Transformers having a primary and multiple secondaries are frequently found in multiple-output Direct Current (DC) power supplies for electronic applications. Distribution transformers used to supply power for domestic purposes usually have two 120 V secondaries connected in series. Circuits for lighting and low-power applications are connected across each of the 120 V windings, while electric ranges, domestic hot-water heaters, clothes-dryers, and other high-power loads are supplied with 240 V power from the series-connected secondaries.

In a similar fashion, a large distribution system can be supplied through a three-phase bank of multiwinding transformers from two or more transmission systems having different voltages. In addition, the three-phase transformer banks used to interconnect two transmission systems of different voltages often have a third, or tertiary, set of windings to provide voltage for auxiliary power purposes in substations or to supply a local distribution system.

Static capacitors or synchronous condensers can be connected to the tertiary windings for power factor correction or voltage regulation.²⁷ Some of the issues arising in the use of multiwinding transformers are associated with the effects of **leakage impedances** on voltage regulation, short-circuit currents, and division of load among circuits. These problems can be solved by an equivalent circuit technique mentioned previously.

Unfortunately equivalent circuits of multi-winding transformers are more complicated as they must take into account the leakage impedances associated with each pair of windings. Typically, in these equivalent circuits, all quantities are referred to a common base by use of the appropriate turns ratios as referring factors.

The exciting current is generally neglected in modelling for simplicity.

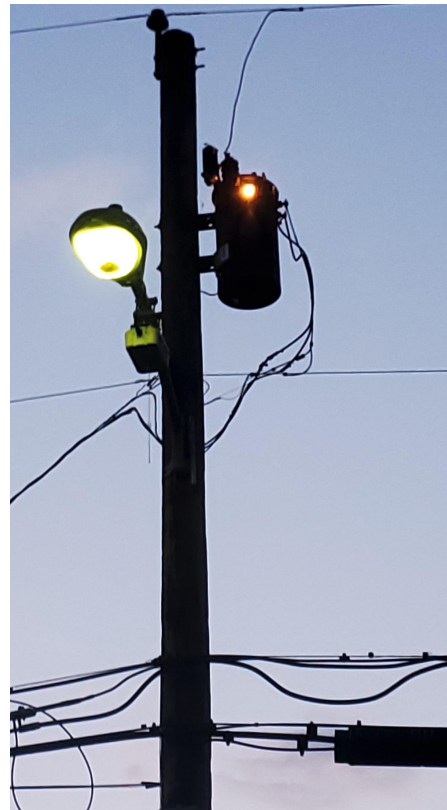


Figure 2.12: An example of a distribution transformer use for street-lighting purposes.

²⁶ For these applications a multiwinding transformer costs less and is more efficient than an equivalent number of two-winding transformers.

²⁷ Sometimes Δ -connected tertiary windings are put on three-phase banks to provide a low-impedance path for third harmonic components of the exciting current to reduce third-harmonic components of the neutral voltage.

2.7 Poly-Phase Transformers

One of the more practical applications of transformers are, of course in the use of the the grid. Given most supplied energy across vast distances are poly-phase;²⁸ we require the use of poly-phase transformers to step-up or step-down voltages as required.

²⁸Disregarding specialty cases, this is mostly three.

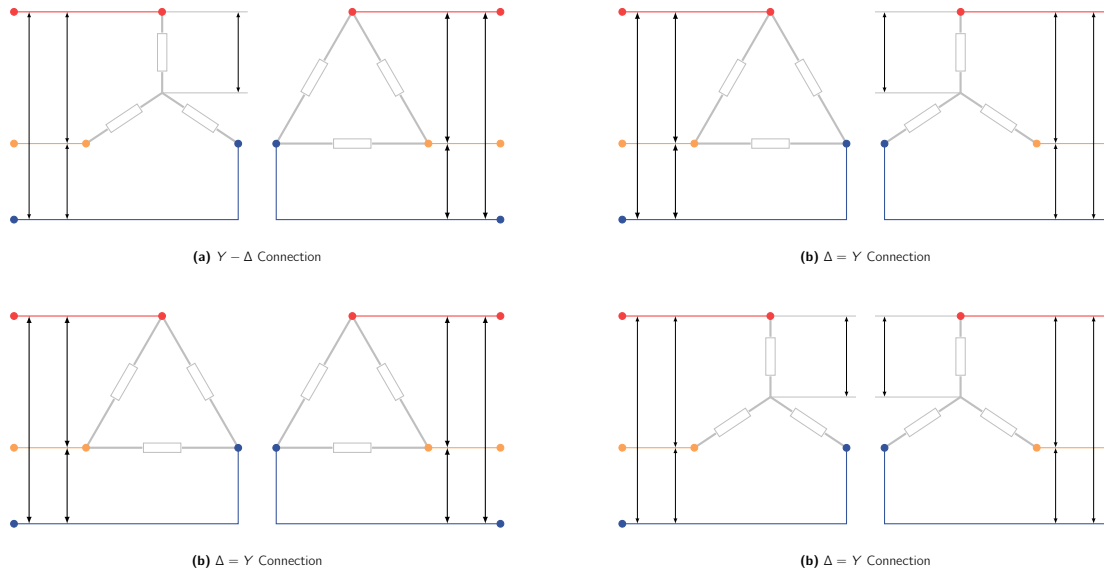


Figure 2.13: Common three-phase transformer connections; the transformer windings are indicated by the heavy lines.

Three single-phase transformers can be connected to form a three-phase transformer bank in any of the four (4) ways shown in **Fig. 2.13**. In all four parts of the figure, the windings at the left are the **primaries**, those at the right are the **secondaries**, and any primary winding in one transformer corresponds to the secondary winding drawn parallel to it.

Also shown are the voltages and currents resulting from balanced impressed primary line-to-line voltages (V_{LL}) and line currents (I_{LL}) when the ratio of primary-to-secondary turns is $N_1/N_2 = a$ and ideal transformer behaviour are assumed.

The rated voltages and currents at the primary and secondary of the three-phase transformer bank depends on the connection used but the rated kVA of the three-phase bank is three times that of the individual single-phase transformers, regardless of the connection.

2.7.1 Connection Types

Star-Delta Connection Commonly used in stepping down from a high voltage to a medium or low voltage. One reason is for this is the presence of a neutral provided for grounding on the high-voltage side, a procedure which can be shown to be desirable in many cases.

Delta-Star Connection Commonly used for stepping up to a high voltage.

Delta-Delta Connection It has the advantage that one transformer can be removed for repair or disconnected for maintenance while the remaining two continue to function as a three-phase bank with the rating reduced to 58 percent of that of the original bank.

This is known as the open-delta, or V, connection.

Star-Star Connection Seldom used because of difficulties with exciting-current phenomena.²⁹

Instead of three single-phase transformers, a three-phase bank may consist of one three-phase transformer having all six (6) windings on a common multi-legged core and contained in a single tank. Advantages of three-phase transformers over connections of three single-phase transformers are that they cost less, weigh less, require less floor space, and have somewhat higher efficiency. A photograph of the internal parts of a large three-phase transformer is shown in **Fig. 2.14**.

Calculations involving three-phase transformer banks **under balanced conditions** can be made by dealing with only one of the transformers or phases and recognizing that conditions are the same in the other two phases except for the phase displacements associated with a three-phase system.

It is usually convenient to carry out the computations on a single-phase per-phase-Y, line-to-neutral basis, given transformer impedances can then be added directly in series with transmission line impedances. The impedances of transmission lines can be referred from one side of the transformer bank to the other by use of the square of the ideal line-to-line voltage ratio of the bank.

In dealing with Y-Δ or Δ-Y banks, all quantities can be referred to the Y-connected side. In dealing with Δ-Δ banks in series with transmission lines, it is convenient to replace the Δ-connected impedances of the transformers by equivalent Y-connected impedances.

A final point to mention is that a balanced Δ-connected circuit of $Z_{\Delta} \Omega \cdot \text{phase}^{-1}$ is **equivalent** to a balanced Y-connected circuit of $Z_Y \Omega \cdot \text{phase}^{-1}$ if

$$Z_Y = \frac{1}{3} Z_{\Delta}$$

²⁹As there is no neutral connection to carry harmonics of the exciting current, harmonic voltages are produced which significantly distort the transformer voltages.



Figure 2.14: A standard three-phase transformer inside its closure. Observe that while it is three-phase all of its six windings were wound onto the three legs, where the low voltage winding is inside and the high voltage winding is outside.

Exercise 2.1 A Poly-phase Transformer

Three standard transformers are reconnected Δ - Δ and supplied with power through a 2400-V (line-to-line) three-phase feeder whose reactance is $0.80 \Omega \cdot \text{phase}^{-1}$.

At its sending end, the feeder is connected to the secondary terminals of a three-phase Y - Δ -connected transformer whose rating is:

$$500 \text{ kV} \cdot \text{A}, \quad 24 \text{ kV} : 2,400 \times 10^3 \text{ V} \quad \text{line-to-line}$$

The equivalent series impedance of the sending-end transformer is:

$$0.17 + j0.92 \Omega \cdot \text{phase}^{-1}$$

referred to the 2400-V side. The voltage applied to the primary terminals of the sending-end transformer is 24.0 kV line-to-line.

A three-phase short circuit occurs at the 240-V terminals of the receiving-end transformers.

Using this information, calculate the steady-state short-circuit current in the 2400-V feeder phase wires, in the primary and secondary windings of the receiving-end transformers, and at the 240-V terminals.

SOLUTION The calculations will be made on an equivalent line-to-neutral basis with all quantities referred to the 2400-V feeder. The source voltage then is:

$$\frac{2400}{\sqrt{3}} = 1385 \text{ V} \quad \text{line-to-neutral}$$

From Eq. (2.35), the single-phase-equivalent series impedance of the Δ - Δ transformer seen at its 2400-V side is

$$\begin{aligned} Z_{eq} &= R_{eq} + jX_{eq} = \frac{1.42 + j1.82}{3} \\ &= 0.47 + j0.61 \Omega \cdot \text{phase}^{-1} \end{aligned}$$

The total series impedance to the short circuit is then the sum of this impedance, that of sending-end transformer and the reactance of the feeder

$$\begin{aligned} Z_{\text{tot}} &= (0.47 + j0.61) + (0.17 + j0.92) + j0.80 \\ &= 0.64 + j2.33 \Omega \cdot \text{phase}^{-1} \end{aligned}$$

which has a magnitude of

$$|Z_{\text{tot}}| = 2.42 \Omega \cdot \text{phase}^{-1}$$

The magnitude of the phase current in the 2400-V feeder can now simply be calculated as the line-neutral voltage divided by the series impedance

$$\text{Current in 2400-V feeder} = \frac{1385}{2.42} = 572 \text{ A}$$

and, the winding current in the 2400-V winding of the receiving-end transformer is equal to the phase current divided by $\sqrt{3}$ or

$$\text{Current in 2400-V windings} = \frac{572}{\sqrt{3}} = 330 \text{ A}$$

while the current in the 240-V windings is 10 times this value

$$\text{Current in 240-V windings} = 10 \times 330 = 3300 \text{ A}$$

Finally, the phase current at the 240-V terminals into the short circuit is given by

$$\text{Current at the 240-V terminals} = 3300\sqrt{3} = 5720 \text{ A}$$

Note of course that this same result could have been computed simply by recognizing that the turns ratio of the Δ - Δ transformer bank is equal to 10:1 and hence, under balanced-three-phase conditions, the phase current on the low voltage side will be 10 times that on the high-voltage side. ■

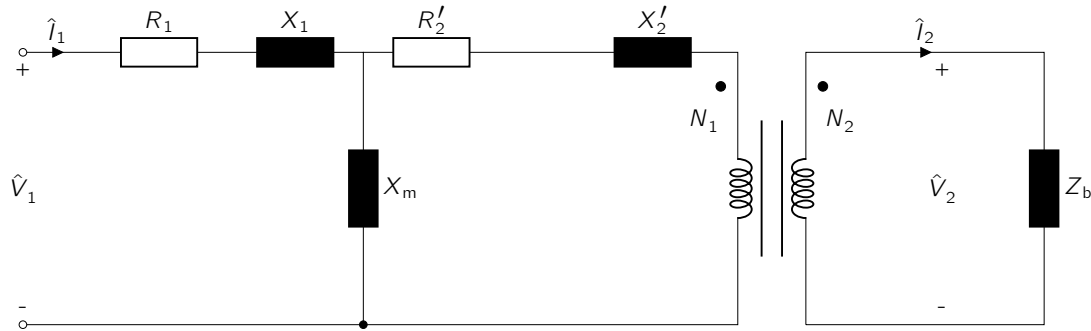


Figure 2.15: Equivalent circuit for an instrumentation transformer.

2.8 Voltage and Current Transformers

Transformers are often used in instrumentation applications to match the magnitude of a voltage or current to the range of a meter or other instrumentation. As an example, most 60 Hz power-systems' instrumentation is based upon voltages in the range of 0 V to 10 V RMS and currents in the range of 0 A to 5 A RMS. Since power system voltages range up to 765 kV line-to-line and currents can be 10's of kA, some method of supplying an accurate, low-level representation of these signals to the instrumentation is needed.

A common technique is through the use of specialized transformers known as **potential transformers** (PT)s and **current transformers** (CT)s. If constructed with a turns ratio of $N_1 : N_2$, an ideal potential transformer would have a secondary voltage equal in magnitude to N_2/N_1 times that of the primary and identical in phase. Similarly, an ideal current transformer would have a secondary output current equal to N_1/N_2 times the current input to the primary, again identical in phase.

Potential and current transformers³⁰ are designed to be ideal transformers as closely as possible.

³⁰also referred to as instrumentation transformers.

The equivalent circuit of **Fig. 2.15** shows a transformer loaded with an impedance $Z_b = R_b + jX_b$ at its secondary. For the sake of this discussion, the core-loss resistance R_c has been neglected, however if there is a need, the analysis presented here can be easily expanded to include its effect. Following conventional terminology, the load on an instrumentation transformer is frequently referred to as the **burden** on that transformer, hence the subscript b . To simplify our discussion, we have chosen to refer all the secondary quantities to the primary side of the ideal transformer.



Figure 2.16: A 120:120 instrument isolation transformer showing two polarity marking conventions.

Voltage Transformer Ideally it should accurately measure voltage while appearing as an open circuit to the system under measurement, meaning it should be drawing negligible current and power. Therefore, its load impedance

should be **large**.

To start, let us assume the transformer secondary is open-circuited, i.e., $|Z_b| = \infty$. In this case:

$$\frac{\hat{V}_2}{\hat{V}_1} = \left(\frac{N_2}{N_1} \right) \frac{jX_m}{R_1 + j(X_1 + X_m)} \quad (2.35)$$

As we see, a potential transformer with an open-circuited secondary has an inherent error, in both magnitude and phase, due to the voltage drop of the **magnetizing current** through the primary resistance and leakage reactance.

The difference of magnitude is high enough to make primary resistance and leakage reactance small compared to the magnetizing reactance, this inherent error can be made quite small. The situation is worsened by the presence of a finite burden. Including the effect of the burden impedance, Eq. (2.35) becomes:³¹

$$\frac{\hat{V}_2}{\hat{V}_1} = \left(\frac{N_2}{N_1} \right) \frac{Z_{eq} Z'_b}{(R_1 + jX_1)(Z_{eq} + Z'_b + R'_2 + jX'_2)} \quad (2.36)$$

where

$$Z_{eq} = \frac{jX_m(R_1 + jX_1)}{R_1 + j(X_m + X_1)} \quad (2.37)$$

and

$$Z'_b = \left(\frac{N_1}{N_2} \right)^2 Z_b \quad (2.38)$$

is the burden impedance referred to the transformer primary. From these equations, it can be seen that the characteristics of an accurate potential transformer include a large magnetizing reactance³² and relatively small winding resistances and leakage reactances. Finally the burden impedance must be kept above a minimum value to avoid introducing excessive errors in the magnitude and phase angle of the measured voltage.

Current Transformer An ideal current transformer would accurately measure current while appearing as a short circuit to the system under measurement, i.e., developing negligible voltage drop and drawing negligible power. Therefore, its load impedance should be “small” in a sense we will now quantify.

Let's start with the assumption that the transformer secondary is short-circuited, meaning $|Z_b| = 0$. In this case we can write that:

$$\frac{\hat{I}_2}{\hat{I}_1} = \left(\frac{N_1}{N_2} \right) \frac{jX_m}{R'_2 + j(X'_2 + X_m)} \quad (2.39)$$

In a fashion quite analogous to that of a potential transformer, Eq. (2.39) shows that a current transformer with a shorted secondary has an inherent error in both magnitude and phase, due to the

³¹Please observe the burden impedance has been referred to the primary.

³²more accurately, a large exciting impedance since the effects of core loss, although neglected in the analysis presented here, must also be minimised.



Figure 2.17: A CT for operation on a 110 kV grid.

fact that some of the primary current is shunted through the magnetizing reactance and does not reach the secondary. To the extent that the magnetizing reactance can be made large in comparison to the secondary resistance and leakage reactance, this error can be made quite small.

A finite burden appears in series with the secondary impedance and increases the error. Including the effect of the burden impedance which rewrites as Eq. (2.39):

$$\frac{\hat{I}_2}{\hat{I}_1} = \left(\frac{N_1}{N_2} \right) \frac{jX_m}{Z'_b + R'_2 + j(X'_2 + X_m)} \quad (2.40)$$

From these equations, it can be seen that an accurate current transformer has a large magnetizing impedance and relatively small winding resistances and leakage reactances. In addition, the burden impedance on a current transformer must be kept below a maximum value to avoid introducing excessive additional magnitude and phase errors in the measured current.

Chapter 3

Electromechanic Energy Conversion

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3.1 The Necessary Definitions

We are now at the precipice of starting to work on rotating machines and analyse their behaviour. But before we get into the fun and interesting part, it is time to discuss the electromechanical-energy-conversion process which takes place through the medium of the electric ($\mathbf{E}$) and/or the magnetic field ($\mathbf{B}$) of the **conversion device** itself¹. While there are different types of conversion devices which operate on similar principles, their structures depend on their function which are broadly classified into three (3) categories:

¹This device can be anything, as long as an energy changes form.

Transducers

Devices meant for measurement and control. They generally operate under linear input-output conditions and with relatively small signals. The many examples include microphones, pickups, sensors, and loudspeakers.

Reactionary

Force-producing devices which includes solenoids, relays, and electromagnets. A key point is

that they create motion when activated but they do not produce continuous motion.

Continuous

Devices which are used as **continuous energy-conversion equipment** such as motors and generators which are designed to be operated continuously.

In this chapter we will focus on the fundamental principles of electromechanical energy conversion and the analysis of the devices which actually carries out the conversion. Emphasis is placed on the analysis of devices which use almost exclusively **magnetic fields** as the conversion medium.²

However, the analytical techniques for electric field systems are quite similar and can be easily extended if required.

²As our focus will be on electric machines which work by using magnetic fields to generate rotational motion.

The purpose of such analysis generalised into three (3) justifications:

1. to understand how the energy conversion process works,
2. to provide the means to design and optimise electro-mechanical devices.
3. to develop models of electro-mechanical energy-conversion devices which can be then used for analysing their performance as components in a bigger engineering infrastructure.

For now we will only focus on **transducers** and **force-producing** devices. Later on, we will shift our focus to continuous energy-conversion devices in the form of rotating electric machinery.



Figure 3.1: One of the significant benefits of energy conversion is the ability to generate electricity from a mechanical process such as a hydro dam where the flowing of water turns turbines which in turn, rotates massive rotor [14].

3.2 Forces and Torque under Magnetic Field

As with anything electromagnetic, it is worth starting with the **Lorentz force law**, which gives the force $\mathbf{F}$ on a particle of charge q in the presence of both **electric** and **magnetic** fields.³

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (3.1)$$

where $\mathbf{F}$ is the force acting on the charge (N), q is the charge travelling in the field(s) (C), $\mathbf{E}$ is the electric field ($\text{V} \cdot \text{m}^{-1}$), $\mathbf{B}$ is the magnetic field (T) and $\mathbf{v}$ is the velocity of the particle relative to the magnetic field ($\text{m} \cdot \text{s}^{-1}$)

Therefore, if we were to assume the charge is travelling in a pure electric-field system ($\mathbf{B} = 0$), the force is determined simply by the charge on the particle and the electric field:

$$\mathbf{F} = q\mathbf{E} \quad (3.2)$$

As we can see from above, the force acts in the direction of the electric field and is **independent** of any particle motion.

In pure magnetic-field systems ($\mathbf{E} = 0$), however, the situation is somewhat more complex. Here the force is represented as:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}), \quad (3.3)$$

and is determined by the magnitude of the charge on the particle and the magnitude of the $\mathbf{B}$ field as well as the velocity of the particle.⁴

Direction of the force is always **perpendicular** to the direction of both the particle motion and that of the magnetic field indicated by the vector cross product $\mathbf{v} \times \mathbf{B}$ in Eq. (3.3).

In Eq. (3.3) The magnitude of this cross product is equal to the product of the magnitudes of $\mathbf{v}$ and $\mathbf{B}$ and the sine of the angle between them:

$$|\mathbf{v} \times \mathbf{B}| = |\mathbf{v}||\mathbf{B}|\sin\theta,$$

with its direction can be found from the right-hand rule, which states that

when the thumb of the right hand points in the direction of $\mathbf{v}$ and the index finger points in the direction of $\mathbf{B}$, the force, which is **perpendicular to the directions** of both $\mathbf{B}$ and $\mathbf{v}$, points in the direction normal to the palm of the hand,

as shown in **Fig. 3.2**.

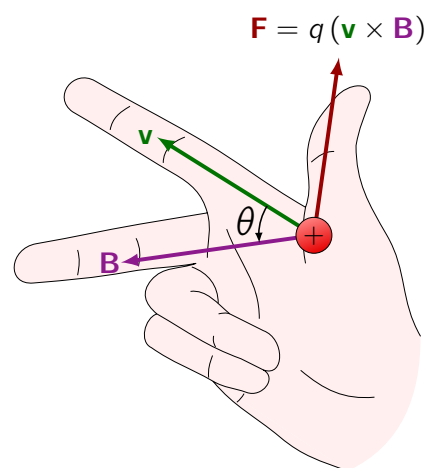


Figure 3.2: Right-hand rule for determining the direction magnetic-field component of the Lorentz force $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$.



³Hendrik Antoon Lorentz (1853 - 1928)

Dutch theoretical physicist who shared the 1902 Nobel Prize in Physics with Pieter Zeeman for their discovery and theoretical explanation of the Zeeman effect. He derived the Lorentz transformation of the special theory of relativity, as well as the Lorentz force.

⁴The $\times$ operator here is called the **cross-product**.

For situations where large numbers of charged particles are in motion, we can rewrite Eq. (3.1) in terms of the **charge density** ρ ($\text{C} \cdot \text{m}^{-3}$) as:

$$\mathbf{F}_v = \rho(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (3.4)$$

where the subscript v indicates $\mathbf{F}_v$ is a **force density** ($\text{N} \cdot \text{m}^{-3}$). A final terminology to consider is the product $\rho\mathbf{v}$ is known as the **current density** ($\text{A} \cdot \text{m}^{-3}$).

$$\mathbf{J} = \rho\mathbf{v} \quad (3.5)$$

The magnetic-system force density corresponding to Eq. (3.3) can then be written as:

$$\mathbf{F}_v = \mathbf{J} \times \mathbf{B} \quad (3.6)$$

For currents flowing in conducting media, Eq. (3.6) can be used to find the force density acting on the material itself.

Considerable amount of physics is hidden in this seemingly simple statement, as the mechanism by which the force is transferred from the moving charges to the conducting medium is a complex one.

Information: Limits on the Method

For situations in which the forces act only on current-carrying elements and which are of simple geometry, Eq. (3.6) is generally the simplest and easiest way to calculate the forces acting on the system. Unfortunately, very few practical situations fall into this class. In fact, most electromechanical-energy-conversion devices contain magnetic material where forces act **directly** on the magnetic material and clearly cannot be calculated from Eq. (3.6)

3.3 The Theory of Energy Balance

Unfortunately, as with everything which is more than an abstract imaginary ball in the land of theory, the techniques for calculating the detailed, localised forces acting in systems with magnetic materials are **extremely complex** and require detailed knowledge of the field distribution throughout the structure.⁵

Fortunately, most electro-mechanical energy conversion devices are constructed of **rigid, non-deforming** structures.

The performance of these devices is typically determined by the net force, or torque, acting on the moving component and it is rarely necessary to calculate the details of the internal force distribution.

For example, in a carefully designed motor, the motor characteristics are determined by the net accelerating torque acting on the rotor. Accompanying forces, such as the forces which squash or deform the rotor, does **NOT** have a significant role in the performance of the motor and generally are not calculated.

Under these assumption and the simplification, to understand the behaviour of rotating machinery, a simple physical picture is quite useful. To start simple, we assume both the rotor and stator have their own magnetic field⁶ and similarly with the stator:

We can picture them as a set of north and south magnetic poles associated with each structure.

Just as a compass needle tries to align with the earth's magnetic field, these two (2) sets of fields⁷ attempt to align, and **torque is associated with their displacement from alignment**.

In a motor, the stator magnetic field rotates ahead of that of the rotor, pulling on it and performing work, whereas the opposite is true for a generator, in which the rotor does work on the stator.

We begin formalising, we start with the principle of **conservation of energy** which states:

Theory 3.1: Conservation of Energy

Energy is neither created nor destroyed; it merely changes in form [15].

For isolated systems with clearly identifiable boundaries, this fact permits us to keep track of energy in a simple fashion:

The net flow of energy into the system across its boundary is equal to the sum of the time rate of change of energy stored in the system.

⁵Solutions may not provide closed form descriptions and for complex geometries, it is better to use FEM software to deal with them.

⁶produced in many machines by currents in windings on the rotor, or through PMs.

⁷To reiterate, these fields are rotor, and stator respectively.

3.3.1 The Energy Method

The technique for calculating forces and torques in the electromechanical energy-conversion process is known as the **energy method** and is based on the principle of conservation of energy. This result, which is a statement of the **first law of thermodynamics**, is quite general and allow us to apply to electro-mechanical systems whose predominant energy-storage mechanism is in **magnetic fields**.

In these systems, we can account for energy transfer as:

$$\left(\begin{array}{c} \text{Energy input} \\ \text{from electric} \\ \text{sources} \end{array} \right) = \left(\begin{array}{c} \text{Mechanical} \\ \text{energy} \\ \text{output} \end{array} \right) + \left(\begin{array}{c} \text{Increase in energy} \\ \text{stored in magnetic} \\ \text{field} \end{array} \right) + \left(\begin{array}{c} \text{Energy} \\ \text{converted} \\ \text{into heat} \end{array} \right) \quad (3.7)$$

Eq. (3.7) is written so, for a motor, both the electric and mechanical energy terms have positive values, and in case of a generator, both the terms have negative values. Either case, the heat generation within the system results in a flow of thermal energy **out** of the system.

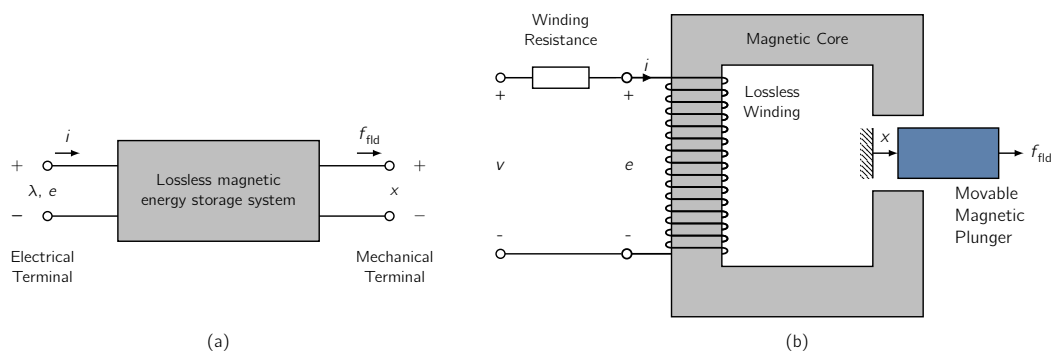


Figure 3.3: (a) Schematic magnetic-field electromechanical-energy-conversion device; (b) simple force-producing device.

In the systems which we consider in this chapter, the conversion of energy into heat occurs by the following mechanisms:

Ohmic heating Caused by current flow in the windings of the electric terminals.

Mechanical friction Caused the motion of components forming the mechanical terminals.

For electromechanical system, it is generally possible to mathematically separate these loss mechanisms from the energy-storage mechanism.

In such cases, the interaction between the electric and mechanical terminals,⁸ occurs through the medium of the magnetic stored energy and the device can be represented as a **lossless energy-storage system** with electric and mechanical terminals as shown in Fig. 3.3a.

⁸i.e., the electromechanical energy conversion.

In systems which can be modelled in this fashion, loss mechanisms can be represented by external

elements connected to these terminals:

Resistances to the electric terminals, and mechanical dampers to the mechanical terminals, and the losses do **NOT** need to be considered in calculations involving the electromechanical-energy-conversion process.

Fig. 3.3b shows an example of such a system:

Here we see a simple force-producing device with a single coil forming the electric terminal, and a movable plunger serving as the mechanical terminal.

Fig. 3.3a can be readily generalised to situations with any number of electric or mechanical terminals.

Fig. 3.3a represents a system in which there is stored magnetic energy and the magnetic field serves as the coupling medium between the electric and mechanical terminals.

This train of thought can be applied equally well to a system with stored electric energy. The ability to identify a lossless-energy-storage system is the essential idea of the energy method and is important to recognise that this is done mathematically as part of the modelling process.

It is **NOT** possible, to take the resistance out of windings or the friction out of bearings. Instead we are making use of the fact that a model in which this is done is a valid representation of the physical system.

For a lossless energy-storage system, Eq. (3.7) can be written as:

$$P_{\text{elec}} = P_{\text{mech}} + \frac{dW_{\text{fld}}}{dt} \quad (3.8)$$

where P_{elec} is the electric input power (W), P_{mech} is the mechanical output power (W) and dW_{fld}/dt is the rate-of-change of the magnetic stored energy with respect to time ($\text{W} \cdot \text{s}^{-1}$).

As can be seen in **Fig. 3.3a**, on the Left Hand Side (LHS), the electric terminal has two (2) terminal variables,

- a voltage source (e), and
- a flowing current (i),

whereas on the Right Hand Side (RHS), the mechanical terminal, also has two (2) terminal variables:

- a force (f_{fld}), and
- a position (x).

The electrical input power (P_{elec}) can be written as the **product** of the voltage e and the current i :

$$P_{\text{elec}} = \frac{dW_{\text{elec}}}{dt} = ei, \quad (3.9)$$

and the mechanical output power can be written as the product of the force f_{fld} and the velocity⁹ as: ⁹the time derivative of the position x .

$$P_{\text{mech}} = f_{\text{fld}} \frac{dx}{dt} \quad (3.10)$$

Re-arranging Eq. (3.8) and substituting Eq. (3.9) and Eq. (3.10) gives:

$$\frac{dW_{\text{fld}}}{dt} = P_{\text{elec}} - P_{\text{mech}} = ei - f_{\text{fld}} \frac{dx}{dt}. \quad (3.11)$$

For a magnetic-energy-storage system, the electrical terminal typically represents a winding such as that shown in **Fig. 3.3b**. Recognising that, from Eq. (1.22), the voltage at the terminals of a lossless winding is given by the time-derivative of the winding flux linkages:

$$e = \frac{d\lambda}{dt}. \quad (3.12)$$

Substitution into Eq. (3.11) and multiplying by dt gives us a useful relation:

$$dW_{\text{fld}} = i d\lambda - f_{\text{fld}} dx \quad (3.13)$$

As we shall see later, Eq. (3.13) permits us to solve for the force simply as a function of the flux linkage λ and the mechanical terminal position x .

This result comes about as a consequence of our assumption that it is possible to separate the losses out of the physical problem, resulting in a lossless energy-storage system, as in **Fig. 3.3a**.

Eq. (3.11) and Eq. (3.13) form the basis for the energy method. This technique is quite powerful in its ability to calculate forces and torques in complex electromechanical-energy-conversion systems.

Information: The Cost of Abstraction

This powerful technique comes at the expense of a detailed picture of the force-producing mechanism as the method merely **abstracts** the forces as the forces themselves are produced by such well-known physical phenomena as the Lorentz force on current carrying elements, described by Eq. (3.6), and the interaction of the magnetic fields with the dipoles in the magnetic material.

3.4 Energy in a Singly Excited Magnetic Circuit

Previously we focused mostly with **fixed-geometry magnetic circuits** such as those used for transformers and inductors. Energy in those devices is stored in the **leakage fields** and to some extent in the core itself.¹⁰

In those circuits, the stored energy does **NOT** enter directly into the transformation process.

In this chapter we are dealing with energy-conversion systems where:

the magnetic circuits have air gaps between the stationary and moving members in which considerable energy is stored in the magnetic field.

This field acts as the energy-conversion medium, and its energy is the **reservoir between the electric and mechanical systems**. To get a better understanding of the above description, consider the electromagnetic relay shown schematically in **Fig. 3.4**.

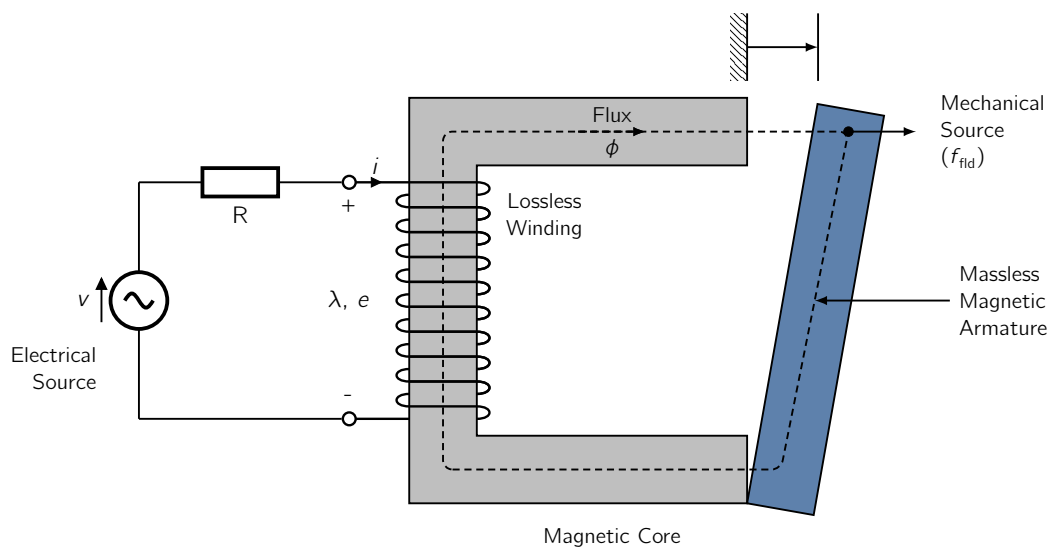
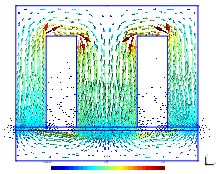


Figure 3.4: An example of an electromagnetic relay.

The resistance of the excitation coil is shown as an external resistance (R) and the mechanical terminal variables are shown as a force (f_{fld}) produced by the magnetic field directed from the relay to the external mechanical system and a displacement (x).

Mechanical losses can be included as external elements connected to the mechanical terminal..

Similarly, the moving armature is shown as being **massless**, its mass represents mechanical energy storage and can be included as an external mass connected to the mechanical terminal. As a result, the magnetic core and armature constitute a lossless magnetic-energy-storage system as is represented schematically in **Fig. 3.3a**.



¹⁰To reiterate, leakage flux is flux which does not flow through the chosen material but through other unwanted environments. In the image above any dots you see which is not within the material is considered leakage flux.

This relay structure is essentially the same as the magnetic structures we looked in Chapter 1 in which we saw that the magnetic circuit of **Fig. 3.4** can be described by an inductance (L) which is a function of the geometry of the magnetic structure and the magnetic permeabilities of the various system components.

Electro-mechanical energy-conversion devices contain air gaps in their magnetic circuits to separate the moving parts and, as discussed in Chapter 1, in most such cases the reluctance of the air gap is much larger than that of the magnetic material. Therefore the predominant energy storage occurs in the air gap, and the properties of the magnetic circuit are determined by the dimensions of the air gap.

Due to the simplicity of the resulting relations, magnetic non-linearity and core losses are often **neglected** in the analysis of practical devices. The final results of such approximate analyses can, if necessary, be corrected for the effects of these neglected factors by semi-empirical methods. Consequently, analyses are carried out under the assumption that the MMF and flux are directly proportional for the entire magnetic circuit.

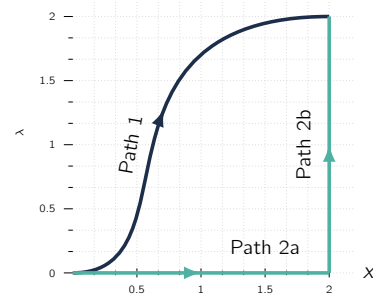


Figure 3.5: Integration paths for W_{fld} .

Therefore the flux linkages (λ) and current (i) are considered to be **linearly related** by an inductance which depends solely on the geometry and hence on the armature position (x).

$$\lambda = L(x) i \quad (3.14)$$

where it can be seen that the inductance (L) is **dependent** on x . Given that the magnetic energy storage system is **lossless**, we can state the system is a **conservative system** and therefore the value of W_{fld} is **uniquely specified** by the values of both λ and x .

λ and x are therefore referred to as **state variables** given their values uniquely determine the state of the system.

Given the magnetic force f_{fld} has been defined as acting from the relay upon the external mechanical system, P_{mech} is defined as the mechanical energy output of the relay. To revisit an equation we have just derived:

$$dW_{fld}(\lambda, x) = i d\lambda - f_{fld} dx, \quad (3.15)$$

we see the stored energy (W_{fld}) being **uniquely** determined by the values of λ and x , is the same regardless of how λ and x are brought to their final values.

Consider **Fig. 3.5**, in which two (2) separate paths are shown over which Eq. (3.15) can be integrated to find W_{fld} at the point (λ_0, x_0) .

Path 1 The general case and is difficult to integrate unless both i and f_{fld} are known explicitly as a function of λ and x .

Path 2 gives the same result and is much easier to integrate as the integration of Eq. (3.15) is path independent,

From Eq. (3.15) we see that:

$$W_{\text{fld}}(\lambda_0, x_0) = \int_{\text{Path 2a}} dW_{\text{fld}} + \int_{\text{Path 2b}} dW_{\text{fld}}. \quad (3.16)$$

Please observe that on Path 2a, $d\lambda = 0$ and $f_{\text{fld}} = 0$.¹¹ Therefore from Eq. (3.15), $dW_{\text{fld}} = 0$ on Path 2a. On Path 2b, $dx = 0$, and, from Eq. (3.15), Eq. (3.16) reduces to the integral of $i d\lambda$ over Path 2b, for which $x = x_0$. Therefore we simplify the equation as:

¹¹since $\lambda = 0$ and there can be no magnetic force in the absence of magnetic fields.

$$W_{\text{fld}}(\lambda_0, x_0) = \int_{\text{Path 2a}} dW_{\text{fld}} + \int_{\text{Path 2b}} dW_{\text{fld}} = \int_0^{\lambda_0} i(\lambda, x_0) d\lambda \quad (3.17)$$

For a linear system in which λ is proportional to i , as in Eq. (3.14), Eq. (3.17) gives

$$W_{\text{fld}}(\lambda_0, x_0) = \int_0^{\lambda_0} i(\lambda, x_0) d\lambda = \int_0^{\lambda_0} \frac{\lambda}{L(x_0)} d\lambda = \frac{1}{2} \frac{\lambda_0^2}{L(x_0)} \quad (3.18)$$

Point (λ_0, x_0) is arbitrary. The expression for W_{fld} of Eq. (3.18) is valid for all points (λ, x) .

To emphasise this point, Eq. (3.18) can be re-written as

$$W_{\text{fld}}(\lambda, x) = \frac{1}{2} \frac{\lambda^2}{L(x)} \quad (3.19)$$

It can be shown, the magnetic stored energy can also be expressed in terms of the energy density of the magnetic field integrated over the volume $\mathcal{V}$ of the magnetic field.

In this case

$$W_{\text{fld}} = \int_{\mathcal{V}} \left(\int_0^B \mathbf{H} \cdot d\mathbf{B}' \right) dV \quad (3.20)$$

For soft magnetic material of constant permeability ($\mathbf{B} = \mu\mathbf{H}$), this reduces to

$$W_{\text{fld}} = \int_{\mathcal{V}} \left(\frac{B^2}{2\mu} \right) dV \quad (3.21)$$

In this section we have seen the relationship between the magnetic stored energy and the electric and mechanical terminal variables for a system which can be represented in terms of a lossless-magnetic-energy-storage element.

If we had chosen for our example a device with a rotating mechanical terminal instead of a linearly displacing one, the results would have been identical except that force would be replaced by torque and linear displacement by angular displacement.

3.5 Determining Force and Torque from Energy

As discussed in the previously, for a lossless magnetic-energy-storage system, the magnetic stored energy (W_{fld}) is a state function,¹² determined uniquely by the values of the **independent state** variables λ and x . This is shown explicitly by Eq. (3.15) and repeated here for reiteration:

$$dW_{fld}(\lambda, x) = i d\lambda - f_{fld} dx \quad (3.22)$$

For any state function of two (2) independent variables, such as $F(x_1, x_2)$, the total differential of F with respect to the two (2) state variables (x_1, x_2) can be written as:

$$dF(x_1, x_2) = \left. \frac{\partial F}{\partial x_1} \right|_{x_2} dx_1 + \left. \frac{\partial F}{\partial x_2} \right|_{x_1} dx_2 \quad (3.23)$$

Now, it is important to recognise the partial derivatives in Eq. (3.23) are each taken by holding the opposite state variable **constant**. Eq. (3.23) is valid for any state function F and hence it is certainly valid for W_{fld} . Which makes:

$$dW_{fld}(\lambda, x) = \left. \frac{\partial W_{fld}}{\partial \lambda} \right|_x d\lambda + \left. \frac{\partial W_{fld}}{\partial x} \right|_{\lambda} dx. \quad (3.24)$$

Given λ and x are both **independent variables**, Eq. (3.22) and Eq. (3.24) must be equal for all values of $d\lambda$ and dx , and so, equating terms, we see that

$$i = \left. \frac{\partial W_{fld}(\lambda, x)}{\partial \lambda} \right|_x \quad (3.25)$$

where the partial derivative is taken while holding x constant and

$$f_{fld} = - \left. \frac{\partial W_{fld}(\lambda, x)}{\partial x} \right|_{\lambda}, \quad (3.26)$$

in this case holding λ constant while taking the partial derivative.

Once we know W_{fld} as a function of λ and x , Eq. (3.25) can be used to solve for $i(\lambda, x)$. More importantly, Eq. (3.26) can be used to solve for the mechanical force $f_{fld}(\lambda, x)$. It cannot be over-emphasised that the partial derivative of Eq. (3.26) is taken while holding the flux linkages λ **constant**. This is easily done provided W_{fld} is a known function of λ and x .

This is purely a mathematical requirement and has nothing to do with whether λ is held constant when operating the actual device.

The force f_{fld} is determined from Eq. (3.26) directly in terms of the electrical state variable λ . If we then want to express the force as a function of i , we can do so by substituting the appropriate expression for λ as a function of i into the expression for f_{fld} that is obtained by using Eq. (3.26).

This substitution must be done only after the partial derivative is taken.

¹²In the thermodynamics of equilibrium, a state function, function of state, or point function for a thermodynamic system is a mathematical function relating several state variables or state quantities (that describe equilibrium states of a system) that depend only on the current equilibrium thermodynamic state of the system [16], not the path which the system has taken to reach that state.

A state function describes equilibrium states of a system, therefore also describing the type of system. A state variable is typically a state function so the determination of other state variable values at an equilibrium state also determines the value of the state variable as the state function at that state.

This basically means the path of integration does not matter.

3.5.1 Linear Motion

For linear magnetic systems for which,

$$\lambda = L(x) i,$$

the energy is expressed by Eq. (3.19) and the force can be found by direct substitution in Eq. (3.26):

$$f_{fld} = -\frac{\partial}{\partial x} \left(\frac{1}{2} \frac{\lambda^2}{L(x)} \right) \bigg|_x = \frac{\lambda^2}{2L(x)^2} \frac{dL(x)}{dx} \quad (3.27)$$

Here, the force can be expressed in terms of the current i simply by substitution of $\lambda = L(x) i$.

$$f_{fld} = \frac{i^2}{2} \frac{dL(x)}{dx} \quad (3.28)$$

3.5.2 Rotating Motion

For a system with a rotating mechanical terminal, the mechanical terminal variables become the angular displacement θ and the torque T_{fld} . In this case, Eq. (3.22) becomes:

$$dW_{fld}(\lambda, \theta) = i d\lambda - T_{fld} d\theta, \quad (3.29)$$

where the explicit dependence of W_{fld} on state variables λ and θ has been indicated. By analogy to the development that led to Eq. (3.26), the torque can be found from the negative of the partial derivative of the energy with respect to θ taken holding λ constant:

$$T_{fld} = -\frac{\partial W_{fld}(\lambda, \theta)}{\partial \theta} \bigg|_{\lambda}. \quad (3.30)$$

For linear magnetic systems for which $\lambda = L(\theta)i$, by analogy to Eq. (3.19) the energy is given by

$$W_{fld}(\lambda, \theta) = \frac{1}{2} \frac{\lambda^2}{L(\theta)}. \quad (3.31)$$

The torque is therefore given by

$$T_{fld} = -\frac{\partial}{\partial \theta} \left(\frac{1}{2} \frac{\lambda^2}{L(\theta)} \right) \bigg|_{\lambda} = \frac{1}{2} \frac{\lambda^2}{L(\theta)^2} \frac{dL(\theta)}{d\theta}, \quad (3.32)$$

which can be expressed indirectly in terms of the current i as:

$$T_{fld} = \frac{i^2}{2} \frac{dL(\theta)}{d\theta} \quad (3.33)$$

3.6 Determination of Magnetic Force and Torque from Co-Energy

A mathematical manipulation of Eq. (3.22) can be used to define a **new state function**, known as the **co-energy**, from which the force can be obtained directly as a function of the current.

The selection of energy or co-energy as the state function is purely a matter of convenience as they both give the same result, but one or the other may be simpler analytically, depending on the desired result and the characteristics of the system being analysed.

The co-energy W'_{fld} is defined as a function of i and x such that

$$W'_{fld}(i, x) = i\lambda - W_{fld}(\lambda, x) \quad (3.34)$$

The desired derivation is carried out by using the differential of $i\lambda$

$$d(i\lambda) = i d\lambda + \lambda di \quad (3.35)$$

and the differential of $dW_{fld}(\lambda, x)$ from Eq. (3.22). From Eq. (3.34)

$$dW'_{fld}(i, x) = d(i\lambda) - dW_{fld}(\lambda, x) \quad (3.36)$$

Substitution of Eq. (3.22) and Eq. (3.35) into Eq. (3.36) results in

$$dW'_{fld}(i, x) = \lambda di + f_{fld} dx \quad (3.37)$$

From Eq. (3.37), the co-energy $W'_{fld}(i, x)$ can be seen to be a state function of the two (2) independent variables i and x . Therefore, its differential can be expressed as

$$dW'_{fld}(i, x) = \left. \frac{\partial W'_{fld}}{\partial i} \right|_x di + \left. \frac{\partial W'_{fld}}{\partial x} \right|_i dx \quad (3.38)$$

Eq. (3.37) and Eq. (3.38) must be equal for all values of di and dx . Therefore:

$$\lambda = \left. \frac{\partial W'_{fld}(i, x)}{\partial i} \right|_{x_i} \quad (3.39)$$

$$f_{fld} = \left. \frac{\partial W'_{fld}(i, x)}{\partial x} \right|_i \quad (3.40)$$

Eq. (3.40) gives the mechanical force directly in terms of both i and x .

The partial derivative in Eq. (3.40) is taken while holding i **constant**. Therefore W'_{fld} must be a known function of i and x .

For any given system, Eq. (3.26) and Eq. (3.40) will give the same result. The choice as to which to use to calculate the force is dictated by user preference and convenience. By analogy to the derivation of Eq. (3.17), the co-energy can be found from the integral of λdi :

$$W'_{fld}(i, x) = \int_0^i \lambda(i', x) di' \quad (3.41)$$

3.6.1 Linear Motion

For linear magnetic systems for which $\lambda = L(x) i$, the co-energy is therefore given by

$$W'_{\text{fld}}(i, x) = \frac{1}{2} L(x) i^2, \quad (3.42)$$

and the force can be found from Eq. (3.40):

$$f_{\text{fld}} = \frac{i^2}{2} \frac{dL(x)}{dx} \quad (3.43)$$

which, as expected, is identical to the expression given by Eq. (3.28). Note that for linear systems, substitution of $L(x) i$ for λ in Eq. (3.19) shows that numerically, $W'_{\text{fld}} = W_{\text{fld}}$.

However, it is important to recognise that when calculating force from energy using Eq. (3.26), the energy must be expressed explicitly in terms of λ in the form of Eq. (3.19). Similarly, when calculating force from co-energy using Eq. (3.40), the energy must be expressed explicitly in terms of i in the form of Eq. (3.42).

3.6.2 Rotating Motion

For a system with a rotating mechanical displacement, the co-energy can be expressed in terms of the current and the angular displacement θ

$$W'_{\text{fld}}(i, \theta) = \int_0^i \lambda(i', \theta) di' \quad (3.44)$$

and the torque is given by

$$T_{\text{fld}} = \left. \frac{\partial W'_{\text{fld}}(i, \theta)}{\partial \theta} \right|_i \quad (3.45)$$

If the system is magnetically linear,

$$W'_{\text{fld}}(i, \theta) = \frac{1}{2} L(\theta) i^2 \quad (3.46)$$

and

$$T_{\text{fld}} = \frac{i^2}{2} \frac{dL(\theta)}{d\theta} \quad (3.47)$$

which is identical to Eq. (3.33).

In field-theory terms, for soft magnetic materials,¹³ it can be shown that

¹³for which $B = 0$ when $H = 0$.

$$W'_{\text{fld}} = \int_V \left(\int_0^{H_0} \mathbf{B} \cdot d\mathbf{H} \right) dV \quad (3.48)$$

For soft magnetic material with constant permeability ($\mathbf{B} = \mu \mathbf{H}$), this reduces to

$$W'_{\text{fld}} = \int \frac{\mu H^2}{2} dV \quad (3.49)$$

For PM materials,¹⁴ for which $B = 0$ when $H = H_c$, the energy and co-energy are equal to zero when $B = 0$ and hence when $H = H_c$. Therefore, although Eq. (3.20) still applies for calculating the energy, Eq. (3.48) must be modified to the form

$$W'_{fld} = \int_V \left(\int_{H_2}^{H_0} \mathbf{B} \cdot d\mathbf{H} \right) dV \quad (3.50)$$

Eq. (3.50) can be considered to apply in general since soft magnetic materials can be considered to be simply hard magnetic materials with $H_c = 0$, in which case Eq. (3.50) reduces to Eq. (3.48).

Information: Finite Element Method

For some cases, magnetic circuit representations may be difficult to realise or may not produce solutions with good accuracy. Often such situations are characterised by complex geometries and/or magnetic materials driven deeply into saturation. In such situations, numerical techniques can be used to evaluate the system energy using Eq. (3.20) or the co-energy using either Eq. (3.48) or Eq. (3.50). One such technique, known as the **finite-element method**, has become widely used, which can be used to calculate the system co-energy for various values of the displacement x of a linear-displacement actuator. The force can then be obtained from Eq. (3.40), with the derivative of co-energy with respect to x being calculated numerically from the results of the finite-element analysis.

Exercise 3.1 Determining from Coenergy

Relooking at the same relay again,

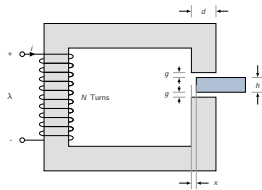


Figure 3.6: Relay with movable plunger. The relay has a depth of l .

Please find the force on the plunger as a function of x when the coil is driven by a controller which produces a current as a function of x of the form:

$$i(x) = \frac{l_0 x}{d} \text{ A}$$

SOLUTION From (TutBook) Ex. ??, we have:

$$L(x) = \frac{N^2 d l \mu_0 \left(1 - \frac{x}{d}\right)}{2g} \text{ H}$$

This is a magnetically-linear system for which the force can be calculated from Eq. (3.43) as:

$$f_{fld} = \frac{i^2}{2} \frac{dL(x)}{dx} = -\frac{N^2 i^2 l \mu_0}{4g} \text{ N}$$

Substituting for $i(x)$, the expression for the force as a function of x can be determined as:

$$f_{fld} = -\frac{l_0^2 N^2 l \mu_0 x^2}{4d^2 g} \text{ N}$$

Note that from Eq. (3.42), the coenergy for this system is equal to

$$W'_{fld}(i, x) = \frac{N^2 d i^2 l \mu_0 \left(1 - \frac{x}{d}\right)}{4g} \text{ J}$$

and the derivative of this expression with respect to x gives the expected expression for the force in terms of the current i . In this example, one might be tempted to introduce the expression for $i(x)$ directly into the coenergy expression, in which case the coenergy would be given by:

$$W'_{fld}(i, x) = \frac{l_0^2 N^2 \mu_0 l d (1 - x/d)}{4\sigma} \left(\frac{x}{d}\right)^2$$

Although this is a perfectly correct expression for the coenergy as a function of x under the specified operating conditions, if one were to attempt to calculate the force from taking the partial derivative of this expression for W'_{fld} with respect to x , the resultant expression would not give the correct expression for the force. The reason for this is quite simple: As seen from Eq. (3.40), the partial

¹⁴These materials are also known as **soft** materials.

derivative must be taken holding the current constant. Having substituted the expression for $i(x)$ to obtain this equation for the coenergy, the current is no longer a

constant and this requirement cannot be met. This illustrates the problems that can arise if the various force and torque expressions developed here are misapplied.

For a magnetically linear system, the energy and co-energy are **numerically equal**:

$$\frac{1}{2} \frac{\lambda^2}{L} = \frac{1}{2} L i^2.$$

The same is true for the energy and co-energy densities:

$$\frac{1}{2} \frac{B^2}{\mu} = \frac{1}{2} \mu H^2$$

For a non-linear system in which λ and i or B and H are **NOT** linearly proportional, the two functions are **NOT** numerically equal. A graphical interpretation of the energy and co-energy for a nonlinear system is shown in **Fig. 3.7**.

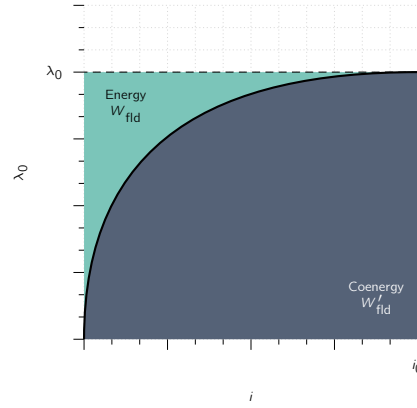


Figure 3.7: Graphical interpretation of energy and co-energy in a singly excited system.

The area between the $\lambda - i$ curve and the vertical axis, equal to the integral of $i d\lambda$, is the **energy**. The area to the horizontal axis given by the integral of λdi is the **co-energy**.

For this singly-excited system, the sum of the energy and co-energy is:

$$W_{fld} + W'_{fld} = \lambda i \quad (3.51)$$

The force produced by the magnetic field in a device such as that of **Fig. 3.4** for some particular value of x and i or λ cannot, of course, depend upon whether it is calculated from the energy or co-energy. A graphical illustration will demonstrate that both methods must give the same result.

For simplicity, assume the relay armature of **Fig. 3.4** is at position x so that the device is operating at point a in **Fig. 3.8a**.

The partial derivative of Eq. (3.26) can be interpreted as the limit of $-\Delta W_{fld}/\Delta x$ with λ constant as $\Delta x \rightarrow 0$. If we allow a change Δx , the change $-\Delta W_{fld}$ is shown by the shaded area in **Fig. 3.8a**.

Therefore, the force:

$$f_{fld} = \frac{\text{shaded area}}{\Delta x} \quad \text{as} \quad \Delta x \rightarrow 0$$

On the other hand, the partial derivative of Eq. (3.40) can be interpreted as the limit of $\Delta W'_{fld}/\Delta x$ with i constant as $\Delta x \rightarrow 0$. This perturbation of the device is shown in **Fig. 3.8b**. The force:

$$f_{fld} = \frac{\text{shaded area}}{\Delta x} \quad \text{as} \quad \Delta x \rightarrow 0$$

The shaded areas differ only by the small triangle abc of sides Δi and $\Delta \lambda$, so that in the limit the shaded areas resulting from Δx at constant λ or at constant i are equal.

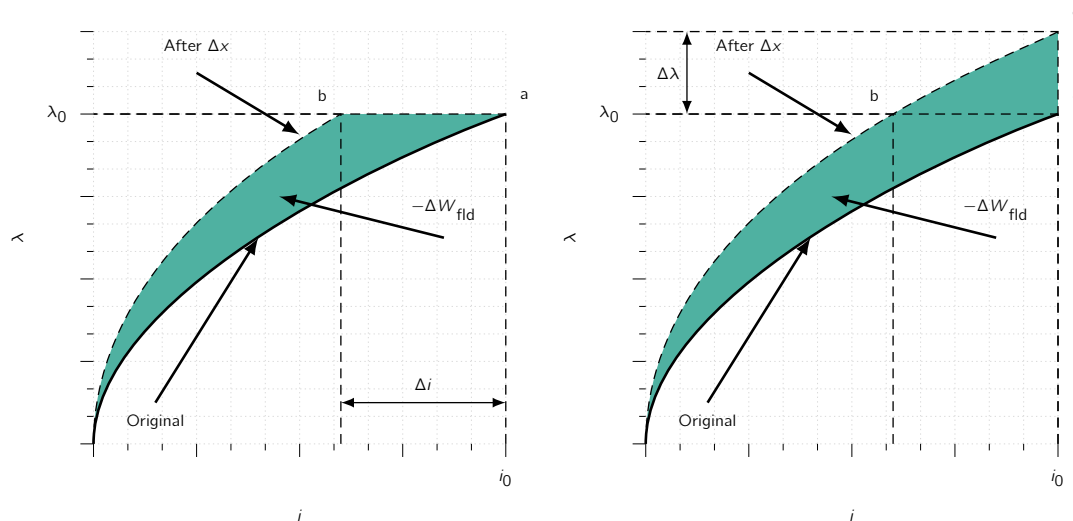


Figure 3.8: Effect of Δx on the energy and co-energy of a singly-excited device: (a) change of energy with λ held constant; (b) change of co-energy with i held constant.

Therefore the force produced by the magnetic field is independent of whether the determination is made with energy or co-energy.

Eq. (3.26) and Eq. (3.40) express the mechanical force of electrical origin in terms of partial derivatives of the energy and co-energy functions $W_{fld}(\lambda, x)$ and $W'_{fld}(i, x)$. It is important to note two (2) things about them:

- the variables in terms of which they must be expressed, and
- their algebraic signs.

Physically, of course, the force depends on the dimension x and the magnetic field. The field¹⁵ can be specified in terms of flux linkage (λ), or current (i), or related variables.

¹⁵ And therefore the energy or co-energy.

To reiterate, the selection of the energy or co-energy function as a basis for analysis is a matter of convenience.

The algebraic signs in Eq. (3.26) and Eq. (3.40) show that the force acts in a direction to decrease the magnetic field stored energy at constant flux or to increase the co-energy at constant current.

In a singly-excited device, the force acts to increase the inductance by pulling on members so as to reduce the reluctance of the magnetic path linking the winding.

3.7 Multiply Excited Magnetic Systems

Many electromechanical devices have multiple electrical terminals. In measurement systems it is often desirable to obtain torques proportional to two (2) electric signals:

For example, a meter which determines power as the product of voltage and current.

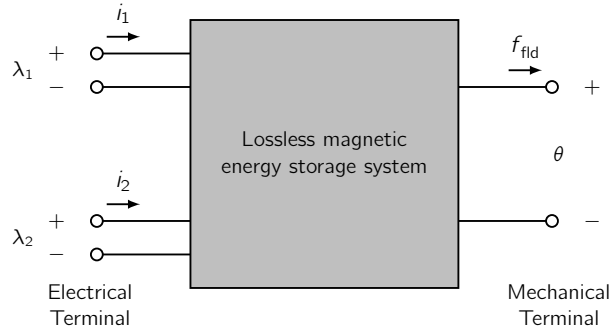


Figure 3.9: Multiply excited magnetic energy storage system.

Similarly, most electromechanical-energy-conversion devices consist of multiply excited magnetic-field systems. Analysis of these systems follows directly from the techniques discussed in previous sections. We will now illustrate these techniques based on a system with two (2) electric terminals. A schematic representation of a simple system with two electrical terminals and one mechanical terminal is shown in **Fig. 3.9**.

In this case it represents a system with rotary motion, and the mechanical terminal variables are torque (T_{fld}) and angular displacement (θ). Given there are three (3) terminals, the system must be described in terms of three (3) independent variables. These can be:

- Mechanical angle (θ),
- Flux linkages (λ_1, λ_2),
- Currents (i_1, i_2), or
- a hybrid set including one current and one flux.

When fluxes are used, differential energy function $dW_{fld}(\lambda_1, \lambda_2, \theta)$ corresponding to Eq. (3.29) is:

$$dW_{fld}(\lambda_1, \lambda_2, \theta) = i_1 d\lambda_1 + i_2 d\lambda_2 - T_{fld} d\theta, \quad (3.52)$$

and in direct analogy to the previous development for a singly-excited system:

$$i_1 = \left. \frac{\partial W_{fld}(\lambda_1, \lambda_2, \theta)}{\partial \lambda_1} \right|_{\lambda_2, \theta} \quad \text{and} \quad i_2 = \left. \frac{\partial W_{fld}(\lambda_1, \lambda_2, \theta)}{\partial \lambda_2} \right|_{\lambda_1, \theta}$$

and

$$T_{fld} = - \left. \frac{\partial W_{fld}(\lambda_1, \lambda_2, \theta)}{\partial \theta} \right|_{\lambda_1, \lambda_2} \quad (3.53)$$

In each of these equations, the partial derivative with respect to each independent variable must be taken holding the other two (2) independent variables constant.

The energy (W_{fld}) can be found by integrating Eq. (3.52). Similarly to the singly-excited case, although the energy at any given point is **independent** of the integration path, this integration is most conveniently done by holding λ_1 and λ_2 fixed at zero and integrating first over θ .

Under these aforementioned conditions, T_{fld} is zero, and therefore this integral is zero. One can then integrate over¹⁶ λ_2 and finally over λ_1 . Therefore:

¹⁶while holding λ_1 zero.

$$W_{fld}(\lambda_{10}, \lambda_{20}, \theta_0) = \int_0^{\lambda_{20}} i_2(\lambda_1 = 0, \lambda_2, \theta = \theta_0) d\lambda_2 + \int_0^{\lambda_{10}} i_1(\lambda_1, \lambda_2 = \lambda_{20}, \theta = \theta_0) d\lambda_1 \quad (3.54)$$

One could, of course, interchange the order of integration for λ_2 and λ_1 .

It is important to recognise however, for the expression of Eq. (3.54), the state variables are integrated over a specific path over which only one state variable is varied at a time.

For example, λ_1 is maintained initially at zero while integrating over λ_2 .

This is explicitly indicated in Eq. (3.54).

Failure to observe the constraints imposed by a chosen integration path is one of the most common errors made in analysing these systems.

In a magnetically-linear system, the relationships between λ and i can be specified in terms of inductances as:

$$\lambda_1 = L_{11}i_1 + L_{12}i_2 \quad \text{and} \quad \lambda_2 = L_{21}i_1 + L_{22}i_2 \quad \text{where} \quad L_{12} = L_{21} \quad (3.55)$$

Here the inductances are, in general, functions of angular position (θ). These equations can be inverted to obtain expressions for the currents (i) as a function of angular position (θ):

$$i_1 = \frac{L_{22}\lambda_1 - L_{12}\lambda_2}{D} \quad i_2 = \frac{-L_{21}\lambda_1 + L_{11}\lambda_2}{D} \quad \text{where} \quad D = L_{11}L_{22} - L_{12}L_{21} \quad (3.56)$$

The energy for this linear system can be found from Eq. (3.54):

$$\begin{aligned} W_{fld}(\lambda_{10}, \lambda_{20}, \theta_0) &= \int_0^{\lambda_{20}} \frac{L_{11}(\theta_0)\lambda_2}{D(\theta_0)} d\lambda_2 \\ &+ \int_0^{\lambda_{10}} \frac{(L_{22}(\theta_0)\lambda_1 - L_{12}(\theta_0)\lambda_{20})}{D(\theta_0)} d\lambda_1 \\ &= \frac{1}{2D(\theta_0)} L_{11}(\theta_0)\lambda_{20}^2 + \frac{1}{2D(\theta_0)} L_{22}(\theta_0)\lambda_{10}^2 - \frac{L_{12}(\theta_0)}{D(\theta_0)} \lambda_{10}\lambda_{20} \end{aligned} \quad (3.57)$$

where the dependence of the inductances and the determinant $D(\theta)$ on the angular displacement (θ) has been explicitly indicated. Previously, the co-energy function was defined to permit determination of force and torque directly in terms of the current for a single-winding system.

A similar co-energy function can be defined in the case of systems with two (2) windings as:

$$dW'_{\text{fld}}(i_1, i_2, \theta) = \lambda_1 di_1 + \lambda_2 di_2 - W_{\text{fld}} \quad (3.58)$$

It is a state function of the two terminal currents and the mechanical displacement. Its differential, following substitution of Eq. (3.52), is given by

$$dW'_{\text{fld}}(i_1, i_2, \theta) = \lambda_1 di_1 + \lambda_2 di_2 + T_{\text{fld}} d\theta \quad (3.59)$$

From Eq. (3.59) we see that

$$\lambda_1 = \left. \frac{\partial W'_{\text{fld}}(i_1, i_2, \theta)}{\partial i_1} \right|_{i_2, \theta} \quad \text{and} \quad \lambda_2 = \left. \frac{\partial W'_{\text{fld}}(i_1, i_2, \theta)}{\partial i_2} \right|_{i_1, \theta} \quad (3.60)$$

Most significantly, the torque can now be determined directly in terms of the currents as

$$T_{\text{fld}} = \left. \frac{\partial W'_{\text{fld}}(i_1, i_2, \theta)}{\partial \theta} \right|_{i_1, i_2} \quad (3.61)$$

Similarly to Eq. (3.54), the co-energy can be found as:

$$W'_{\text{fld}}(i_{10}, i_{20}, \theta_0) = \int_0^{i_{20}} \lambda_2(i_1 = 0, i_2, \theta = \theta_0) di_2 + \int_0^{i_{10}} \lambda_1(i_1, i_2 = i_{20}, \theta = \theta_0) di_1 \quad (3.62)$$

For the linear system of described in Eq. (3.55):

$$W'_{\text{fld}}(i_1, i_2, \theta) = \frac{1}{2} L_{11}(\theta) i_1^2 + \frac{1}{2} L_{22}(\theta) i_2^2 + L_{12}(\theta) i_1 i_2 \quad (3.63)$$

For such a linear system, the torque can be found either from the energy of Eq. (3.57) using Eq. (3.53) or from the co-energy of Eq. (3.63) using Eq. (3.61). It is at this point, the utility of the co-energy function becomes apparent.

The energy expression of Eq. (3.57) is a complex function of displacement, and its derivative is even more so. Alternatively, the co-energy function is a relatively simple function of displacement, and from its derivative a straightforward expression for torque can be determined as a function of the winding currents i_1 and i_2 as:

$$T_{\text{fld}} = \left. \frac{\partial W'_{\text{fld}}(i_1, i_2, \theta)}{\partial \theta} \right|_{i_1, i_2} \quad (3.64)$$

$$= \frac{i_1^2}{2} \left(\frac{dL_{11}(\theta)}{d\theta} \right) + \frac{i_2^2}{2} \left(\frac{dL_{22}(\theta)}{d\theta} \right) + i_1 i_2 \left(\frac{dL_{12}(\theta)}{d\theta} \right) \quad (3.65)$$

Systems with more than two (2) electrical terminals are handled in analogous fashion. As with the two-terminal-pair system above, the use of a co-energy function of the terminal currents greatly simplifies the determination of torque or force.

Exercise 3.2 Multiply Excited System

Consider the following system with multiple excitation.

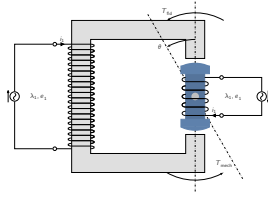


Figure 3.10: A system with multiple excitation. In this system the rotor is connected to a spring, which is then connected to a wall for stabilisation.

The inductances in henrys are given as:

$$L_{11} = 0.001 \cos(2\theta) + 0.003 \text{ H}$$

$$L_{12} = 0.3 \cos(\theta) \text{ H}$$

$$L_{22} = 10 \cos(2\theta) + 30 \text{ H}$$

Find and plot the torque $T_{fld}(\theta)$ for current $i_1 = 0.8 \text{ A}$ and $i_2 = 0.01 \text{ A}$

SOLUTION The torque can be determined from Eq. (3.64).

$$T_{fld} = \frac{i_1^2}{2} \left(\frac{dL_{11}(\theta)}{d\theta} \right) + \frac{i_2^2}{2} \left(\frac{dL_{22}(\theta)}{d\theta} \right) + i_1 i_2 \left(\frac{dL_{12}(\theta)}{d\theta} \right).$$

For $i_1 = 0.8 \text{ A}$ and $i_2 = 0.01 \text{ A}$, the torque is:

$$T_{fld} = -0.0024 \sin(\theta) - 0.00164 \sin(2\theta) \text{ N} \cdot \text{m}$$

Notice the torque expression consists of terms of two (2) types:

One term, proportional to $i_1 i_2 \sin \theta$, is due to the mutual interaction between the rotor and stator currents; it acts in a direction to align the rotor and stator so as to maximize their mutual inductance. Alternately, it can be thought of as being due to the tendency of two magnetic fields (in this case those of the rotor and stator) to align.

The torque expression also has terms proportional to $\sin 2\theta$ and to the square of the individual coil currents. These terms are due to the action of the individual winding currents alone and correspond to the torques one sees in singly-excited systems. Here each torque component acts in a direction to maximize its corresponding inductance so as to maximize the coenergy. The 2θ torque variation is due to the 2θ variation in the self inductances (exactly as was seen previously in Example 3.4), which in turn is due to the variation of the air-gap reluctance; notice that rotating the rotor by 180° from any given position gives the same air-gap reluctance (hence the twice-angle variation). This torque component is known as the reluctance torque. The two (2) torque components (mutual and reluctance), along with the total torque, are plotted below.

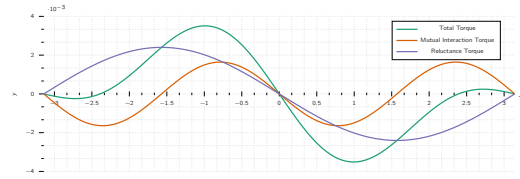


Figure 3.11: Plot of torque components for the multiply excited system of the exercise Multiply Excited System.

3.7.1 Linear Motion

The we looked previously for angular displacement can be repeated in an analogous fashion for the systems with linear displacement. If this is done, the expressions for energy and co-energy will be found to be

$$W_{fld}(\lambda_{10}, \lambda_{20}, x_0) = \int_0^{\lambda_{20}} i_2(\lambda_1 = 0, \lambda_2, x = x_0) d\lambda_2 + \int_0^{\lambda_{10}} i_1(\lambda_1, \lambda_2 = \lambda_{20}, x = x_0) d\lambda_1$$

$$W'_{fld}(i_{10}, i_{20}, x_0) = \int_0^{i_{20}} \lambda_2(i_1 = 0, i_2, x = x_0) di_2 + \int_0^{i_{10}} \lambda_1(i_1, i_2 = i_{20}, x = x_0) di_1 \quad (3.66)$$

Similarly the force can be found from

$$f_{\text{fld}} = - \left. \frac{\partial W_{\text{fld}}(\lambda_1, \lambda_2, x)}{\partial x} \right|_{\lambda_1, \lambda_2} \quad \text{or} \quad f_{\text{fld}} = \left. \frac{\partial W'_{\text{fld}}(i_1, i_2, x)}{\partial x} \right|_{i_1} \quad (3.67)$$

For a magnetically linear system, the co-energy expression of Eq. (3.63) becomes

$$W'_{\text{fld}}(i_1, i_2, x) = \frac{1}{2} L_{11}(x) i_1^2 + \frac{1}{2} L_{22}(x) i_2^2 + L_{12}(x) i_1 i_2 \quad (3.68)$$

and the force is therefore given by:

$$f_{\text{fld}} = \frac{i_1^2}{2} \left(\frac{dL_{11}(x)}{dx} \right) + \frac{i_2^2}{2} \left(\frac{dL_{22}(x)}{dx} \right) + i_1 i_2 \left(\frac{dL_{12}(x)}{dx} \right) \quad (3.69)$$

Information: Buchholz Relay

In electric power distribution and transmission, a Buchholz relay is a safety device mounted on some oil-filled power transformers and reactors, equipped with an external overhead oil reservoir called a “conservator”.

The Buchholz relay is used as a protective device sensitive to the effects of dielectric failure inside the equipment. The relay was first developed by Max Buchholz (1875 - 1956) in 1921.

Operation

Depending on the model, the relay has multiple methods to detect a failing transformer. On a slow accumulation of gas, gas produced by decomposition of insulating oil **accumulates** in the top of the relay and forces the oil level down. A float switch in the relay is used to initiate an alarm signal. Depending on design, a second float may also serve to detect slow oil leaks.

If an electrical arc forms, gas accumulation is rapid, and oil flows rapidly into the conservator. This flow of oil operates a switch attached to a vane located in the path of the moving oil. This switch normally will operate a circuit breaker to isolate the apparatus before the fault causes additional damage. Buchholz relays have a test port to allow the accumulated gas to be withdrawn for testing. Flammable gas found in the relay indicates some internal fault such as overheating or arcing, whereas air found in the relay may only indicate low oil level or a leak.

Through a connected gas sampling device the control can also be made from the ground. Depending on the requirements, the Buchholz relay has a flange or threaded connection.

The classic Buchholz relay has to comply with the requirements of the DIN EN 50216-2 standard. Depending on the requirements, it is equipped with up to four (2 per float) switches or change-over switches, which can either send a light signal or switch off the transformer.

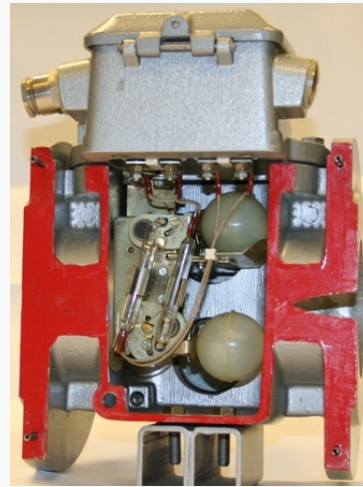


Figure 3.12: Two ball-shaped floats and two glass-enclosed reed switches are visible inside this cutaway view of a Buchholz relay.



Part II

Rotating Machines

Invention is the most important product of man's creative brain. The ultimate purpose is the complete mastery of mind over the material world, the harnessing of human nature to human needs.

(Nikola Tesla, My Inventions)

Chapter 4

Rotating Magnetic Fields

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4.1 Fundamental Concepts

The primary goal of this lecture is to introduce and understand the principles impacting the performance of electric machines. As we will see, the principles mentioned here are common for both AC and DC machines.¹

We will start with the following equation, which we have looked before:

$$e = \frac{d\lambda}{dt},$$

which can be used to determine the voltages induced by **time-varying magnetic fields**.

Electromagnetic energy conversion occurs when changes in the flux linkage λ result from mechanical motion, which we should know.

In rotating machines, voltages are generated in windings or groups of coils by rotating these windings

¹This part of the lecture plays an important part of understanding how electrical machines operate and therefore it is suggested to take a closer look.

mechanically through a magnetic field, by mechanically rotating a magnetic field past the winding, or by designing the magnetic circuit so that the **reluctance** varies with rotation of the rotor.

By either or both of these methods, the flux linking a specific coil is changed cyclically, and a time-varying voltage is generated.

Information: Armature Winding

A set of coils connected together is typically referred to as an **armature** winding. In general, the term armature winding is used to refer to a winding or a set of windings on a rotating machine which carry AC. In AC machines such as Synchronous Machine (SynM) or Induction Machine (IM), the armature winding is typically on the **stationary portion** of the motor referred to as the **stator**, in which case these windings may also be referred to as **stator windings**. In DC machines, however, the armature winding is found on the rotating member, referred to as the **rotor**. **Fig. 4.1** shows a DC machine rotor with its armature windings. As we will see, the armature winding of a DC machine consists of numerous coils connected together to form a closed loop. A rotating mechanical contact is used to supply current to the armature winding as the rotor rotates.

SynM and DC machines typically include a second set of winding² which carry DC current and which are used to produce the main operating flux in the machine. Such a winding is typically referred to as **field winding**. The field winding on a DC machine is found on the stator, while that on a SynM is found on the rotor, in which case current is typically supplied to the field winding via a rotating mechanical contact, with the exception being machines for which a rotating excitation system supplies the field current.

PMs also produce DC magnetic flux and are used in the place of field windings in some machines.

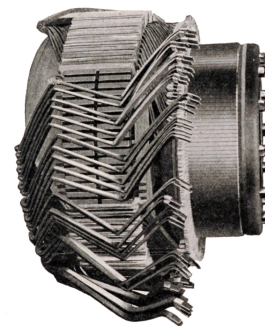


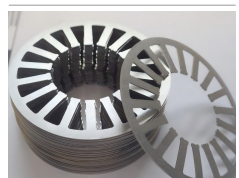
Figure 4.1: A partially-constructed DC armature, showing the (incomplete) windings.

For most rotating machines, the stator and rotor are made of electrical steel, and the windings are inserted in slots on these structures.

Electrical steel is a speciality steel used in the cores of electromagnetic devices such as motors, generators, and transformers because it reduces power loss. It is an iron alloy with silicon as the main additive element, instead of carbon.

The use of such high-permeability material maximises the coupling between the coils and increases the magnetic energy density associated with the electromechanical interaction. It also enables the engineer to shape and distribute the magnetic fields according to the requirements of each particular machine design.

The time-varying flux present in the armature structures of these machines tends to induce currents, known as **eddy currents**, in the electrical steel. Eddy currents can be a large source of loss in these machines and can significantly inhibit machine performance. To minimise the effects of eddy currents, the armature structure is typically built from thin laminations³ of electrical steel which are



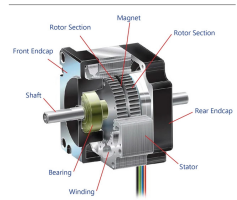
³ These laminations tend to be 0,55 mm thick, which we have also looked at in the previous chapters.

insulated from each other.

Information: Winding-less Rotors

Machines, such as variable reluctance machines and stepper motors,⁴ have no windings on the rotor. Operation of these machines depends on the **non-uniformity of air-gap reluctance** associated with variations in rotor position in conjunction with time-varying currents applied to their stator windings.

In such machines, both the stator and rotor structures are subjected to time-varying magnetic flux and, as a result, both **require lamination to reduce eddy-current losses**.



⁴An schematic view of a stepper motor. As we can see the rotor has no windings [17].

Rotating electric machines take many forms and are known by many names:

DC, SynM, PM, IM, variable reluctance, hysteresis, brush-less, and so on.

While these machines appear to be quite different, the physical principles governing their behaviour are quite similar, and it is often helpful to think of them in terms of the same physical picture. For example:

DC Machine Analysis shows that associated with both the rotor and the stator are magnetic flux distributions which are fixed in space and the torque-producing characteristic of stems from the tendency of these flux distributions to align.

Induction Machine Works on exactly the same principle. We can identify flux distributions associated with the rotor and stator. While they are **NOT** stationary but rather rotate in synchronism, just as in a DC motor they are displaced by a constant angular separation, and torque is produced by the tendency of these flux distribution to align.⁵

⁵A more important aspect is to create an analogy between AC and DC machines, which allows high-efficient control. This will be one of the main topics in **M.Sc Drive Systems**.

4.1.1 A Look at AC and DC Drives

Traditionally AC machines fall into one of two (2) categories:

1. Synchronous,
2. Induction

In SynM, rotor-winding currents are supplied **directly** from the stationary frame through a rotating contact whereas in IM, rotor currents are **induced** in the rotor windings by a combination of the time-variation of the stator currents and the motion of the rotor **relative to the stator**.

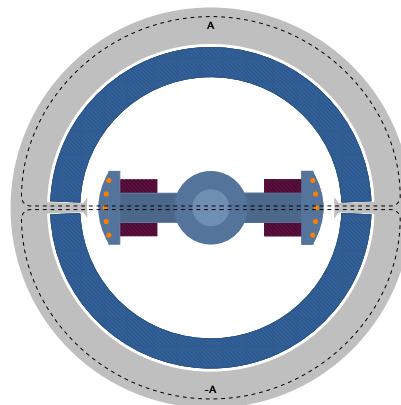


Figure 4.2: A schematic view of the first quadrant of a simple, two-pole, single-phase synchronous generator.

Synchronous Machines

A rough picture of SynM performance can be obtained by discussing the voltage induced in the armature of the noticeably simplified **salient-pole** AC synchronous generator shown schematically in **Fig. 4.2**. The field-winding of this machine produces a single pair of magnetic poles,⁶ and therefore this machine is referred to as a **two-pole machine** or **one-pole pair machine**.

⁶You can think of this as a bar magnet.

With rare exceptions, the armature winding of a SynM is on the stator, and the field winding is on the rotor.

The field winding is excited by DC, typically conducted to it by means of stationary **carbon brushes** which contact rotating slip rings or collector rings.⁷ Practical factors usually dictate this orientation of the two windings:

It is advantageous to have the single, low-power field winding on the rotor while having the high-power, typically multiple-phase, armature winding on the stator.

The armature winding, consisting here of only a single coil of N turns, is indicated in cross section by the two coil sides a and $-a$ placed in diametrically opposite narrow slots on the inner periphery of the stator of **Fig. 4.2**. The conductors forming these coil sides are parallel to the shaft of the machine and are connected in series by end connections.

The rotor is turned at a constant speed by a source of mechanical power connected to its shaft. The armature winding is assumed to be **open-circuited** and therefore the flux in this machine is produced by the field winding alone. Flux paths are shown schematically by dashed lines in **Fig. 4.2**.

A highly idealised analysis of this machine would be to assume a **sinusoidal distribution** of magnetic flux in the air gap. The resultant idealised radial distribution of air-gap flux density B is shown in **Fig. 4.3a** as a function of the spatial angle θ_a around the rotor periphery.⁸

⁷For cost-effective systems, carbon brushes are generally used. However in some cases the field winding may be supplied from a rotating excitation system, typically referred to as a brush-less excitation system.

⁸measured with respect to the magnetic axis of the armature winding

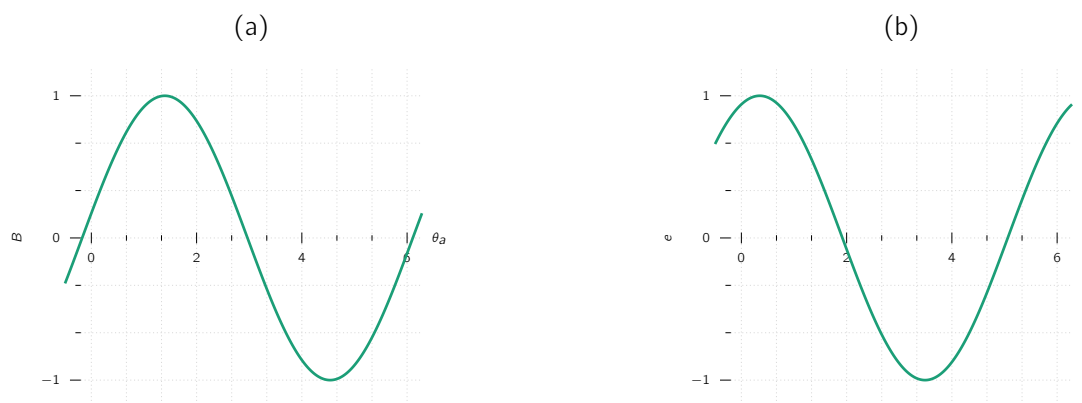


Figure 4.3: (a) Idealised sinusoidal space distribution of the air-gap radial flux density and (b) corresponding waveform of the generated voltage for the single-phase generator of **Fig. 4.2**.

In practice, the air-gap flux-density of practical salient-pole machines can be made to approximate a sinusoidal distribution by properly shaping the pole faces.

As the rotor rotates, the flux-linkages of the armature winding **change with time**. Under the assumption of a sinusoidal flux distribution and constant rotor speed, the resulting coil voltage will be sinusoidal in time as shown in **Fig. 4.3b**. The coil voltage passes through a complete cycle for each revolution of the two-pole machine of **Fig. 4.2**.

Its frequency in cycles per second (Hz) is the same as the speed of the rotor in revolutions per second: the electric frequency of the generated voltage is synchronised with the mechanical speed, and this is the reason for the designation “synchronous” machine.

Therefore a two-pole synchronous machine must revolve at 3000 revolutions per minute to produce a 50 Hz voltage.

It is worth mentioning most SynMs have more than just two poles. As a specific example, **Fig. 4.4** shows in schematic form a **four-pole** single-phase generator. The field coils are connected so that the poles are of alternate polarity. There are two complete wavelengths, or cycles, in the flux distribution around the periphery, as shown in **Fig. 4.5**.

The armature winding now consists of two (2) coils: $a_1, -a_1$ and $a_2, -a_2$ which can be connected either in **series** or **parallel**. The span of each coil is one wavelength of flux. The generated voltage now goes through two (2) complete cycles per revolution of the rotor. The frequency in Hz will therefore be twice the speed in revolutions per second.

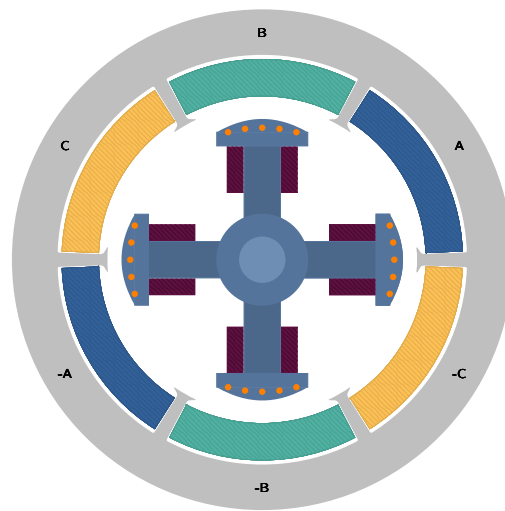


Figure 4.4: Schematic view of a simple, four-pole, single-phase synchronous generator.

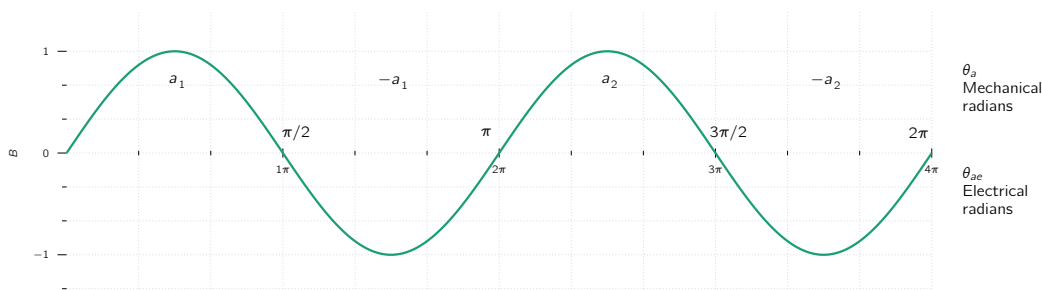


Figure 4.5: Space distribution of the air-gap flux density in an idealised, four-pole synchronous generator.

When a machine has more than two (2) poles, it is convenient to concentrate on a single pair of poles⁹ and to recognise the electric, magnetic, and mechanical conditions associated with every other pole pair are repetitions of those for the pair under consideration. For this reason it is convenient to express angles in **electrical** degrees or radians rather than in physical units.

⁹Or what we call pole-pairs.

One pair of poles in a multi-pole machine or one cycle of flux distribution corresponds to 360 electrical degrees or 2π electrical radians. Given there are poles/2 amount of complete wavelengths, or cycles, in one complete revolution, we can state:

$$\theta_a = \left(\frac{\text{poles}}{2} \right) \theta_{ae} \quad (4.1)$$

where θ_{ae} is the angle in electrical units, θ_a is the spatial angle. This same relationship applies to all angular measurements in a multi-pole machine.

Values in electrical units will be equal to (poles/2) times their actual spatial values.

The coil voltage of a multi-pole machine passes through a complete cycle every time a pair of poles sweeps by, or (poles/2) times each revolution. The electrical frequency f_e of the voltage generated in a SynM is therefore:

$$f_e = \left(\frac{\text{poles}}{2} \right) \frac{n}{60} \text{ Hz} \quad (4.2)$$

where n is the mechanical speed in revolutions per minute (rpm), which means $n/60$ is the speed in revolutions per second. The electrical frequency of the generated voltage in radians per second is

$$\omega_e = \left(\frac{\text{poles}}{2} \right) \omega_m \quad (4.3)$$

where ω_m is the mechanical speed in radians per second ($\text{rad} \cdot \text{s}^{-1}$).

The rotors shown in **Fig. 4.2** and **Fig. 4.4** have **salient**, or **projecting**, poles with concentrated windings. **Fig. 4.6** shows diagrammatically a non-salient-pole, or cylindrical, rotor. The field winding on the rotor is a two-pole distributed winding; the coil sides are distributed in multiple slots around the rotor periphery and arranged to produce an approximately sinusoidal distribution of radial air-gap flux.

The relationship between electrical frequency and rotor speed of Eq. (4.2) can serve as a basis for understanding why some synchronous generators have salient-pole rotor structures while others have cylindrical rotors.

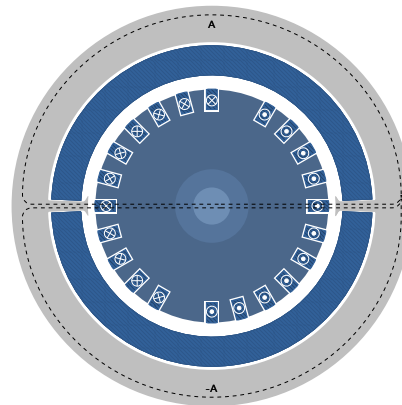


Figure 4.6: Simple two-pole round-rotor field winding.

Most power systems in the world operate at frequencies of either 50 or 60 Hz.

- A salient-pole construction is characteristic of hydro-electric generators because hydraulic turbines operate at relatively low speeds, and hence a relatively large number of poles is required to produce the desired frequency. A salient-pole construction is better adapted mechanically to this situation.
- Steam turbines and gas turbines operate best at relatively high speeds, and turbine-driven alternators or turbine generators are commonly two- or four-pole cylindrical-rotor machines.

These cylindrical rotors are made from a single steel forging or from several forgings, as shown in **Fig. 4.7**.

Most of the world's power systems are three-phase systems and, as a result, with very few exceptions, synchronous generators are three-phase machines. For the production of a set of three voltages phase-displaced by 120 electrical degrees in time, a minimum of three coils phase-displaced 120 electrical degrees in space must be used. A simplified schematic view of a three-phase, two-pole machine with one coil per phase is shown in **Fig. 4.8a**. The three phases are designated by the letters a, b, and c. In an elementary four-pole machine, a minimum of two such sets of coils must be used, as illustrated in **Fig. 4.8b**; in an elementary multi-pole machine, the minimum number of coil sets is given by one half the number of poles.

The two coils in each phase of **Fig. 4.8b** are connected in series so that their voltages add, and the three phases may then be either Y- or Δ -connected.

When a synchronous generator supplies electric power to a load, the armature current creates a magnetic flux wave in the air gap which rotates at synchronous speed. This flux reacts with the flux created by the field current, and electromechanical torque results from the tendency of these two magnetic fields to align. In a generator this torque opposes rotation, and mechanical torque must be applied from the prime mover to sustain rotation. This electromechanical torque is the

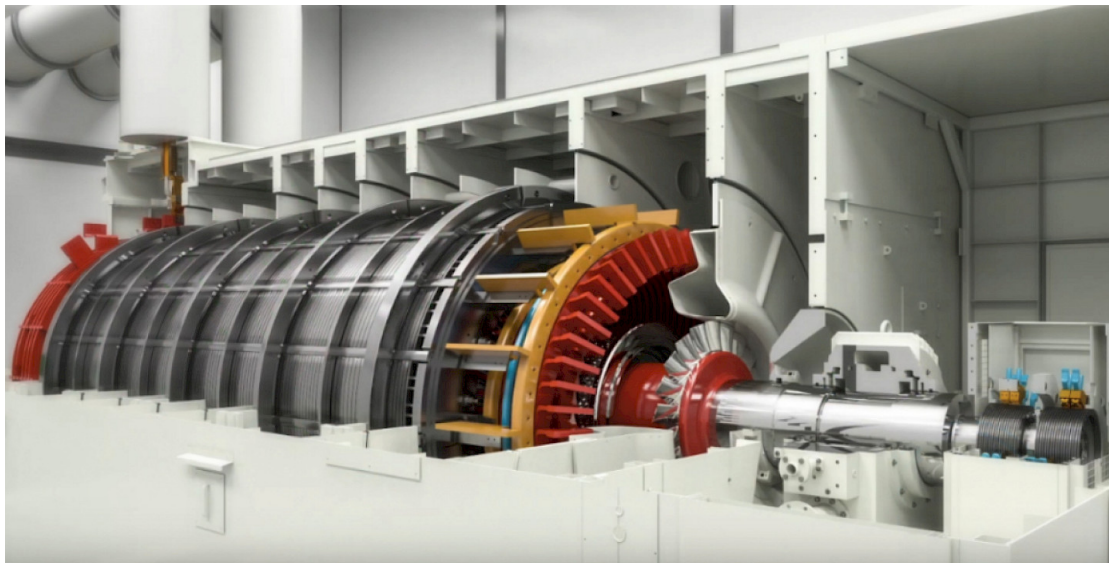


Figure 4.7: Round rotor synchronous generator used in grid regulation [18].

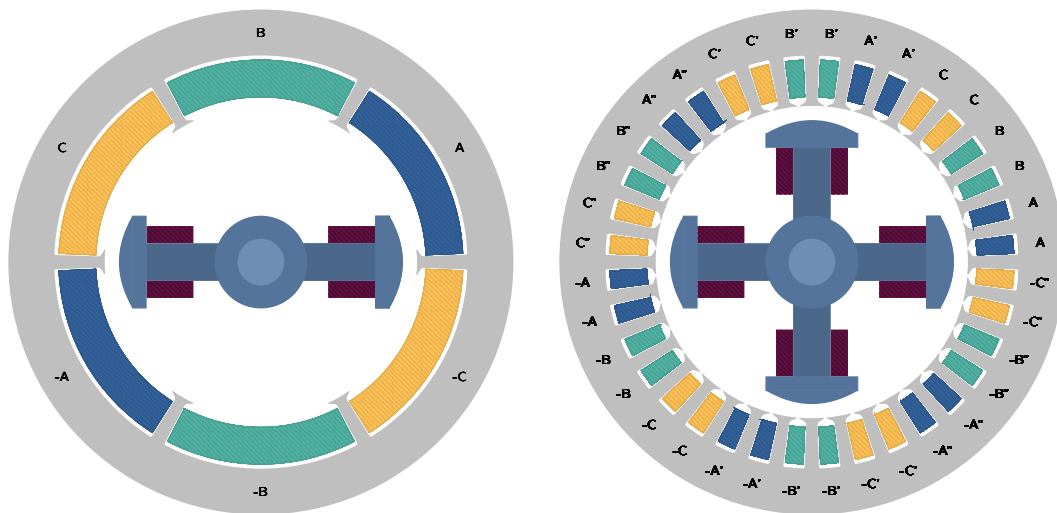


Figure 4.8: Schematic views of three-phase generators: (a) two-pole, (b) four-pole, and (c) Y connection of the windings.

mechanism through which the synchronous generator converts mechanical to electric energy.

The opposite of the synchronous generator is the SynM. AC is supplied to the armature winding on the stator, and DC excitation is supplied to the field winding on the rotor. The magnetic field produced by the armature currents rotates at synchronous speed.¹⁰ A steady electromechanical torque is produced when the rotor rotates in synchronism with the magnetic field produced by the armature currents.

¹⁰equal to 2/poles times the electrical frequency of the armature currents

Therefore the steady-state speed of a SynM is determined by the number of poles and the frequency of the armature current and a SynM operated from a constant-frequency ac source will operate at a constant steady-state speed.

In a motor the electromechanical torque is in the direction of rotation and balances the opposing torque required to drive the mechanical load. The flux produced by currents in the armature of a SynM rotates **ahead** of that produced by the rotor field winding, thus pulling on the field winding¹¹ and doing work. This is the opposite of the situation in a synchronous generator, where the field does work as its flux pulls on that of the armature. In both generators and motors, in addition to an electromechanical torque, a speed voltage (emf) is induced in the armature by the rotating field winding.

¹¹and therefore on the rotor

Production of both force or torque and a speed voltage are both essential components of electromechanical energy conversion.

Induction Drives

A second type of AC machine is the IM. In an IM, the stator windings are essentially the same as those of a SynM. However, the rotor windings are electrically short-circuited and frequently have **no external connections**. Currents are induced by transformer action from the stator winding. A cutaway view of a squirrel-cage induction motor is shown in **Fig. 4.9**. Here the rotor “windings” are actually solid aluminium bars which are cast into the slots in the rotor and which are shorted together by cast aluminium rings at each end of the rotor. This type of rotor construction results in IM which are relatively inexpensive and highly reliable, factors contributing to their immense popularity and widespread application.



Figure 4.9: A cutaway of an explosion-proof induction motor [19].

In contrast to a SynM in which a field winding on the rotor is excited with DC current and the rotor rotates in synchronism with the flux wave produced by AC armature currents, the rotor windings of an IM are **NOT** excited by an external source. Rather, currents are induced in the shorted rotor windings as the rotor slips past the synchronously-rotating armature flux wave. Therefore, IMs are **asynchronous machines** and produce torque only when the rotor speed differs from synchronous speed.

Interestingly, although the rotor operates **asynchronously**, the flux wave produced by the induced rotor currents rotates **in synchronism** with the stator flux wave. This in fact is a requirement for, and consistent with, the ability of an IM to produce net torque. IMs operate at speeds less than the synchronous mechanical speed, in which case the armature flux in the IM leads that of the rotor and produces an electromechanical torque which pulls on the rotor just as is the case in a SynM. A typical speed-torque characteristic for an IM is shown in **Fig. 4.10**.

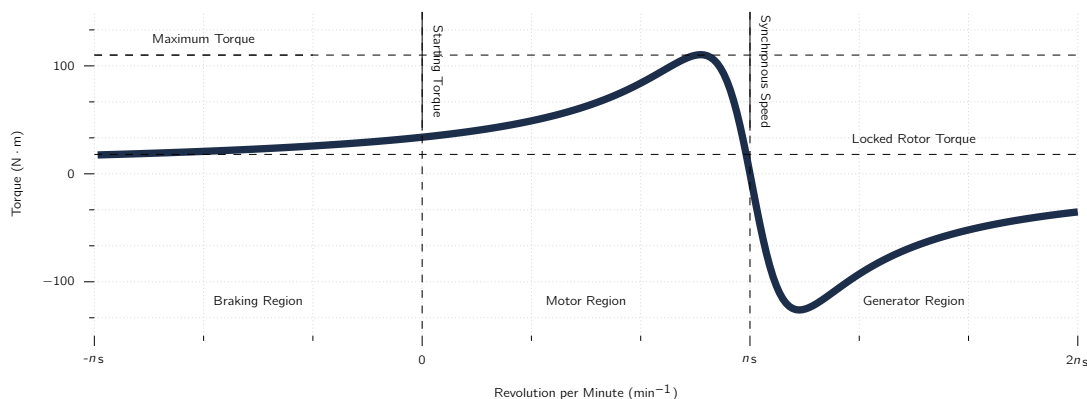


Figure 4.10: The speed-torque plot of an induction machine during steady-state operation.

Information: Rotating Transformers

Because rotor currents are produced by induction, an IM may be regarded as a generalised transformer in which electric power is transformed between rotor and stator together with a change of frequency and a flow of mechanical power.

Although IMs are primarily used as motors, in recent years induction generators have been found to be well suited for some wind-power applications.

4.1.2 DC Machines

As discussed previously, the armature winding of a DC machine is on the rotor with current conducted to it by means of carbon brushes. The field winding is on the stator and is excited by direct current. A cutaway view of a DC motor is shown in **Fig. 4.11** and a simple DC armature is shown in **Fig. 4.13**.

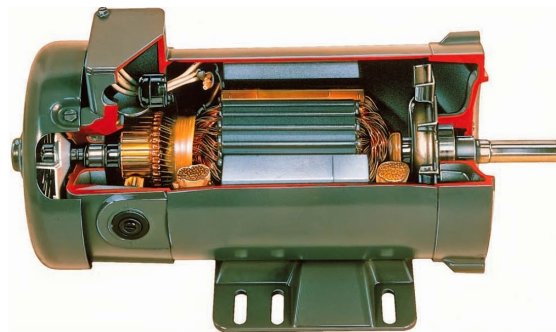


Figure 4.11: A cutaway diagram of a standard DC machine.

The armature winding, as can be seen consisting of coils of, say N , turns, placed at diametrically opposite points on the rotor with the conductors parallel to the shaft. The rotor is normally turned at a constant speed by a source of mechanical power connected to the shaft. The air-gap flux distribution usually approximates a **flat-topped wave**, rather than the sine wave found in AC machines.

Rotation of the coil, generates a coil voltage which is a time function having the same waveform as the spatial flux-density distribution.

The function of a DC generator is the production of DC voltage and current. Therefore the AC voltages and currents induced in the armature winding must be rectified.

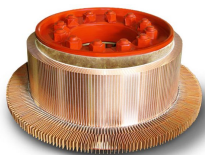


Figure 4.12: An example of a mechanical commutation ring. Please observe that each segment is electrically isolated.

In a DC machine, rectification is produced mechanically by means of a **commutator** which is a cylinder formed of copper segments to which the armature coils are connected, shown in **Fig. 4.12**. These segments are otherwise insulated from each other by mica or some other highly insulating material and mounted on, but insulated from, the rotor shaft. Stationary carbon brushes held against the commutator surface connect the winding to the external armature terminals. The commutator and brushes can readily be seen in **Fig. 4.11**

To understand its function as a rectifier, the commutator at all times

connects the coil side which is closest to the south pole to the positive brush and the coil side closest to the north pole is connected to the negative brush.

Therefore, each half rotation of the rotor, the brushes switch their polarity with respect to the coil polarity. As a result, although the coil voltage is an alternating voltage, the commutator provides full-wave rectification, transforming the coil voltage to the voltage between brushes and making available a **unidirectional** voltage to the external circuit. In a later chapter, we shall look at DC machines in more detail.

The direct current in the field winding of a DC machine creates a magnetic flux distribution which is stationary with respect to the stator. Similarly, the effect of the commutator is such that when direct current flows through the brushes, the armature creates a magnetic flux distribution which is also fixed in space and whose axis, determined by the design of the machine and the position of the brushes, is typically perpendicular to the axis of the field flux.

Therefore, just as in the AC machines discussed previously, it is the interaction of these two (2) flux distributions that creates the torque of the DC machine:

- If the machine is acting as a generator, the electro-mechanical torque opposes rotation.
- If it is acting as a motor, the torque acts in the direction of rotation. Remarks similar to those already made concerning the roles played by the generated voltage and electromechanical torque in the energy conversion process in SynMs apply equally well to DC machines.

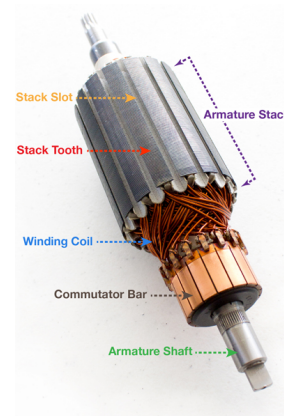


Figure 4.13: Example of a DC armature.

4.2 MMF of Distributed Windings

Most armatures have **distributed windings**, such as windings spread over a number of slots around the air-gap periphery. The individual coils are **inter-connected** so the result is a magnetic field having the same number of poles as the field winding.

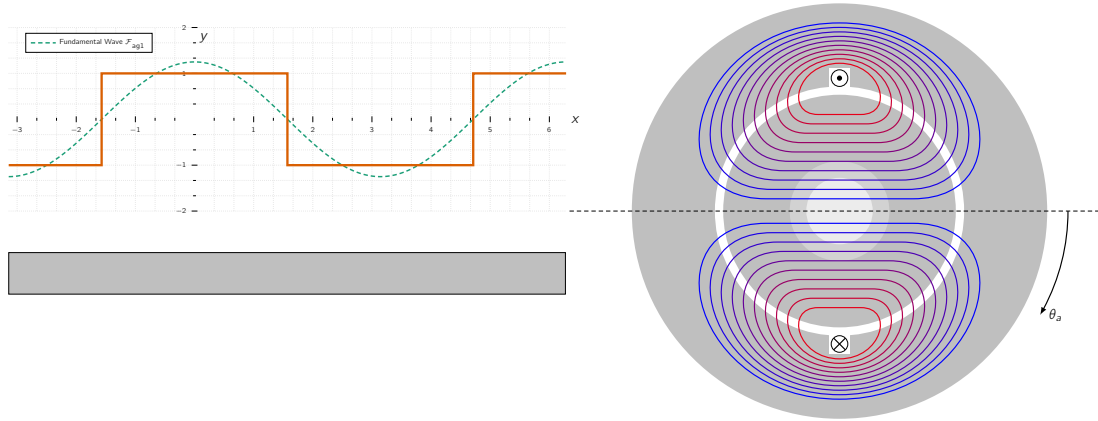


Figure 4.14: (a) Flux produced by a concentrated, full-pitch winding in a machine with a uniform air gap. (b) The air-gap MMF produced by current in this winding.

The study of the magnetic fields of distributed windings can be approached by examining the magnetic field produced by a winding consisting of a single N-turn coil which spans 180 electrical degrees, as shown in **Fig. 4.14a**.

A coil which spans 180 electrical degrees is known as a **full-pitch coil**.

The dot and cross indicate current flow toward and away from the reader, respectively.

For simplicity, a concentric cylindrical rotor is shown. The dipole nature of the magnetic field produced by the current in the coil is shown by the flux lines in **Fig. 4.14a**. Given the permeability of the armature and field iron is much greater than that of air, it is sufficiently accurate for our present purposes to assume that all the reluctance of the magnetic circuit is in the air gap.

From symmetry of the structure it is evident that the magnetic field intensity (H_{ag}) in the air gap at angle θ_a under one pole is the same in magnitude as that at angle $\theta_a + \pi$ under the opposite pole, but the fields are in the opposite direction.

Around any of the closed paths shown by the flux lines in **Fig. 4.14a** the MMF is Ni . Because we have assumed that the reluctance of the magnetic circuit is predominantly that of the air gap, the MMF drop in the iron can be neglected and all of the MMF drop will appear across the air gap. By symmetry we argued that the air-gap fields H_{ag} on opposite sides of the rotor are equal in magnitude but opposite in direction. It follows that the air-gap MMF should be similarly distributed given each flux line crosses the air gap twice, the MMF drop across the air gap must be equal to half of the total or $Ni/2$.

Fig. 4.14b shows the air gap and winding in developed form.¹² The air-gap MMF distribution is shown by the step-like distribution of amplitude $Ni/2$. On the assumption of narrow slot openings, the MMF jumps abruptly by Ni in crossing from one side to the other of a coil.

¹²i.e., laid out flat

4.2.1 AC Machines

Fourier analysis¹³ can be used to show that the air-gap MMF produced by a single coil such as the full-pitch coil of **Fig. 4.14** consists of a fundamental space-harmonic component as well as a series of higher-order harmonic components. In the design of AC machines, serious efforts are made to distribute the coils making up the windings so as to minimise the higher-order harmonic components and to produce an air-gap MMF wave which consists predominantly of the space-fundamental sinusoidal component. It is therefore appropriate here to assume that this has been done and to focus our attention on the fundamental component.

The rectangular air-gap MMF wave of the concentrated two-pole, full-pitch coil of **Fig. 4.14b** can be resolved into a Fourier series comprising a fundamental component and a series of **odd** harmonics. The fundamental component $\mathcal{F}_{ag1}$ is:

$$\mathcal{F}_{ag1} = \frac{4}{\pi} \left(\frac{Ni}{2} \right) \cos \theta_a \quad (4.4)$$

where θ_a is measured from the **magnetic axis of the stator coil**, as shown by the dashed sinusoid in **Fig. 4.14b**. It is a sinusoidal space wave with an amplitude of:

$$(\mathcal{F}_{ag1})_{\text{peak}} = \frac{4}{\pi} \left(\frac{Ni}{2} \right) \quad (4.5)$$

with its peak **aligned with the magnetic axis of the coil**.

Let's change our focus and consider a **distributed winding**, consisting of coils distributed in several slots. For example, Fig4.17a shows phase a of the armature winding of a somewhat simplified two-pole, three-phase AC machine.

The windings of the three phases are identical and are located with their magnetic axes 120° apart. We direct our attention to the air-gap MMF of phase *a* alone for now. The winding is arranged in two layers, each full-pitch coil of N_c turns having one side in the top of a slot and the other coil side in the bottom of a slot a pole pitch away.

This two-layer arrangement simplifies the geometric problem of getting the end turns of the individual coils past each other when the winding is inserted in the stator.

Figure 4.17b shows one pole of this winding laid out flat. With the coils connected in series and therefore carrying the same current, the MMF wave is a series of steps each of height $2N_c i_a$,¹⁴ where i_a is the winding current.

¹⁴equal to the ampere-turns in the slot.

Its space-fundamental component is shown by the sinusoid. It can be seen that the distributed



¹³ Jean-Baptiste Joseph Fourier
(1768 - 1830)

French mathematician and physicist born in *Auxerre, Burgundy* and best known for initiating the investigation of Fourier series, which eventually developed into Fourier analysis and harmonic analysis, and their applications to problems of heat transfer and vibrations. The Fourier transform and Fourier's law of conduction are also named in his honour.

winding produces an MMF wave which is a closer approximation to a sinusoidal MMF wave than that of the concentrated coil of **Fig. 4.14**.

The modified form of Eq. (4.4) for a distributed multi-pole winding having N_{ph} series turns per phase is as follows:

$$\mathcal{F}_{ag1} = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{\text{poles}} \right) i_a \cos \left(\frac{\text{poles}}{2} \theta_a \right) = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{\text{poles}} \right) i_a \cos \theta_{ae} \quad (4.6)$$

in which the factor $4/\pi$ arises from the Fourier-series analysis of the rectangular MMF wave of a concentrated **full-pitch coil**, as in Eq. (4.4), and the **winding factor** (k_w) takes into account the distribution of the winding.

This factor is required because the MMFs produced by the individual coils of any one phase group have different magnetic axes and hence don't sum directly. When they are connected in series to form the phase winding, their phasor sum is then less than their numerical sum.

For most three-phase windings, k_w typically falls in the range of 0.85 to 0.95.

The factor $k_w N_{ph}$ is called the **effective series turns per phase** for the fundamental MMF. The peak amplitude of the MMF wave of Eq. (4.6) is

$$(\mathcal{F}_{ag1})_{\text{peak}} = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{\text{poles}} \right) i_a \quad (4.7)$$

Exercise 4.1 Studying MMF Waves

A single armature winding phase of a two-pole machine can be considered to consist of 8 N_c -turn, full-pitch coils connected in series, with each slot containing two coils. There are a total of 24 armature slots, and thus each slot is separated by $360^\circ / 24 = 15^\circ$.

Define the angle θ_a as being measured from the magnetic axis of phase a such that the four (4) slots containing the coil sides labeled a are at $\theta_a = 67.5^\circ, 82.5^\circ, 97.5^\circ$, and, 112.5° .

The opposite sides of each coil are thus found in the slots found at $-112.5^\circ, -97.5^\circ, -82.5^\circ$ and -67.5° , respectively. Assume this winding to be carrying current i_a .

- Write an expression for the space-fundamental MMF produced by the two coils whose sides are in the slots at $\theta_a = 112.5^\circ$ and -67.5° .
- Write an expression for the space-fundamental MMF produced by the two coils whose sides are

in the slots at $\theta_a = 67.5^\circ$ and -112.5° .

- Write an expression for the space-fundamental MMF of the complete armature winding.
- Determine the winding factor k_w for this distributed winding.

SOLUTION

- Noting that the magnetic axis of this pair of coils is at

$$\theta_a = (112.5^\circ - 67.5^\circ) / 2 = 22.5^\circ$$

and that the total ampere-turns in each slot is equal to $2N_c i_a$, the MMF produced by this pair of coils can be found from analogy with Eq. (4.4) to be

$$(\mathcal{F}_{ag1})_{22.5^\circ} = \frac{4}{\pi} \left(\frac{2N_c i_a}{2} \right) \cos(\theta_a - 22.5^\circ)$$

- This pair of coils produces the same space-fundamental mmf as the pair of part (a) with

the exception that this mmf is centered at $\theta_a = -22.5^\circ$. Thus

$$(\mathcal{F}_{ag1})_{-22.5^\circ} = \frac{4}{\pi} \left(\frac{2N_c i_a}{2} \right) \cos(\theta_a + 22.5^\circ)$$

■ By analogy with parts (a) and (b), the total space-fundamental mmf can be written as

$$\begin{aligned} (\mathcal{F}_{ag1})_{total} &= (\mathcal{F}_{ag1})_{-22.5^\circ} + (\mathcal{F}_{ag1})_{-7.5^\circ} \\ &\quad + (\mathcal{F}_{ag1})_{7.5^\circ} + (\mathcal{F}_{ag1})_{22.5^\circ} \\ &= \frac{4}{\pi} \left(\frac{2N_c}{2} \right) i_a \\ &\quad \times \left[\cos(\theta_a + 22.5^\circ) \right. \end{aligned}$$

$$\begin{aligned} &\quad + \cos(\theta_a + 7.5^\circ) + \cos(\theta_a - 7.5^\circ) \\ &\quad \left. + \cos(\theta_a - 22.5^\circ) \right] \\ &= \frac{4}{\pi} \left(\frac{7.66N_c}{2} \right) i_a \cos \theta_a \\ &= 4.88N_c i_a \cos \theta_a \end{aligned}$$

■ Recognizing that, for this winding $N_{ph} = 8N_c$, the total MMF of part (c) can be rewritten as

$$(\mathcal{F}_{ag1})_{total} = \frac{4}{\pi} \left(\frac{0.958N_{ph}}{2} \right) i_a \cos \theta_a$$

Comparison with Eq. (4.6) shows that for this winding, the winding factor is $k_w = 0.958$.

Eq. (4.6) describes the space-fundamental component of the MMF wave produced by current in phase a of a distributed winding. If the phase- a current is sinusoidal in time, e.g., $i_a = I_{max} \cos \omega t$, the result will be an MMF wave which is stationary in space and varies sinusoidally both with respect to θ_a and in time.

In a directly analogous fashion, rotor windings are often distributed in slots to reduce the effects of space harmonics. Figure 4.18a shows the rotor of a typical two pole round-rotor generator. Although the winding is symmetric with respect to the rotor axis, the number of turns per slot can be varied to control the various harmonics.

In Fig. 4.18b it can be seen that there are fewer turns in the slots nearest the pole face.¹⁵ In addition, ¹⁵pole axis the designer can vary the spacing of the slots. As for distributed armature windings, the fundamental air-gap MMF wave of a multi-pole rotor winding can be expressed in the form of Eq. (4.6) in terms of the total number of series turns N_r , the winding current I_r , and a winding factor k_r as

$$\mathcal{F}_{ag1} = \frac{4}{\pi} \left(\frac{k_r N_r}{\text{poles}} \right) I_r \cos \left(\frac{\text{poles}}{2} \theta_r \right) \quad (4.8)$$

where θ_r is the spatial angle measured with respect to the rotor magnetic axis, as shown in Fig. 4.18b. Its peak amplitude is:

$$(\mathcal{F}_{ag1})_{peak} = \frac{4}{\pi} \left(\frac{k_r N_r}{\text{poles}} \right) I_r \quad (4.9)$$

4.3 Magnetic Field in Rotating Machinery

We base our preliminary investigations of both AC and DC machines on the basic assumption of sinusoidal spatial distributions of MMF. This assumption will be found to give very satisfactory results for most problems involving AC machines because their windings are commonly distributed so as to minimise the effects of space harmonics.

Because of the restrictions placed on the winding arrangement by the commutator, the MMF waves of DC machines inherently approach more nearly a **saw-tooth waveform**. Nevertheless, an analysis based on a sinusoidal model brings out the essential features of DC machine performance. The results can readily be modified whenever necessary to account for any significant discrepancies.

It is often easiest to begin by examination of a two-pole machine, in which the electrical and mechanical angles and velocities are equal. The results can immediately be extrapolated to a multi-pole machine when it is recalled that electrical angles and angular velocities are related to mechanical angles and angular velocities by a factor of poles/2.¹⁶

¹⁶see, for example, Eq. (4.1)

The behaviour of electric machinery is determined by the magnetic fields created by currents in the various windings of the machine which we will now look at how these magnetic fields and currents are related.

4.3.1 Machines with Uniform Air-Gaps

Figure 4.22a shows a single full-pitch, N-turn coil in a high-permeability magnetic structure ($\mu \rightarrow \infty$), with a concentric, cylindrical rotor. The air-gap MMF $\mathcal{F}_{ag}$ of this configuration is shown plotted versus angle θ_a in Fig. 4.22b. For such a structure, with a uniform air gap of length g at radius r_r ,¹⁷ it is quite accurate to assume that the magnetic field $\mathbf{H}$ in the air gap is predominantly radially directed and that it is constant in magnitude across the air gap.

¹⁷very much larger than g .

The air-gap MMF distribution of Fig. 4.22b is equal to the line integral of H_{ag} across the air gap. For this case of constant radial H_{ag} , this integral is simply equal to the product of the air-gap radial magnetic field H_{ag} times the air-gap length g , and thus H_{ag} can be found simply by dividing the air-gap MMF by the air-gap length:

$$H_{ag} = \frac{\mathcal{F}_{ag}}{g} \quad (4.10)$$

Thus, in Fig. 4.22c, the radial field H_{ag} and the MMF can be seen to be identical in form, simply related by a factor of $1/g$. The fundamental space-harmonic component of H_{ag} can be found directly from the fundamental component $\mathcal{F}_{ag1}$, given by Eq. (4.4).

$$H_{ag1} = \frac{\mathcal{F}_{ag1}}{g} = \frac{4}{\pi} \left(\frac{Ni}{2g} \right) \cos \theta_a \quad (4.11)$$

It is a sinusoidal space wave of amplitude

$$(H_{ag1})_{\text{peak}} = \frac{4}{\pi} \left(\frac{Ni}{2g} \right) \quad (4.12)$$

For a distributed winding such as that of Fig. 4.17 with winding factor k_w , the air-gap magnetic field intensity is easily found once the air-gap MMF is known. Therefore, generalising for the case of a multi-pole machine with N_{ph} series turns per pole, the fundamental component of H_{ag} can be found by dividing the fundamental component of the air-gap MMF (Eq. (4.6)) by the air-gap length g .

$$H_{ag1} = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{g \times \text{poles}} \right) i_a \cos \theta_{ae} \quad (4.13)$$

Note that the space-fundamental air-gap MMF $\mathcal{F}_{ag1}$ and air-gap magnetic field H_{ag1} produced by a distributed winding of winding factor k_w and N_{ph}/poles series turns per pole is equal to that produced by a concentrated, full-pitch winding of $(k_w N_{ph})/\text{poles}$ turns per pole. In the analysis of machines with distributed windings, this result is useful since in considering space-fundamental quantities it permits the distributed solution to be obtained from the single N-turn-per-pole, full-pitch coil solution simply by replacing N by the effective number of turns per pole, $k_w N_{ph}/\text{poles}$, of the distributed winding.

Exercise 4.2 Uniform Air Gap

A four-pole synchronous AC generator with a smooth air gap has a distributed rotor winding with 264 series turns, a winding factor of 0.935, and an air gap of length 0.7 mm. Assuming the MMF drop in the electrical steel to be negligible, find the rotor-winding current required to produce a peak, space-fundamental magnetic flux density of 1.6 T in the machine air gap.

SOLUTION

The space-fundamental air-gap magnetic field can be found from the space-fundamental component of the air-gap MMF by dividing by the air-gap length (g). Multiplying by the permeability of free space (μ_0) then gives

the space-fundamental air-gap magnetic flux density. Therefore, from Eq. (4.9):

$$(B_{ag1})_{\text{peak}} = \frac{\mu_0 (\mathcal{F}_{ag1})_{\text{peak}}}{g} = \frac{4\mu_0}{\pi g} \left(\frac{k_r N_r}{\text{poles}} \right) I_r$$

Solving for I_r gives:

$$\begin{aligned} I_r &= \left(\frac{\pi g \times \text{poles}}{4\mu_0 k_r N_r} \right) (B_{ag1})_{\text{peak}} \\ &= \left(\frac{\pi \times 7 \times 10^{-4} \times 4}{4 \times 4\pi \times 10^{-7} \times 0.935 \times 264} \right) 1.6 \\ &= 11.4 \text{ A} \quad \blacksquare \end{aligned}$$

4.4 Describing Rotating MMF Waves in AC Machines

To get an understanding of the theory and operation of poly-phase AC machines, it is important to analyse the nature of the MMF wave produced by a **poly-phase** winding. To aid in this study, attention will be focused on a two-pole machine.

We will begin with an analysis of a single-phase winding.

4.4.1 MMF Wave of a Single-Phase Winding

Fig. 4.15a shows the space-fundamental MMF distribution of a single-phase winding. From Eq. (4.6),

$$\mathcal{F}_{ag1} = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{\text{poles}} \right) i_a \cos(\theta_{ae}) \quad (4.14)$$

where θ_{ae} is given by Eq. (4.1). When this winding is excited by a current varying sinusoidally in time at electrical frequency ω_e

$$i_a = I_a \cos \omega_e t \quad (4.15)$$

the MMF distribution is given by:

$$\mathcal{F}_{ag1} = F_{\max} \cos(\theta_{ae}) \cos \omega_e t \quad (4.16)$$

Eq. (4.16) has been written in a form to emphasise the fact that the result is an MMF distribution of maximum amplitude.

$$F_{\max} = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{\text{poles}} \right) I_a \quad (4.17)$$

This MMF distribution remains fixed in space with an amplitude that varies sinusoidally in time at frequency ω_e , as shown in **Fig. 4.15a**. Note that, to simplify the notation, Eq. (4.1) has been used to express the MMF distribution of Eq. (4.16) in terms of the electrical angle θ_{ae} .

Information: Useful Trigonometric Identity

$$\cos \alpha \cos \beta = \frac{1}{2} \cos(\alpha - \beta) + \frac{1}{2} \cos(\alpha + \beta)$$

Use of a common trigonometric identity permits Eq. (4.16) to be rewritten in the form

$$\mathcal{F}_{ag1} = F_{\max} \left[\frac{1}{2} \cos(\theta_{ae} - \omega_e t) + \frac{1}{2} \cos(\theta_{ae} + \omega_e t) \right] \quad (4.18)$$

which shows that the MMF of a single-phase winding can be resolved into two (2) rotating MMF waves, each of amplitude equal to one-half the maximum amplitude of $\mathcal{F}_{ag}$ with one, $\mathcal{F}_{ag1}^+$, rotating in the $+\theta_{ae}$ direction and the other, $\mathcal{F}_{ag1}^-$, rotating in the $-\theta_{ae}$ direction

$$\mathcal{F}_{ag1}^+ = \frac{1}{2} F_{\max} \cos(\theta_{ae} - \omega_e t) \quad (4.19)$$

$$\mathcal{F}_{ag1}^- = \frac{1}{2} F_{\max} \cos(\theta_{ae} + \omega_e t) \quad (4.20)$$

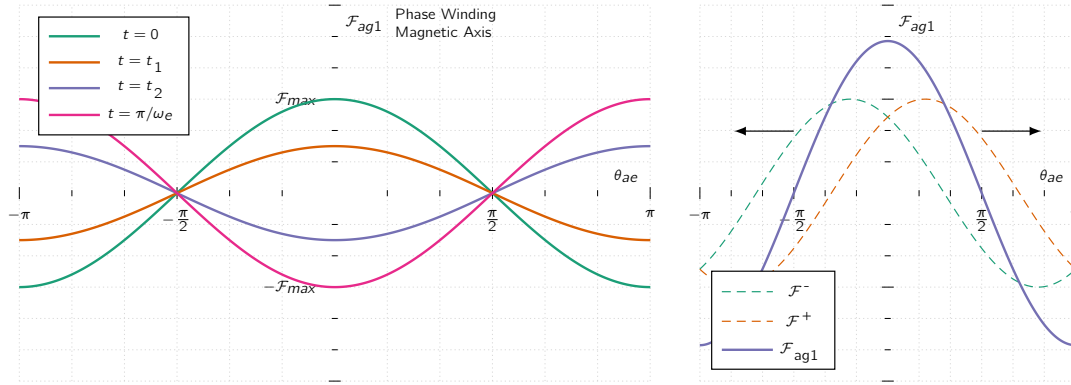


Figure 4.15: Single-phase-winding space-fundamental air-gap mmf: (a) MMF distribution of a single-phase winding at various times; (b) total MMF F_{ag1} decomposed into two travelling waves F^- and F^+ .

Both flux waves rotate in their respective directions with electrical angular velocity ω_e , corresponding to a mechanical angular velocity ω_m where

$$\omega_m = \left(\frac{2}{\text{poles}} \right) \omega_e = \left(\frac{\pi}{30} \right) n \quad (4.21)$$

where n is the rotational speed in rpm. The fact that the air-gap MMF of a single-phase winding excited by a source of alternating current can be resolved into rotating travelling waves is an important conceptual step in understanding AC machinery. In single-phase AC machinery, the positive-travelling flux wave produces useful torque while the negative-travelling flux wave produces both negative and pulsating torque as well as losses.

Although single-phase machines are designed so as to minimise the effects of the negative-travelling flux wave, they cannot be totally eliminated. On the other hand, in poly-phase AC machinery, the windings are equally displaced in space phase, and the winding currents are similarly displaced in time phase, with the result that the negative-travelling flux waves of the various windings sum to zero while the positive-travelling flux waves reinforce, giving a single positive-travelling flux wave.

4.4.2 MMF Wave of a Polyphase Winding

It is time to study the MMF distributions of three-phase windings such as those found on the stator of three-phase IM and SynMs. The analyses we will work here can be readily extended to a poly-phase winding with any number of phases.

Once again attention is focused on a two-pole machine or one pair of poles of a multi-pole winding.

In a three-phase machine, the windings of the individual phases are displaced from each other by **120 electrical degrees** in space around the air-gap circumference, as shown by coils (A, -A), (B, -B), and (C, -C) in **Fig. 4.16**.

The concentrated full-pitch coils shown here may be considered to represent distributed windings producing sinusoidal MMF waves centered on the magnetic axes of the respective phases. The space-fundamental sinusoidal MMF waves of the three phases are accordingly displaced 120 electrical degrees in space. Each phase is excited by AC which varies in magnitude sinusoidally with time.

Under balanced three-phase conditions, the instantaneous currents are as follows:

$$i_a = I_{\max} \cos \omega_c t \quad (4.22)$$

$$i_b = I_{\max} \cos (\omega_c t - 120^\circ) \quad (4.23)$$

$$i_c = I_{\max} \cos (\omega_c t + 120^\circ) \quad (4.24)$$

where $I_{\max}$ is the maximum value of the current and the time origin is arbitrarily taken as the instant when the phase-a current is at its positive maximum. The phase sequence is assumed to be abc. The instantaneous currents are shown in Fig. 4.27. The dots and crosses in the coil sides, shown in **Fig. 4.16**, indicate the reference directions for positive phase currents.

The MMF of phase a has been shown to be:

$$\mathcal{F}_{a1} = \mathcal{F}_{a1}^+ + \mathcal{F}_{a1}^- \quad (4.25)$$

where:

$$\mathcal{F}_{a1}^+ = \frac{1}{2} F_{\max} \cos (\theta_{ae} - \omega_e t) \quad \text{and} \quad \mathcal{F}_{a1}^- = \frac{1}{2} F_{\max} \cos (\theta_{ae} + \omega_e t) \quad (4.26)$$

and

$$F_{\max} = \frac{4}{\pi} \left(\frac{k_w N_{ph}}{\text{poles}} \right) I_{\max} \quad (4.27)$$

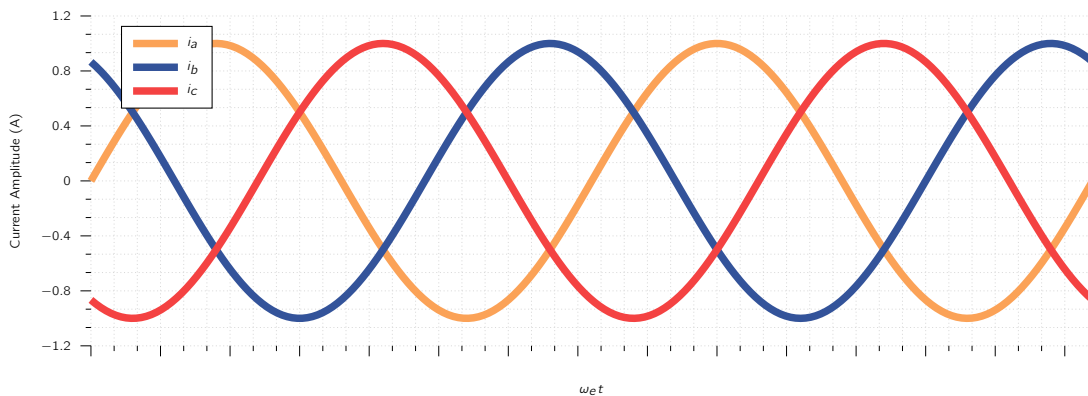


Figure 4.17: Instantaneous phase currents under balanced three-phase conditions.

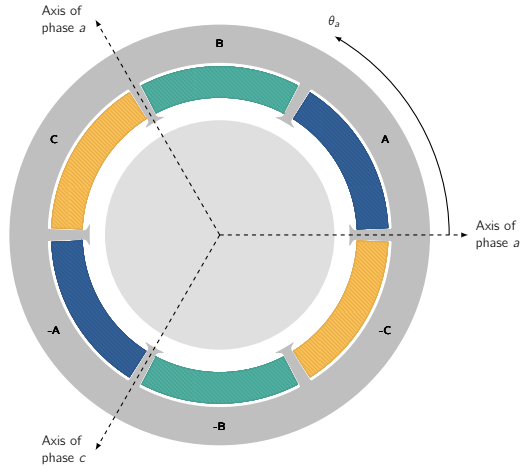


Figure 4.16: Simplified two-pole three-phase stator winding.

To avoid excessive notational complexity, the subscript **ag** has been dropped.

Here the subscript **a1** indicates the space-fundamental component of the phase-a air-gap MMF.

Similarly, for phases *b* and *c*, whose axes are at $\theta_{ae} = 120^\circ$ and $\theta_{ae} = -120^\circ$ respectively, their air-gap MMFs can be shown to be

$$\mathcal{F}_{b1} = \mathcal{F}_{b1}^+ + \mathcal{F}_{b1}^- \quad (4.28)$$

$$\mathcal{F}_{b1}^+ = \frac{1}{2} F_{\max} \cos(\theta_{ae} - \omega_e t) \quad (4.29)$$

$$\mathcal{F}_{b1}^- = \frac{1}{2} F_{\max} \cos(\theta_{ae} + \omega_e t + 120^\circ) \quad (4.30)$$

and

$$\mathcal{F}_{c1} = \mathcal{F}_{c1}^+ + \mathcal{F}_{c1}^- \quad (4.31)$$

$$\mathcal{F}_{c1}^+ = \frac{1}{2} F_{\max} \cos(\theta_{ae} - \omega_e t) \quad (4.32)$$

$$\mathcal{F}_{c1}^- = \frac{1}{2} F_{\max} \cos(\theta_{ae} + \omega_e t - 120^\circ) \quad (4.33)$$

The total MMF is the **sum** of the contributions from each of the three phases as:

$$\mathcal{F}(\theta_{ae}, t) = \mathcal{F}_{a1} + \mathcal{F}_{b1} + \mathcal{F}_{c1} \quad (4.34)$$

This summation can be performed quite easily in terms of the positive and negative travelling waves. The negative-travelling waves sum to zero:

$$\begin{aligned} \mathcal{F}^-(\theta_{ae}, t) &= \mathcal{F}_{a1}^- + \mathcal{F}_{b1}^- + \mathcal{F}_{c1}^- \\ &= \frac{1}{2} F_{\max} \left[\cos(\theta_{ae} + \omega_e t) + \cos(\theta_{ae} + \omega_e t - 120^\circ) + \cos(\theta_{ae} + \omega_e t + 120^\circ) \right] \\ &= 0 \end{aligned} \quad (4.35)$$

while the positive-travelling waves reinforce

$$\mathcal{F}^+(\theta_{ae}, t) = \mathcal{F}_{a1}^+ + \mathcal{F}_{b1}^+ + \mathcal{F}_{c1}^+ = \frac{3}{2} F_{\max} \cos(\theta_{ae} - \omega_e t) \quad (4.36)$$

Therefore, the result of displacing the three windings by 120° in space phase and displacing the winding currents by 120° in time phase is a single positive-travelling MMF wave

$$\mathcal{F}(\theta_{ae}, t) = \frac{3}{2} F_{\max} \cos(\theta_{ae} - \omega_e t) \quad (4.37)$$

The air-gap MMF wave described by Eq. (4.37) is a space-fundamental sinusoidal function of the electrical space angle θ_{ae} .¹⁸ It has a constant amplitude of $(3/2) F_{\max}$, which is 1.5 times the peak amplitude of the air-gap MMF wave produced by the individual phases alone. It has a positive peak at angle $\theta_a = (2\omega_e t / \text{poles})$.

¹⁸Which means the space angle $\theta_a = 2\theta_{ae} / \text{poles}$.

Therefore, under balanced three-phase conditions, the three-phase winding produces an air-gap MMF wave which rotates at synchronous angular velocity ω_s :

$$\omega_s = \left(\frac{2}{\text{poles}} \right) \omega_e \quad (4.38)$$

where ω_e is the angular frequency of the applied electrical excitation ($\text{rad} \cdot \text{s}^{-1}$), and ω_s is synchronous spatial angular velocity of the air-gap MMF wave ($\text{rad} \cdot \text{s}^{-1}$). The corresponding synchronous speed n_s in r/min can be expressed in terms of the applied electrical frequency

$$f_e = \frac{\omega_e}{2\pi} \quad \text{and} \quad n_s = \left(\frac{120}{\text{poles}} \right) f_e \quad (4.39)$$

In general, a rotating field of constant amplitude will be produced by a q -phase winding ($q \geq 3$) excited by balanced q -phase currents of frequency f_e when the respective phase axes are located $2\pi/q$ electrical radians apart in space. The amplitude of this flux wave will be $q/2$ times the maximum contribution of any one phase, and the synchronous angular velocity will remain $\omega_s = (2 \omega_e / \text{poles})$ radians per second.

For a two phase machine, the phase axes are located $\pi/2$ electrical radians apart in space and the amplitude of the rotating flux wave will be equal to that of the individual phases.

In this section, we have seen that a poly-phase winding excited by balanced poly-phase currents produces a rotating MMF wave. Production of a rotating MMF wave and the corresponding rotating magnetic flux wave is key to the operation of poly-phase rotating electrical machinery. It is the interaction of this magnetic flux wave with that of the rotor which produces torque. Constant torque is produced when rotor-produced magnetic flux wave rotates in synchronism with that of the stator.

4.4.3 Graphical Analysis of Polyphase MMF

For balanced three-phase currents as given by Eq. (4.22) to Eq. (4.24), the production of a rotating MMF can also be shown graphically.

Consider the event at $t = 0$, as seen in **Fig. 4.16**, the moment when the phase-a current is at its maximum value I_{max} . The MMF of phase a then has its maximum value F_{max} , as shown by the

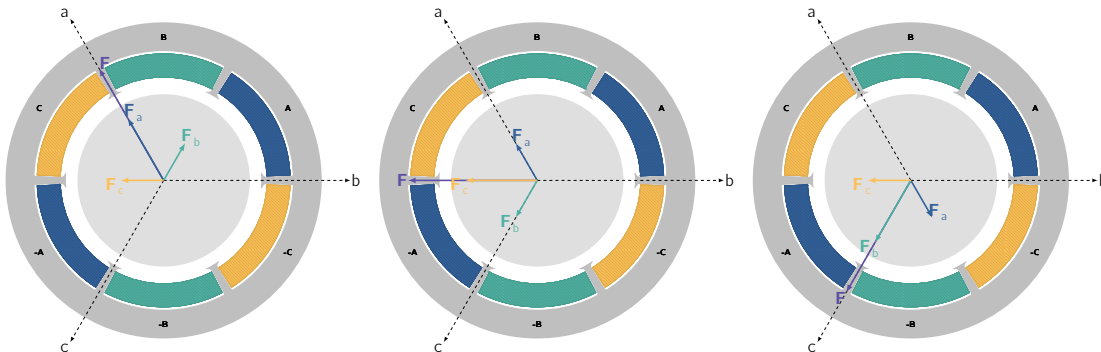


Figure 4.18: The production of a rotating magnetic field by means of three-phase currents.

vector F_a drawn along the magnetic axis of phase a in the two-pole machine shown schematically in Fig. 4.28a. At this moment, currents i_b and i_c are both $I_{\max}/2$ in the negative direction, as shown by the dots and crosses in Fig. 4.28a indicating the actual instantaneous directions. The corresponding MMFs of phases b and c are shown by the vectors F_b and F_c , both of magnitude $F_{\max}/2$ drawn in the negative direction along the magnetic axes of phases b and c, respectively. The resultant, obtained by adding the individual contributions of the three phases, is a vector of magnitude $F = 3/2 F_{\max}$ centered on the axis of phase a. It represents a sinusoidal space wave with its positive peak centered on the axis of phase a and having an amplitude 3/2 times that of the phase-a contribution alone.

At a later time $\omega_e t = \pi/3$ **Fig. 4.16**, the currents in phases a and b are a positive half maximum, and that in phase c is a negative maximum. The individual MMF components and their resultant are now shown in Fig. 4.28b. The resultant has the same amplitude as at $t = 0$, but it has now rotated counter-clockwise 60 electrical degrees in space. Similarly, at $\omega_e t = 2\pi/3$ (when the phase-b current is a positive maximum and the phase-a and phase-c currents are a negative half maximum) the same resultant MMF distribution is again obtained, but it has rotated counter-clockwise 60 electrical degrees still farther and is now aligned with the magnetic axis of phase b (see Fig. 4.28c). As time passes, the resultant MMF wave retains its sinusoidal form and amplitude but rotates progressively around the air gap; the net result can be seen to be an MMF wave of constant amplitude rotating at a uniform angular velocity. In one cycle the resultant MMF must be back in the position of Fig. 4.28a. The MMF wave therefore makes one revolution per electrical cycle in a two-pole machine. In a multi-pole machine the MMF wave travels one pole-pair per electrical cycle and hence one revolution in $(\text{poles}/2)$ electrical cycles.

4.5 Generated Voltage

Previously we've looked at the general nature of the induced voltage. Now we will use quantitative expressions for the induced voltage and derive its mathematical expression.

4.5.1 AC Machines

An elementary AC machine is shown in cross section in **Fig. 4.19**. The coils on both the rotor and the stator have been shown as concentrated, multiple-turn, full-pitch coils.

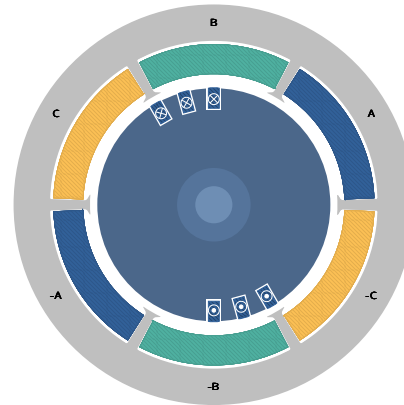


Figure 4.19: Cross-sectional view of an elementary three-phase ac machine.

The performance of a machine with distributed windings can be determined from that of a machine with concentrated windings by multiplying the number of series turns in the winding by a winding factor

Under the assumption of a small air gap, the field winding on the rotor can be assumed to produce an essentially radial-directed space-fundamental sinusoidal air-gap flux of peak flux density (B_{peak}).

If we were to assume the air gap to be **uniform**, B_{peak} can be found from:

$$B_{peak} = \frac{4\mu_0}{\pi g} \left(\frac{k_f N_f}{poles} \right) I_f \quad (4.40)$$

where:

- g = air-gap length
- N_f = total series turns in the field winding
- k_f = field-winding winding factor
- I_f = field current

When the rotor poles are in line with the magnetic axis of a stator phase winding, the flux linkage with that winding (of N_{ph} series turns per phase and winding factor k_w) is:

$$k_w N_{ph} \Phi_p,$$

where Φ_p is the air-gap flux per pole. For the assumed sinusoidal air-gap flux-density of form:

$$B = B_{peak} \cos \left(\frac{poles}{2} \theta_r \right) \quad (4.41)$$

Φ_p can be found as the integral of the flux density over the pole area

$$\Phi_p = l \int_{-\pi/\text{poles}}^{+\pi/\text{poles}} B_{\text{peak}} \cos\left(\frac{\text{poles}}{2}\theta_r\right) d\theta_r = \left(\frac{2}{\text{poles}}\right) 2lrB_{\text{peak}}, \quad (4.42)$$

where θ_r is angle measured from the rotor magnetic axis, r is radius to air gap, and l axial length of the stator/rotor iron.

Although **Fig. 4.19** shows a two-pole machine, the analysis presented here is for the general case of a multi-pole machine.

As the rotor turns, the flux linkage of each stator phase varies cosinusoidally with the angle between the magnetic axes of that phase and that of the rotor field winding. With the rotor spinning at constant angular velocity (ω_m):

$$\theta_r = \omega_m t$$

and the flux linkage with the phase-a stator winding is thus:

$$\begin{aligned} \lambda_a &= k_w N_{ph} \Phi_p \cos\left(\left(\frac{\text{poles}}{2}\right) \omega_m t\right) \\ &= k_w N_{ph} \Phi_p \cos \omega_e t \end{aligned} \quad (4.43)$$

where time (t) is arbitrarily chosen as zero when the field winding magnetic axis coincides with that of phase a and ω_e is the electrical frequency in rad/sec given by Eq. (4.3).

By Faradays law, the voltage induced in a given phase (i.e., phase-a) is found from Eq. (4.43) as:

$$e_a = \frac{d\lambda_a}{dt} = k_w N_{ph} \frac{d\Phi_p}{dt} \cos(\omega_e t) - \omega_e k_w N_{ph} \Phi_p \sin(\omega_e t) \quad (4.44)$$

The polarity of this induced voltage is such that if the stator coil were short-circuited, the induced voltage would cause a current to flow in the direction that would oppose any change in the flux linkage of the stator coil. Although Eq. (4.44) is derived on the assumption that only the field winding is producing air-gap flux, the equation applies equally well to the general situation where Φ_p is the net air-gap flux per pole produced by currents on both the rotor and the stator.

The first term on the right-hand side of Eq. (4.44) is a transformer voltage and is present only when the amplitude of the air-gap flux wave changes with time. The second term is the speed voltage generated by the relative motion of the air-gap flux wave and the stator coil. In the normal steady-state operation of most rotating machines, the amplitude of the air-gap flux wave is **constant**; under these conditions the first term is zero and the generated voltage is simply the speed voltage. The term electromotive force (abbreviated emf) is often used for the speed voltage. Thus, for constant air-gap flux,

$$e_a = -\omega_e k_w N_{ph} \Phi_p \sin(\omega_e t) \quad (4.45)$$

In the normal steady-state operation of AC machines, we are usually interested in the rms values of voltages and currents rather than their instantaneous values. From Eq. (4.45) the maximum value of the induced voltage is:

$$E_{\max} = \omega_e k_w N_{ph} \Phi_p = 2\pi f_e k_w N_{ph} \Phi_p \quad (4.46)$$

where f_e is the electrical frequency of the generated voltage in Hz. Its rms value is

$$E_{rms} = \frac{2\pi f_e k_w N_{ph} \Phi_p}{\sqrt{2}} = \sqrt{2} \pi f_e k_w N_{ph} \Phi_p \quad (4.47)$$

If you remember from previous lectures, you will notice these equation to be identical in form to the corresponding EMF equations for a transformer.

Rotation introduces the element of time variation and transforms a space distribution of flux density into a time variation of flux linkage and voltage. The voltage induced in a single winding is a single-phase voltage. For the production of a set of balanced, three-phase voltages, it follows that three windings displaced 120 electrical degrees in space must be used.

4.6 Torque in Non-Salient-Pole Machines

The behaviour of **any** electromagnetic device as a component in an electromechanical system can be described in terms of:

- its electrical-terminal equations, and
- its displacement and electromechanical force or torque.

The purpose of this section is to derive the terminal relationships and torque equations for an idealized elementary rotating machine, results which can be readily extended later to more complex machines.

We derive these equations from two viewpoints and show their equality

The first viewpoint is essentially the same as we discussed in **Electromechanical Energy Conversion**. The machine will be regarded as a circuit element whose inductances depend on the angular position (θ) of the rotor. The flux linkages (λ) and magnetic field co-energy will be expressed in terms of the currents and inductances. The torque can then be found from the partial derivative of the co-energy with respect to the rotor position and the terminal voltages from the sum of the resistance drops Ri and the Faraday's law voltages $d\lambda/dt$.

The result will be a set of non-linear differential equations describing the dynamic performance of the machine.

The second viewpoint regards the machine as two groups of windings producing magnetic flux in the air gap, one group on the stator, and the other on the rotor. By making suitable assumptions regarding these fields,¹⁹ simple expressions can be derived for the flux linkages and the coenergy in the air gap in terms of the field quantities.

¹⁹similar to those used to derive analytic expressions for the inductances

The torque and generated voltage can then be found from these expressions.

In this fashion, torque can be expressed explicitly as the tendency of two (2) magnetic fields²⁰ to align, in the same way that permanent magnets tend to align, and generated voltage can be expressed in terms of the relative motion between a field and a winding.²¹

²⁰Which are stator and rotor fields.

²¹These expressions lead to a simple physical picture of the normal steady-state behavior of rotating machines.

4.6.1 Coupled-Circuit Viewpoint

Consider an elementary smooth-air-gap machine shown in **Fig. 4.20a**.

Here with one winding on the stator and one on the rotor and with θ_m being the mechanical angle between the axes of the two (2) windings.

These windings are distributed over a number of slots so that their MMF waves can be approximated by space sinusoids.

We assume, for simplicity, the stator and rotor are **concentric cylinders** and slot openings are neglected. Consequently, our elementary model does not include the effects of salient poles.²² We also assume, the reluctances of the stator and rotor iron are assumed negligible.

²²Which are covered in M.Sc Drive Systems.

Finally, although **Fig. 4.20a** shows a two-pole machine, we will write the derivations that follow for the general case of a multi-pole machine, replacing θ_m by the electrical rotor angle

$$\theta_{me} = \left(\frac{\text{poles}}{2} \right) \theta_m \quad (4.48)$$

Based upon these assumptions, the stator and rotor self-inductances L_{ss} and L_{rr} can be seen to be constant, but the stator-to-rotor mutual inductance **depends on the electrical angle** (θ_{me}) between the magnetic axes of the stator and rotor windings. The mutual inductance is at its positive maximum when $\theta_{me} = 0$ or 2π , is zero (0) when $\theta_{me} = \pm\pi/2$, and is at its negative maximum when $\theta_{me} = \pm\pi$.

On the assumption of sinusoidal MMF waves and a uniform air gap, the space distribution of the air-gap flux wave is sinusoidal, and the mutual inductance will be of the form:

$$\mathcal{L}_{sr}(\theta_{me}) = L_{sr} \cos \theta_{me} \quad (4.49)$$

where the script letter $\mathcal{L}$ denotes an inductance which is a function of the electrical angle θ_{me} . Here the italic capital letter L denotes a **constant value**. Therefore L_{sr} is the magnitude of the mutual inductance; its value when the magnetic axes of the stator and rotor are aligned ($\theta_{me} = 0$). In terms of the inductances, the stator and rotor flux linkages λ_s and λ_r are

$$\lambda_s = L_{ss}i_s + \mathcal{L}_{sr}(\theta_{me})i_r = L_{ss}i_s + L_{sr} \cos(\theta_{me})i_r \quad (4.50)$$

$$\lambda_r = \mathcal{L}_{sr}(\theta_{me})i_s + L_{rr}i_r = L_{sr} \cos(\theta_{me})i_s + L_{rr}i_r \quad (4.51)$$

In matrix notation, we can write the aforementioned equations as:

$$\begin{bmatrix} \lambda_s \\ \lambda_r \end{bmatrix} = \begin{bmatrix} L_{ss} & \mathcal{L}_{sr}(\theta_{me}) \\ \mathcal{L}_{sr}(\theta_{me}) & L_{rr} \end{bmatrix} \begin{bmatrix} i_s \\ i_r \end{bmatrix} \quad (4.52)$$

The terminal voltages (v_s , v_r) are:

$$v_s = R_s i_s + \frac{d\lambda_s}{dt} \quad (4.53)$$

$$v_r = R_r i_r + \frac{d\lambda_r}{dt} \quad (4.54)$$

where R_s and R_r are the resistances of the stator and rotor windings respectively.

When the rotor is revolving, θ_{me} varies with time.

Differentiation of Eq. (4.50) and Eq. (4.51) and substitution of the results in Eq. (4.53) and Eq. (4.54) then give:

$$v_s = R_s i_s + L_{ss} \frac{di_s}{dt} + L_{sr} \cos \theta_{me} \frac{di_r}{dt} - L_{sr} i_r \sin \theta_{me} \frac{d\theta_{me}}{dt} \quad (4.55)$$

$$v_r = R_r i_r + L_{rr} \frac{di_r}{dt} + L_{sr} \cos \theta_{me} \frac{di_s}{dt} - L_{sr} i_s \sin \theta_{me} \frac{d\theta_{me}}{dt} \quad (4.56)$$

where

$$\frac{d\theta_{me}}{dt} = \left(\frac{\text{poles}}{2} \right) \omega_m = \omega_e \quad (4.57)$$

is the instantaneous speed in electrical radians per second. In a two-pole machine,²³ θ_{me} and ω_e are equal to the instantaneous shaft angle θ_m and the shaft speed ω_m respectively.

²³such as that of Fig. 4.20a.

In a multi-pole machine, they are related by Eq. (4.48) and Eq. (4.3).

The second and third terms on the right-hand sides of Eq. (4.55) and Eq. (4.56) are $L(di/dt)$ induced voltages like those induced in stationary coupled circuits such as the windings of transformers. The fourth terms are caused by mechanical motion and are proportional to the instantaneous speed.

These are the speed voltage terms which are associated with the interchange of power between the electric and mechanical systems.

The electromechanical torque can be found from the co-energy. From Eq. (3.63):

$$W'_{fid} = \frac{1}{2} L_{ss} i_s^2 + \frac{1}{2} L_{rr} i_r^2 + L_{sr} i_s i_r \cos \theta_{me} \quad (4.58)$$

$$= \frac{1}{2} L_{ss} i_s^2 + \frac{1}{2} L_{rr} i_r^2 + L_{sr} i_s i_r \cos \left(\left(\frac{\text{poles}}{2} \right) \theta_m \right) \quad (4.59)$$

Please observe here, the co-energy of Eq. (4.58) has been expressed specifically in terms of the shaft angle (θ_m) because the torque expression of Eq. (3.61) requires that the torque be obtained from the derivative of the coenergy with respect to the spatial angle θ_m and not with respect to the electrical angle θ_{me} .

Therefore, from Eq. (3.61):

$$\begin{aligned} T &= \left. \frac{\partial W'_{fid}(i_s, i_r, \theta_m)}{\partial \theta_m} \right|_{i_s, i_r} = - \left(\frac{\text{poles}}{2} \right) L_{sr} i_s i_r \sin \left(\frac{\text{poles}}{2} \theta_m \right) \\ &= - \left(\frac{\text{poles}}{2} \right) L_{sr} i_s i_r \sin \theta_{me} \end{aligned} \quad (4.60)$$

where T is the electromechanical torque acting to accelerate the rotor.²⁴ The negative sign in Eq. (4.60) means that the electromechanical torque acts in the direction to bring the magnetic fields of the stator and rotor into alignment. Eq. (4.55), Eq. (4.56), and Eq. (4.60) are a set of three equations relating the electrical variables v_s , i_s , v_r , i_r and the mechanical variables T and θ_m .

²⁴i.e., a positive torque acts to increase θ_m

Information: A Handbrake of Deriving Further

These equations, together with the constraints imposed on the electrical variables by the networks connected to the terminals²⁵ and the constraints imposed on the rotor (applied torques and inertial, frictional, and spring torques), determine the performance of the device and its characteristics as an energy-conversion device between the external electrical and mechanical systems. These are non-linear differential equations and are difficult to solve except under special circumstances. We are not specifically concerned with their solution here; rather we are using them merely as steps in the development of the theory of rotating machines.

However, these equations will pop-up again in **M.Sc Drive Systems** to get a better understanding of machine performance and simulation of transient behaviour.

²⁵sources or loads and external impedances

Now consider a uniform-air-gap machine with several stator and rotor windings. The same general principles that apply to the elementary model of **Fig. 4.20a** also apply to the multi-winding machine.

Each winding has its own self inductance as well as mutual inductances with other windings. The self inductances and mutual inductances between pairs of windings on the same side of the air gap are constant on the assumption of a uniform gap and negligible magnetic saturation. However, the mutual inductances between pairs of stator and rotor windings vary as the cosine of the angle between their magnetic axes. Torque results from the tendency of the magnetic field of the rotor windings to line up with that of the stator windings.

It can be expressed as the sum of terms like that of Eq. (4.60).

As we have seen from the above example, under balanced conditions, a four-pole SynM will produce constant torque at a rotational angular velocity equal to half of the electrical excitation frequency. This result can be generalized to show that, under balanced operating conditions, a multi-phase, multi-pole SynM will produce constant torque at a rotor speed, at which the rotor rotates in synchronism with the rotating flux wave produced by the stator currents.

Therefore, this is known as the synchronous speed of the machine. From Eq. (4.38) and Eq. (4.39), the synchronous speed is equal to:

$$\omega_s = \left(\frac{2}{\text{poles}} \right) \omega_e \text{ rad} \cdot \text{s}^{-1} \quad \text{or} \quad n_s = \left(\frac{120}{\text{poles}} \right) f_e \text{ rpm.} \quad (4.61)$$

4.6.2 Magnetic Field Viewpoint

In this part, we will explore an alternative formulation in terms of the **interacting magnetic fields**.

As we have seen, currents in the rotor and stator windings each produce their **own** MMF distributions which in turn produce their own magnetic fields in the machine. The rotor and stator MMF waves are shown schematically in **Fig. 4.20a** for a two-pole machine with a **smooth** air gap.

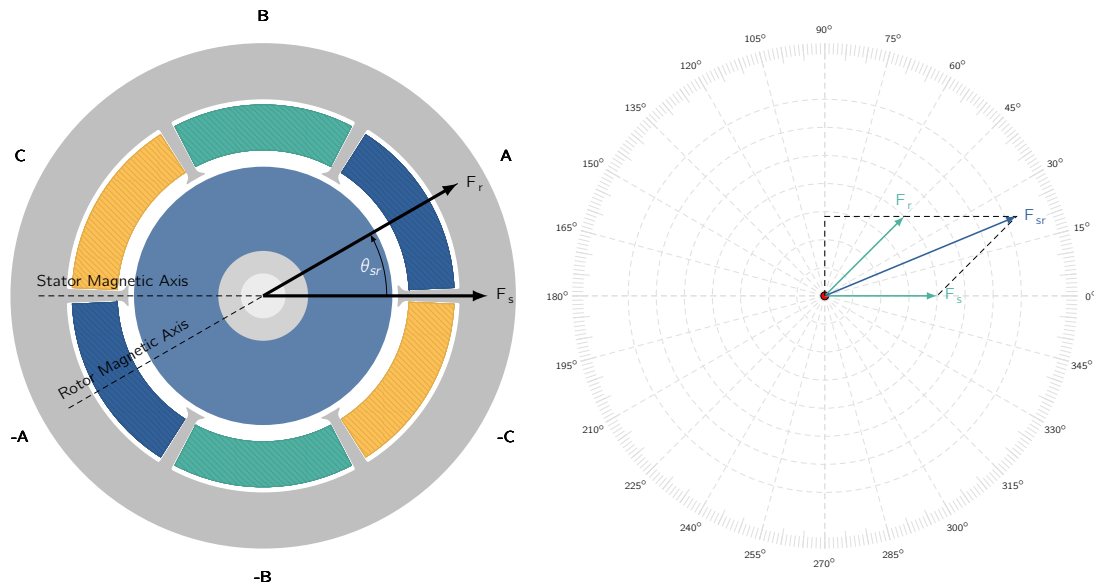


Figure 4.20: Simplified two-pole machine: (a) elementary model and (b) vector diagram of MMF waves. Torque is produced when both the rotor and stator magnetic fields align. Please observe, these figures are drawn with δ_{sr} positive, i.e., with the rotor MMF wave F_r leading that of the stator F_s .

Torque is produced by the tendency of rotor and stator magnetic fields to align their magnetic axes.

In the machine of shown in **Fig. 4.20a**, the torque is proportional to the product of the amplitudes of the stator and rotor MMF waves and is also a function of the angle (δ_{sr}) measured from the axis of the stator MMF wave to that of the rotor.

In fact, we will see, for a smooth-air-gap machine, the torque is proportional to $\sin \delta_{sr}$.

In a typical machine, most of the flux produced by the stator and rotor windings crosses the air gap and links both windings which is termed the **mutual flux**.²⁶ However, some of the flux produced by the rotor and stator windings does **NOT** cross the air gap, which is analogous to the leakage flux in a transformer, are referred to as the **rotor leakage flux** and the **stator leakage flux**.

²⁶This is directly analogous to the mutual or magnetizing flux in a transformer.

Components of this leakage flux include slot and tooth-tip leakage, end-turn leakage, and space harmonics in the air-gap field.

Only the mutual flux is of direct concern in torque production. The leakage fluxes do affect machine performance however, by virtue of the voltages they induce in their respective windings. Their effect on the electrical characteristics is accounted for by means of leakage inductances, analogous to the use of inclusion of leakage inductances in the transformer models.

When expressing torque in terms of the winding currents or their resultant MMFs, the resulting expressions do **NOT** include terms containing the leakage inductances.

Our analysis here, then, will be in terms of the **resultant mutual flux**. We will derive an expression for the magnetic co-energy stored in the air gap in terms of the stator and rotor MMFs and the angle (δ_{sr}) between their magnetic axes. The torque can then be found from the partial derivative of the co-energy with respect to angle δ_{sr} .

For analytical simplicity, we will assume the radial length of the air gap²⁷ (g) is small compared with the radius of the rotor or stator.

²⁷the radial clearance between the rotor and stator

For a smooth-air-gap machine constructed from electrical steel with high magnetic permeability, this will result in air-gap flux which is primarily radially directed and that there is relatively little difference between the flux density at the rotor surface, at the stator surface, or at any intermediate radial distance in the air gap.

The air-gap field then can be represented as a radial field H_{ag} or B_{ag} whose intensity varies with the angle around the periphery. The line integral of H_{ag} across the gap then is simply $H_{ag}g$ and equal to the **resultant air-gap** MMF $\mathcal{F}_{sr}$ produced by the stator and rotor windings, which means we can write:

$$H_{ag}g = \mathcal{F}_{sr} \quad (4.62)$$

where $\mathcal{F}$ denotes the MMF wave as a function of the angle around the periphery.

The MMF waves of both stator and rotor are **spatial sine waves** with δ_{sr} being the phase angle between their magnetic axes in electrical degrees. They can be represented by the space vectors $\mathbf{F}_s$ and $\mathbf{F}_r$ drawn along the magnetic axes of the stator- and rotor MMF waves respectively, as in **Fig. 4.20b**.

The resultant MMF acting to produce flux across the air gap is their vector sum and is represented by the space vector $\mathbf{F}_{sr}$. From the trigonometric formula for the diagonal of a parallelogram, its peak value is found from:

$$F_{sr}^2 = F_s^2 + F_r^2 + 2F_sF_r \cos \delta_{sr}, \quad (4.63)$$

in which the F_s are the peak values of the MMF waves. The resultant radial H_{ag} field is a sinusoidal space wave whose peak value $H_{ag,peak}$ is, from Eq. (4.62),

$$(H_{ag})_{peak} = \frac{F_{sr}}{g} \quad (4.64)$$

Now, let us, consider the magnetic field co-energy stored in the air gap. The coenergy density at a point where the magnetic field intensity (H) is calculated as $(\mu_0/2)H^2$. Therefore, the co-energy density averaged over the volume of the air gap is $\mu_0/2$ times the average value of H_{ag}^2 .

The average value of the square of a sine wave is half its peak value, which makes:

$$\text{Average co-energy density} = \frac{\mu_0}{2} \left(\frac{(H_{ag})_{peak}^2}{2} \right) = \frac{\mu_0}{4} \left(\frac{F_{sr}}{g} \right)^2 \quad (4.65)$$

Based upon the small-gap approximation,²⁸ the volume of the air gap is given by:

$$\text{Air-gap Volume} = \pi D l g$$

where l is the axial length of the air gap and D is its average diameter. The total co-energy can be found by multiplying the average coenergy density by the air-gap volume

$$W'_{\text{fld}} = \frac{\mu_0}{4} \left(\frac{F_{\text{sr}}}{g} \right)^2 \pi D l g = \frac{\mu_0 \pi D l}{4g} F_{\text{sr}}^2 \quad (4.66)$$

From Eq. (4.63) the coenergy stored in the air gap can now be expressed in terms of the peak amplitudes of the stator- and rotor-MMF waves and the space-phase angle between them. This allows us to write the following:

$$W'_{\text{fld}} = \frac{\mu_0 \pi D l}{4g} (F_s^2 + F_r^2 + 2F_s F_r \cos \delta_{\text{sr}}) \quad (4.67)$$

Recognizing that holding MMF constant is equivalent to holding current constant, an expression for the electromechanical torque T can now be obtained in terms of the interacting magnetic fields by taking the partial derivative of the field coenergy with respect to angle. For a two-pole machine

$$T = \left. \frac{\partial W'_{\text{fld}}}{\partial \delta_{\text{sr}}} \right|_{F_s, F_r} = - \left(\frac{\mu_0 \pi D l}{2g} \right) F_s F_r \sin \delta_{\text{sr}} \quad (4.68)$$

The general expression for the torque for a multi-pole machine is

$$T = - \left(\frac{\text{poles}}{2} \right) \left(\frac{\mu_0 \pi D l}{2g} \right) F_s F_r \sin \delta_{\text{sr}} \quad (4.69)$$

In this equation, δ_{sr} is the electrical space-phase angle between the rotor and stator MMF waves and the torque T acts in the direction to accelerate the rotor.

Therefore when δ_{sr} is negative, the torque is positive²⁹ and the machine is operating as a motor. Similarly, a positive δ_{sr} corresponds to negative torque tending to de-accelerate the rotor as is the case when the machine is operating as a generator.

This important equation states the torque is **proportional** to the peak values of the stator- and rotor-MMF waves (F_s , F_r) and to the sine of the electrical space-phase angle δ_{sr} between them. Equal and opposite torques are exerted on the stator and rotor.

The minus sign means that the fields tend to align themselves.

One can now compare the results of Eq. (4.69) with that of Eq. (4.60). Recognizing F_s is proportional to i_s and F_r is proportional to i_r , one sees that they are similar in form. **In fact, they must be equal.**

These results have been derived with the assumption that the iron reluctance is negligible. However, the two techniques are equally valid for finite iron permeability.

²⁸A method of simplification used if the two diameters of a co-axial cylinder have negligible difference.

²⁹acting in a direction to accelerate the rotor

On referring to **Fig. 4.20b**, we can see $F_r \sin \delta_{sr}$ is the component of the F_r wave in electrical space quadrature with the F_s wave. Similarly $F_s \sin \delta_{sr}$ is the component of the F_s wave in quadrature with the F_r wave. Thus, the torque is proportional to the product of one magnetic field and the component of the other in quadrature with it, much like the cross product of vector analysis. Also note that in **Fig. 4.20b**:

$$F_s \sin \delta_{sr} = F_{sr} \sin \delta_r \quad (4.70)$$

$$F_r \sin \delta_{sr} = F_{sr} \sin \delta_s \quad (4.71)$$

where, as seen in **Fig. 4.20**, δ_r is the angle measured from the axis of the resultant MMF wave to the axis of the rotor MMF wave. Similarly, δ_s is the angle measured from the axis of the stator MMF wave to the axis of the resultant MMF wave.

The torque, acting to accelerate the rotor, can then be expressed in terms of the resultant MMF wave F_{sr} by substitution of either Eq. (4.70) or Eq. (4.71) in Eq. (4.69). Writing these substitutions down, we get:

$$T = - \left(\frac{\text{poles}}{2} \right) \left(\frac{\mu_0 \pi D l}{2g} \right) F_s F_{sr} \sin \delta_s \quad (4.72)$$

$$T = - \left(\frac{\text{poles}}{2} \right) \left(\frac{\mu_0 \pi D l}{2g} \right) F_r F_{sr} \sin \delta_r \quad (4.73)$$

Comparison of Eq. (4.69), Eq. (4.72), and Eq. (4.73) shows the torque can be expressed in terms of the component magnetic fields due to each current acting alone, as in Eq. (4.69), or in terms of the **resultant** field and **either** of the components, as in Eq. (4.72) and Eq. (4.73), provided that we use the corresponding angle between the axes of the fields.³⁰

³⁰ Ability to reason in any of these terms is a convenience in machine analysis.

In Eq. (4.69), Eq. (4.72), and Eq. (4.73), the fields have been expressed in terms of the peak values of their MMF waves. When magnetic saturation is neglected, the fields can, of course, be expressed in terms of the peak values of their flux-density waves or in terms of total flux per pole.

Therefore, the peak value B_{ag} of the field due to a sinusoidally distributed MMF wave in a uniform-air-gap machine is $\mu_0 F_{ag,peak}/g$, where $F_{ag,peak}$ is the peak value of the MMF wave.

For example, the resultant MMF F_{sr} produces a resultant flux-density wave whose peak value is $B_{sr} = \mu_0 F_{sr}/g$.

Therefore, $F_{sr} = g B_{sr} / \mu_0$ and substitution in Eq. (4.73) gives

$$T = - \left(\frac{\text{poles}}{2} \right) \left(\frac{\pi D l}{2} \right) B_{sr} F_r \sin \delta_r \quad (4.74)$$

One of the inherent limitations in the design of electromagnetic apparatus is the saturation flux density of magnetic materials. Due to saturation in the armature teeth the peak value B_{sr} of the resultant flux-density wave in the air gap is limited to about 1.5 to 2.0 T. The maximum permissible value of the winding current, and hence the corresponding MMF wave, is limited by the temperature rise of the winding and other design requirements. Because the resultant flux density and MMF appear explicitly in Eq. (4.74), this equation is in a convenient form for design purposes and can be used to estimate the maximum torque which can be obtained from a machine of a given size.

Exercise 4.3 Estimating Maximum Torque

An $1,800 \times 10^3$ rpm, four-pole, 60 Hz SynM has an air-gap length of 1.2 mm. The average diameter of the air-gap is 27 cm, and its axial length is 32 cm.

The rotor winding has 786 turns and a winding factor (k_w) of 0.976. Assuming that thermal considerations limit the rotor current to 18 A, please estimate the maximum torque and power output one can expect to obtain from this machine.

SOLUTION First, we can determine the maximum rotor mmf from Eq. (4.9)

$$\begin{aligned}(F_r)_{\max} &= \frac{4}{\pi} \left(\frac{k_r N_r}{\text{poles}} \right) (I_r)_{\max} \\ &= \frac{4}{\pi} \left(\frac{0.976 \times 786}{4} \right) 18 = 4395 \text{ A}\end{aligned}$$

Assuming the peak value of the resultant air-gap flux is limited to 1.5 T, we can estimate the maximum torque

from Eq. (4.74) by setting δ_r equal to $-\pi/2$ (recognising negative values of δ_r , with the rotor MMF lagging the resultant MMF, correspond to positive, motoring torque).

$$\begin{aligned}T_{\max} &= \left(\frac{\text{poles}}{2} \right) \left(\frac{\pi D l}{2} \right) B_{sr} (F_r)_{\max} \\ &= \left(\frac{4}{2} \right) \left(\frac{\pi \times 0.27 \times 0.32}{2} \right) 1.5 \times 4395 \\ &= 1,789 \times 10^3 \text{ N} \cdot \text{m}\end{aligned}$$

For a synchronous speed of $1,800 \times 10^3$ rpm:

$$\omega_m = n_s \left(\frac{\pi}{30} \right) = 1800 \left(\frac{\pi}{30} \right) = \frac{60}{\pi} \text{ rad} \cdot \text{s}^{-1}$$

and therefore the corresponding power can be calculated as:

$$P_{\max} = \omega_m T_{\max} = 337 \text{ kW} \quad \blacksquare$$

Alternative forms of the torque equation arise when it is recognized that the resultant flux per pole is

$$\Phi_p = (\text{average value of } B \text{ over a pole})(\text{pole area}) \quad (4.75)$$

and the average value of a sinusoid over one-half wavelength is $2/\pi$ times its peak value, which means

$$\Phi_p = \frac{2}{\pi} B_{\text{peak}} \left(\frac{\pi D l}{\text{poles}} \right) = \left(\frac{2 D l}{\text{poles}} \right) B_{\text{peak}} \quad (4.76)$$

where B_{peak} is the peak value of the corresponding flux-density wave.

For example, using the peak value of the resultant flux B_{sr} and substitution of Eq. (4.76) into Eq. (4.74) gives

$$T = -\frac{\pi}{2} \left(\frac{\text{poles}}{2} \right)^2 \Phi_{sr} F_r \sin \delta_r \quad (4.77)$$

where Φ_{sr} is the resultant flux per pole produced by the combined effect of the stator and rotor MMFs.

To tie this section in a nice bow, we now have several forms in which the torque of a uniform air-gap machine can be expressed in terms of its magnetic fields.

All are merely statements that the torque is proportional to the product of the magnitudes of the interacting fields and to the sine of the electrical space angle between their magnetic axes.

The negative sign indicates that the electromechanical torque acts in a direction to decrease the displacement angle between the fields. In our preliminary discussion of machine types, Eq. (4.77) will be the preferred form.

One further remark can be made concerning the torque equations and the thought process leading to them. There was no restriction in the derivation that the MMF wave or flux-density wave remain stationary in space. They may remain stationary, or they may be traveling waves.

As we have seen, if the magnetic fields of the stator and rotor are constant in amplitude and travel around the air gap at the same speed, a steady torque will be produced by the tendency of the stator and rotor fields to align themselves in accordance with the torque equations.

4.7 Magnetic Saturation

It goes without saying but to reiterate:

The characteristics of electric machines depend heavily upon the use of magnetic materials.

These materials are required to form the magnetic circuit and are used by the machine designer to obtain specific machine characteristics. However, as we can imagine, **magnetic materials are less than ideal**. As the magnetic flux they carry increases, they begin to **saturate**, with the result being a decrease in the materials magnetic permeability, along with its effectiveness in contributing to the overall flux density in the machine.

Both electromechanical torque and generated voltage in all machines depend on the winding flux linkages. For specific MMFs in the windings, the fluxes depend on the reluctances of the iron portions of the magnetic circuits and on those of the air gaps. Saturation may therefore appreciably influence the characteristics of the machines.

Another aspect of saturation which is important is how we approach to model an electrical machine. Specifically, relations for the air-gap MMF are generally based on the assumption of negligible reluctance in the **iron**.

When these relations are applied to practical machines with varying degrees of saturation in the iron, significant errors in the analytical results can be expected.

To improve these analytical relationships, in one approach the actual machine is replaced by an equivalent machine whose iron has negligible reluctance but whose air-gap length is increased by an amount sufficient to absorb the MMF drop in the iron of the actual machine.

This means that to model the effect of saturation, we increase the air-gap in our idealised machine to take the effect into account.

Similar to the above premise, the effects of air-gap non-uniformities such as **slots** and **ventilating ducts** are also incorporated by increasing the effective air-gap length.³¹

Information: Limits of Analytical Technique

In cases where such simple techniques are found to be inadequate or if the motor geometry proves itself to be unwieldy, detailed analyses, such finite-element or other numerical techniques, can be used.

³¹However, from a manufacturing point of view, techniques must be verified and confirmed experimentally before we can call a product ready.

Saturation characteristics of rotating machines are often presented in the form of an open-circuit characteristic, also called a magnetization curve or saturation curve. For a SynM, the open-circuit saturation curve is obtained by operating the machine at constant speed and measuring the open-circuit armature voltage as a function of the field current. A typical open-circuit saturation curve for a synchronous machine takes the form shown in **Fig. 4.21**.

It is worth mentioning here that the nature of this curve is determined both by the geometry of the machine under consideration as well as by the magnetization characteristics of the electrical steel used in the machine. The straight line tangent to the lower portion of the curve is the **air-gap line**, corresponding to low levels of flux density within the machine. Under these conditions the reluctance of the machine iron is typically negligible, and the MMF required to excite the machine is simply that required to overcome the reluctance of the air gap. If it were not for the effects of saturation, the air-gap line and open-circuit characteristic would coincide. Therefore, the departure of the curve from the air-gap line as the field current is increased is an indication of the degree of saturation in the machine.³²

At the design stage, the open-circuit characteristic can be calculated using techniques such as finite-element analyses. A typical finite-element solution for the flux distribution around the pole of a salient-pole machine is shown in **Fig. 4.22**. The distribution of the air-gap flux found from this

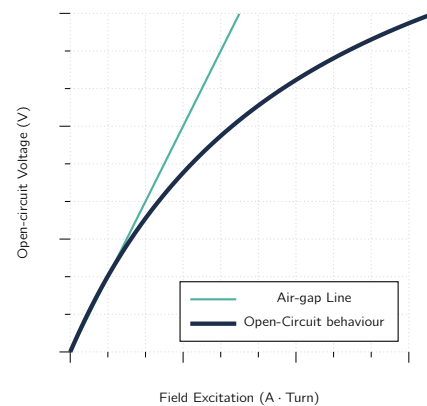


Figure 4.21: Typical open-circuit characteristic and air-gap line.

³²In typical machines the ratio at rated voltage of the total MMF to that required by the air gap alone is between 1.1 and 1.25.

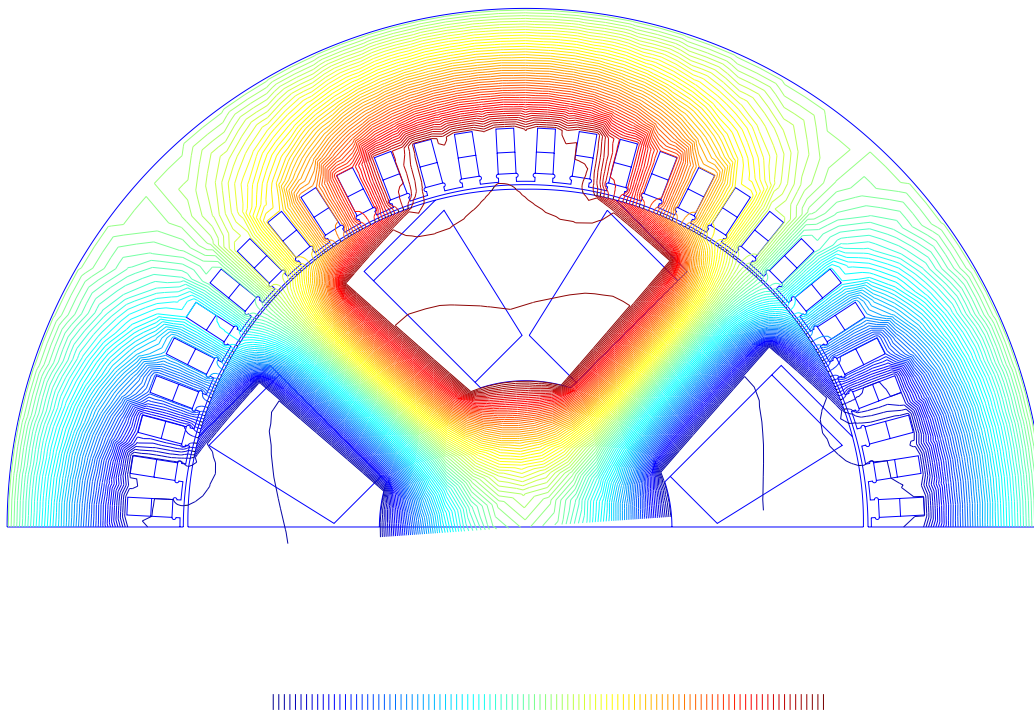


Figure 4.22: A FEM simulation of salient pole. The simulation is done using getDP.

solution, together with the fundamental and third-harmonic components, is shown in Fig. 4.36.

In addition to saturation effects, Fig. 4.36 clearly illustrates the effect of a non-uniform air gap. As we would expect, the flux density over the pole face, where the air gap is small, is much higher than that away from the pole.³³

As we have seen by now, the magnetization curve for an existing SynM can be determined by operating the machine as an unloaded generator and measuring the values of terminal voltage corresponding to a series of values of field current.

If we want to do the same thing for an IM, the machine is operated at or close to synchronous speed,³⁴ and the stator current is measured as a function of stator terminal voltage. The magnetization curve is produced by plotting the stator voltage as a function of stator current.

It should be emphasized that saturation in a fully loaded machine occurs as a result of the total MMF acting on the magnetic circuit. Since the flux distribution under load generally differs from that of no-load conditions, the details of the machine saturation characteristics may vary from the open-circuit curve of Fig. 4.34.

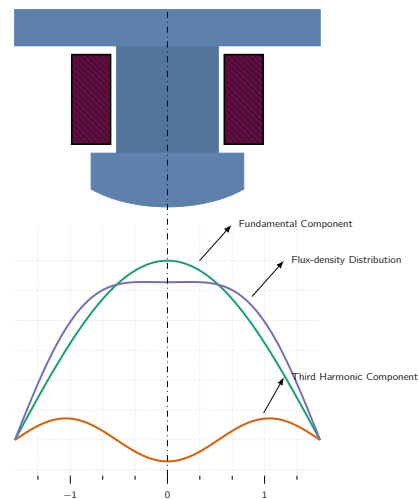


Figure 4.23: Flux-density wave with its fundamental and third-harmonic components.

³³This type of detailed analysis is of great use to a designer in obtaining specific machine characteristics.

³⁴in which case very little current will be induced in the rotor windings.

4.8 Leakage Flux

When we studied transformers in a previous chapter, we have seen that in a two-winding transformer, the flux created by each winding can be separated into two (2) components:

Mutual Flux

One component consists of flux which links both windings, causing coupling between the two coils.

Leakage Flux

the other consists of flux which links only the winding creating the flux. Contributes only to the self-inductance of each coil.

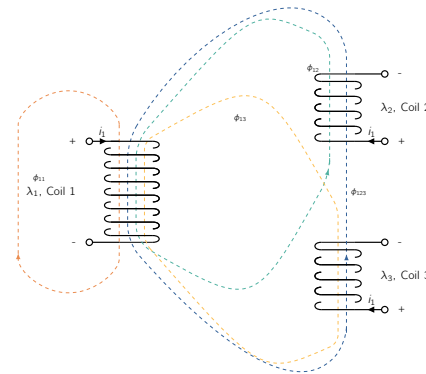


Figure 4.24: Three-coil system showing components of mutual and leakage flux produced by current in coil 1.

Note that the concept of mutual and leakage flux is meaningful only in the context of a multi-winding system. For systems of three or more windings, the bookkeeping must be done very carefully. Consider, for example, the three-winding system of Fig. 4.37. Shown schematically are the various components of flux created by a current in winding 1. Here ϕ_{123} is clearly mutual flux that links all three windings, and ϕ_{1l} is clearly leakage flux since it links only winding 1. However, ϕ_{12} is mutual flux with respect to winding 2 yet is leakage flux with respect to winding 3, while ϕ_{13} mutual flux with respect to winding 3 and leakage flux with respect to winding 2.

Electric machinery is a system of multiple windings, which requires a careful overview to account for the flux contributions of different types windings.

Air-Gap Space-Harmonic Fluxes Although single distributed coils create air-gap flux with a significant amount of space-harmonic content, it is possible to distribute these windings so the space-fundamental component is accentuated while the harmonic effects are greatly reduced. As a result, we can neglect harmonic effects and consider only space-fundamental fluxes.

Though often small, the space-harmonic components of air-gap flux do exist.

- In DC machines they are useful torque-producing fluxes and therefore can be counted as mutual flux between the rotor and stator windings.
- In AC machines, however, they may generate time-harmonic voltages or asynchronously rotating flux waves.

Slot-Leakage Flux Fig. 4.25 shows the flux created by a single coil side in a slot. Please observe that, in addition to flux which crosses the air gap, which contributes to the air-gap flux, there are

flux components which **cross the slot**. In a slot with coils from a single phase, this flux links only the coil that is producing it. It also forms a component of the leakage inductance of the winding producing it. In other cases, coils of two phases share a single slot and some of the slot flux is mutual between the phases. However, since this flux does not cross the air gap, it remains leakage flux with respect to any windings on the rotor.

End-Turn Fluxes Fig. 4.26 shows the stator end windings of an AC machine. The magnetic field distribution created by end turns is extremely complex. In general these fluxes do not contribute to useful rotor-to-stator mutual flux, and thus they, too, contribute to leakage inductance.

From this discussion we can see the self-inductance expression needs to be modified by an additional term L_l , which represents the winding leakage inductance. This leakage inductance corresponds directly to the leakage inductance of a transformer winding as discussed in Chapter 1.

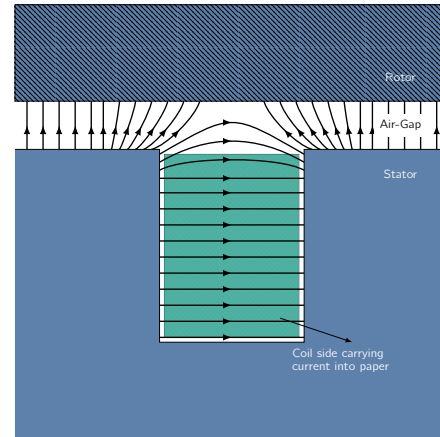


Figure 4.25: Flux created by a single coil side in a slot.

Although the leakage inductance is usually difficult to calculate analytically and must be determined by approximate or empirical techniques, it plays an important role in machine performance.



Figure 4.26: The end windings of an induction machine.

Glossary

AC An electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction.. 19, 20, 22, 26, 77–80, 84–87, 89, 92–96, 100, 102, 116, 117

DC One-directional flow of electric charge. Direct current may flow through a conductor such as a wire, but can also flow through semiconductors, insulators, or even through a vacuum as in electron or ion beams. The electric current flows in a constant direction, distinguishing it from alternating current (AC). A term formerly used for this type of current was galvanic current.. 44, 77–80, 84–87, 92, 116

EMF An energy transfer to an electric circuit per unit of electric charge, measured in volts. Devices called electrical transducers provide an EMF by converting other forms of energy into electrical energy. Other types of electrical equipment also produce an emf, such as batteries, which convert chemical energy, and generators, which convert mechanical energy. This energy conversion is achieved by physical forces applying physical work on electric charges. However, electromotive force itself is not a physical force. 28–30, 36, 102

IM an AC electric motor in which the electric current in the rotor that produces torque is obtained by electromagnetic induction from the magnetic field of the stator winding.. 78, 79, 85, 86, 95, 115

LHS Left Hand Side. 57

MMF The magnetic pressure which drives magnetic flux in a magnetic circuit, and is the work done to move a unit magnetic pole.. 7–11, 13–15, 19, 21, 28, 29, 31, 32, 35, 60, 88–99, 104, 106–115

PM An object made from a material that is magnetized and creates its own persistent magnetic field. 5, 16, 55, 66, 78, 79

RHS Right Hand Side. 57

RMS Equivalent direct current (DC) voltage of an alternating current (AC) source.. 19–21, 29, 30, 40, 48

SynM Synchronous Machine. 78–82, 84, 85, 87, 95, 106, 111, 113, 115

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